Development of a Model Climatology Tool to Assess Ensemble Variability and Uncertainty Using East Coast Cyclones

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Taylor Mandelbaum

We, the thesis committee for the above candidate for the

Master of Science degree, hereby recommend

acceptance of this thesis

Brian A. Colle - Thesis Advisor
Professor, School of Marine and Atmospheric Sciences, Stony Brook University

---

Sultan Hameed
Professor, School of Marine and Atmospheric Sciences, Stony Brook University

---

Trevor Alcott
Earth Systems Research Laboratory, Global Systems Division

This thesis is accepted by the Graduate School

Dean of the Graduate School

Abstract of the Thesis

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Probabilistic forecasting is an important tool for both public and private sectors. The use of ensemble models increases the awareness of uncertainty and errors in the output of a model. A common tool to diagnose uncertainty in an ensemble model is ensemble spread, or standard deviation. Ensemble spread is most commonly viewed in a geographic reference frame with implications of the most likely areas of uncertainty, and by extension, errors. Although the use of ensembles has increased, there exist many opportunities for better visualization of ensemble model output, which is a major objective of the Stony Brook University CSTAR project. The Ensemble Situational Awareness Table (ESAT), managed by the National Weather Service and Weather Prediction Center, compares forecasts from the North American Ensemble Forecast System (NAEFS) and Global Ensemble Forecast System (GEFS) to reanalysis (R-Climate) and model reforecast (M-Climate) climatologies. While M-Climate output from the GEFS Reforecast can place the current ensemble mean forecast in context, it does not yet assess the ensemble spread relative to similarly anoma-

lous events. We attempt to take the M-Climate diagnostic a step further by proposing a new index for spread, using a restricted climatology based on ensemble mean anomaly. We also look at the relationship between ensemble spread and the error for East Coast cyclones with the intent of providing context for the tool as well as the performance of EPS's ability to "predict" their own error.

Our goal is to output and justify an operational spread anomaly product that will complement the existing ESAT. To justify the tool, an analysis of 90 verified East Coast cyclones from 2007-2014 was performed for the GEFS, CMC, and ECMWF ensembles. Cases restricted to the winter (DJF) timeframe over the contiguous United States were chosen for this project. Mid-latitude synoptic cyclones are the most prevalent high impact events, especially for southern New York and New England. Initial test variables include mean sea-level pressure (SLP), surface temperature, and precipitable water, but SLP will be the focus of this study. By analyzing the ensemble spread for all members of the GEFS, a geographic and forecast cyclone-relative perspective of the magnitude and co-location of the spread and error can be assessed. Spread and mean absolute error (MAE) expand at different rates and are consistently different magnitudes, with maximum values of the GEFS MAE 270% greater than the spread for days 1-3 (short range) and 188% greater for days 5-7 (long range). RMSE values for the GEFS are similar. A method is developed which identifies the centroids of spread and MAE for the upper quartile of cyclone-relative data. The ECMWF does significantly better in both short and long range when looking at the distance between centroids, as well as the percentage of points on the cyclone-relative grid which overlap. The GEFS tends to perform the worst in both metrics. 500 hPa geopotential height patterns are also compared between the best and worst performing cyclone cases with respect to centroid distance and percent overlap where significantly different wave patterns are identified.

GEFS Reforecast cyclone forecasts are compared to GEFS operational forecasts to assess the similarity of products, where it is found that there are statistically significant differences in spread and MAE for most lead times. GEFS Reforecast cases at each point within a threshold of comparable standardized anomalies ($\pm 5\%$) to the forecast are utilized to construct a new spread M-Climate. Using this method, a subset standardized spread anomaly (SSA) can be calculated for each point on the domain. The resulting plot, together with the ensemble mean pressure contours of that forecast, displays a filtered, historical reference of uncertainty in the GEFS forecast. The SSA provides context for each lead time while identifying areas of anomalous spread, which shows a correlation with standardized error for short and long ranges with r values $\sim$0.54 and $\sim$0.64, respectively for forecast cyclone relative domain-averaged SSA and standardized MAE.

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List of Abbreviations

ACR: Analysis Cyclone Relative

CFSR: Climate Forecast System Renanalysis

CMC: Canadian Meteorological Centre

ECMWF: European Centre for Medium-Range Weather Forecasts (model)

ESAT: Ensemble Situational Awareness Table

GEFS: Global Ensemble Forecast System

FCR: Forecast Cyclone Relative

KISP: Islip Airport, Bohemia, NY

MAE: Mean Absolute Error

ME: Mean Error

MOS: Model Output Statistics

NCEP: National Centers for Environmental Prediction

NOAA: National Oceanic and Atmospheric Administration

NWS: National Weather Service

PDF: Probability Density Function

RMSE: Root Mean Squared Error

SSA: Standardized Spread Anomaly

STTP: Stochastic Total Tendency Perturbation

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1 Introduction

1.1 Background

Numerical weather prediction (NWP) models with one output solution, or deterministic models, have been a fundamental component of the weather enterprise since the mid 20th century (Charney 1948, Lewis 2005). Deterministic models follow the idea that the future state of a system is completely determined by its present state. The models then and now that are single-solution are limited by errors in initial conditions and physical parameterizations which propagate temporally and spatially. A lack of an exact representation of the atmosphere leads to an expansion of errors and, ostensibly, an incalculable number of potential solutions as a result of its non-periodic nature (Lorenz 1963). Probabilistic forecasting, otherwise known as ensemble modeling, was developed to address the limitations in single-solution models in a system which lacks periodicity (Leith 1974). The stochastic nature of ensemble models produce smaller errors in comparison to deterministic solutions at the expense of processing power (Epstein 1969).

Probabilistic forecasting is important for both public and private sectors. Methods of viewing predictability and uncertainty of high impact weather events such as cyclones are desired in the operational forecasting community (Novak et al. 2008). Improvements in relevant technology (processing power, data storage, etc.) have provided multiple organizations, such as the National Centers for Environmental Prediction (NCEP), European Centre for Medium-Range Weather Forecasts (ECMWF), and Canadian Meteorological Centre (CMC), the ability to run probabilistic models alongside deterministic products (Toth and Kalnay 1993, Molteni et al. 1996, Houtekamer et al. 1996). Probabilistic forecasts employ the use of statistical analysis to assess predictability (Ehrendorfer 1997, Kolczynski Jr et al. 2011) by attempting to address all sources of uncertainty in a

forecast. An ensemble produces a probability density function (PDF) for an atmospheric forecast parameter that ideally covers the range of potential solutions (Leith 1974). The use of ensembles, such as the Global Ensemble Forecast System (GEFS, Toth and Kalnay 1993), increases the awareness of such uncertainty and errors in the output of a model to forecasters, decision support services, and researchers.

Methods of viewing anomalies and uncertainties in a forecast are an important component to the forecast process. The sharpness, or variability, of the suite of solutions provides an understanding of the uncertainty within a forecast variable. The standard deviation of an ensemble model, or spread (Whitaker and Loughe 1998, WL98), is a common metric to understand the uncertainty of the model output. The 96 hour forecast of the operational GEFS ensemble mean and spread for 30 November 2007, valid 4 December 2007 00z, provides an example of a typical spatial ensemble mean and spread map (Fig. 1.1). The spread of a forecast illustrates how wide (or narrow) the distribution of solutions is for the forecasted variables. Spread can be visualized and represented in a geographical context or through ensemble model plume diagrams (Fig. 1.2), which graphically represent the variability of ensemble members over the length of the model's forecast run. Other probability visualizations and methods are utilized by end users, such as the Weather Prediction Center (WPC) and the National Weather Service (NWS) through methods such as AWIPS and statistical analysis (e.g. model output statistics (MOS), probability matched mean, and predictability horizon diagrams) (Novak et al. 2008).

1.2 Previous Work on the Role of Probabilistic Forecasting

Many studies have been performed assessing probabilistic forecasting using ensemble models. Hamill and Colucci (1998) assessed short range forecasts of the Eta-Regional Spectral Model against MOS. They found that the ensemble, corrected and fit to gamma distributions, performed at the same skill as MOS and occasionally outperformed with a small training sample. The study also stressed the need for proper dispersion amongst the ensemble members to accurately predict the skill of the forecast. Methods such as stochastic total tendency perturbation (STTP), which perturbs prognostic variables in the GEFS such as temperature and wind stochastically, are used to develop a suite of initial conditions that addresses such model uncertainties (Zhou et al. 2016, Hou et al. 2008) considered by Hamill and Colucci (1998). Buizza (1997) assessed the ECMWF 32-member ensemble (currently 50 perturbed members and 1 control member) against a perfect ensemble. They found that locations with comparatively small spread correlate with skill, but the spread did not grow as much as the control skill did. The results were indicative of an underdispersed ensemble which could be addressed by an increase in resolution or increase in ensemble size.

Studies focusing on uncertainty of ensemble modeling have focused largely on idealized models and variables which represent a more Gaussian distribution over temporal and spatial frameworks (WL98, Hopson 2012, Grimit and Mass 2007, Barker 1991, Kolczynski Jr et al. 2011), though there has been an increase in studies investigating less symmetric distributions such as that of sea level pressure, precipitation, and 10-meter winds (Van Schaeybroeck and Vannitsem 2016 (VSV16), McMurdie and Ancell 2014, Greybush et al. 2017). VSV16 utilized a host of different error and spread metrics to assess the error as a function of spread. The study detailed the construction of a method to determine if spread can be used as an accurate indicator of forecast uncertainty,

by using a regression of log-transformed spread and error to account for changes in variability. It was found that the spread could be used without skill degradation using the log-transformed spread. The study also noted that replacing the raw ensemble with a normal distribution improved the skill scores that were being used to evaluate the method. It is also noted in the study that there has been criticism of comparing the spread-error Pearson correlation coefficient to a perfectly-reliable model (Kumar et al. 2014). Biases were observed at medium-range forecasts when there were model errors, which correspond to model under-dispersion.

Grimit and Mass (2007) examined a probabilistic approach to the spread-error relationship by using probability forecasts based on events of similar spread, noting that the temporal variability affects its usage as a predictor. Other studies (Barker 1991, Hopson 2012) note the spread-error relationship in various iterations using "perfect" models that circumvent the issue of initial condition errors. Barker (1991) detailed a method of correlating mean square spread of the streamfunction averaged over the northern hemisphere to mean square forecast which produced low correlations at initialization due to the Monte Carlo approach taken in the study.

Hopson (2012) explored the ability for an ensemble system to accurately predict its own skill. The study argues that the Pearson correlation coefficient is inadequate due to multiple reasons, including model performance and the various disjoint definitions of spread and error being employed to assess their own performance. Hopson (2012) also introduced two metrics which assessed an ensemble model's ability to forecast its own error based on skill scores. These metrics were tested on forecasts of 10-meter temperatures in Utah and streamflow of the Brahmaputra River in Bangladesh. The results showed an increase in in reliability but a decrease in sharpness, i.e. an increase in variance.

WL98 showed that large values of spread can act as a predictor of error. By observing the

variations in spread over two winter seasons using the natural logarithm of spread as a predictor of day-to-day variability (β), it was shown that the error would increase with increased variability. Using streamfunction at 250 hPa, WL98 found the root mean squared error between the NCEP/NCAR reanalysis (Kalnay et al. 1996) and the NCEP ensemble mean (Toth and Kalnay 1993) showed a maximum correlation in the northern hemisphere (NH) (20°N to 90°N) between day 5 and day 6 (averaged about the NH domain), as well as a maximum of β around day 4 and 5. It was also shown that the variability, and subsequently the correlation, increased pole-ward and reach a maximum in the extreme northern regions about the North Pole.

1.3 Probabilistic Methods of Analyzing Ensembles

Novak et al. (2014) detailed an objective measure to utilize ensemble PDFs to compute percentiles of snowfall possibilities. The PDF is generated from a multimodel ensemble generated by the WPC with members from the NCEP Short Range Ensemble Forecast (SREF), North American Model (NAM), GFS, ECMWF, CMC, and GEFS to combine for 28 members during the 2012-2013 season and 32 during the 2013-2014 season. Statistical verification was performed during the 2013-2014 season by estimating the percentile range for snowfall amount between the 10th and 90th percentiles for 2693 cases from point forecasts. 83% of observations for the day-1 forecast fell within the 10th and 90th percentiles, where a perfect score would be 90-10, or 80%. The resulting verification shows the efficacy and possibility of novel probabilistic methods in the operational forecasting enterprise.

Toth et al. (2001) utilized the NCEP GEFS ensemble to assess the climatological uncertainty of 500mb height forecasts. 10 equally likely bins of heights was generated using NCEP-NCAR

reanalysis data (Kalnay et al. 1996), where the 14 GEFS ensemble members were binned into for each lead time. The bin with the most ensemble members is the mode, where the larger value of the mode corresponds to a sharper PDF and a more "confident" or certain forecast. The metric developed in this study utilized the ensemble mean and ensemble members to identify how many members fall into the bin of the ensemble mean, corresponding to predictability. A percentile is calculated by comparing the forecast to the preceding 30 days level of predictability. Results indicated that longer range forecasts (days 6-15) identified as highly predictable had similar accuracy as shorter range forecasts with predictability below average. The authors also noted the importance of the interpretation of ensemble spread. When using spread in an operational context, large values do not directly correspond with large forecast uncertainty for "extreme" cases (Toth 1992, Ziehmann 2001). The tool is available at: $http://www.emc.ncep.noaa.gov/gmb/yluo/htmlpqpf/rmop.html$.

1.4 Climatological Context for Ensemble Forecasts

A method to analyze the significance of a forecast is by utilizing climatological anomalies, standardized based on the variability of the climatology. Research on normalized and standardized anomalies has focused on reanalysis climatologies (R-Climate). Standardized anomalies utilize the mean and variability of a climatology to provide a first-order value describing the anomalous nature of a forecasted variable. Hart and Grumm (2001) used the 2.5° x 2.5° NCEP reanalysis dataset (Kalnay et al. 1996) between 1948 and 2000 at 10 pressure levels (8 for specific humidity) with a domain centered about the eastern portion of the United States to develop a climatology of multiple fields from 1961-1990. Variables which were used in the calculations included sea level pressure, multiple thickness levels, pressure-level heights, and pressure-level temperatures.

Multiple case studies within the paper indicated that significant snowstorms exhibit significant standardized anomalies compared against their R-Climate when plotted geographically. The standardized and normalized anomalies did not exhibit Gaussian characteristics, but fell within the limits set by Chebyshev's Theorem.

Graham and Grumm (2010) also performed this analysis on the west coast of the United States, using 1000 to 200 hPa height, temperature, wind, and precipitable water to calculate a metric $M_{variable}$ over a 6 hourly 30 year reanalysis climatology from 1971-2000, where storms identified included cutoff lows that produced tornadoes in southern Arizona and record rainfalls in Montana (Lussky 1989). Storms were ranked based on the magnitude of $M_{variable}$, where the top events were significantly anomalous in all variables examined. Return intervals were also calculated for the anomalies to provide a statistical context of the frequency of the events. Forecasters can utilize these anomalies to understand the potential significance of the verifying event.

Forecasters and decision support services can utilize the Ensemble Situational Awareness Table (ESAT), managed by the National Weather Service, Western Region (NWS WR) and the National Oceanic and Atmospheric Administration's Weather Prediction Center (NOAA WPC). The ESAT places the ensemble mean in a climatological context to aid forecasters in identifying significant forecasted events (Graham et al. 2013). The table compares forecasts from the North American Ensemble Forecast System (NAEFS, Candille 2009) and the current operational GEFS (Wei et al. 2008) to R-Climate (Saha et al. 2010) and model reforecast climatologies (M-Climate), generated from the Global Ensemble Forecast System Reforecast, version 2 (GEFS-R, Hamill et al. 2013). Standardized anomalies, percentiles, and return intervals are calculated to assist in identifying potentially significant weather events (Graham et al. 2013).

1.5 Motivation

While current M-Climate products generated from NOAA's 2nd generation global ensemble reforecast dataset (Hamill et al. 2013) can place the current ensemble mean forecast in context, there has been few studies which assess the ensemble spread relative to similarly anomalous events in an M-Climate. The ESAT provides a method of assessing the anomaly of a forecast but there is not a method that assesses the anomaly of the uncertainty of a forecast. We have attempted to take the M-Climate diagnostic a step further by assessing whether confidence in the developing anomaly is unusually high or low. By comparing a forecast to a modified climatology, it is possible that the model spread for that forecast can be identified as greater or less than normal. It is also important to compare the forecasts in the new climatology, generated using the GEFS Reforecast, version 2 (Hamill et al. 2013) to the same forecasts in the operational model to assess the similarity in distributions.

Although model spread provides valuable information in the output of ensemble models, it is inherently limited by multiple factors. Spread climatologies exhibit a log-normal distribution (WL98) due to the incidence of extreme events (e.g. mid-latitude cyclones) with strong spatial and temporal gradients (Fig. 1.3a, b). The composite climatology of spread exhibits a south-to-north gradient (Fig. 1.4), which obfuscates the significant anomalies exhibited around mid-latitude cyclones insert south to north gradient map, thus there exists a fundamental issue when examining spread in an historical context. By accounting for the distribution, more relevant information can be gathered when putting the forecast in the context of the M-Climate. Comparing a forecast to similarly anomalous events can assist forecasters and decision support services with risk assessment relative to confidence of a forecast. If the spread (relative to other events of similar anomaly)

is average or lower, this should translate to less uncertainty.

With the current state of ensemble forecasting, there exists many underutilized opportunities for visualization of ensemble variation and uncertainty beyond spread. New and novel products to convey uncertainty to end users have been of interest to the community (Novak et al. 2008). Communicating uncertainty has continued to be a challenge to forecasters and the weather enterprise at large, where barriers to entry (i.e. complexity or the ability to process large amounts of data) have been a limitation to furthering the state of the science.

It is also important to understand the immediate impacts of analyzing spread in an operational context. The spread of a forecast is often thought of to denote the likelihood of errors in the forecast. For example, an area with a large spread value near a cyclone is more likely to exhibit large errors, where an area in a ridge with a lesser spread value will likely have smaller errors (Sivillo et al. 1997). There has been little research which looks at the colocationality between ensemble spread and ensemble error, which are often considered to be spatially related with each other.

Given the idealized nature of some previous studies analyzing spread and error characteristics, more accessible uncertainty metrics would add valuable information to the forecasting process. Focusing on high impact events, specifically winter cyclones near the east coast of the United States, can provide new information on the nature and interpretation of spread for decision makers and forecasters. Mid-latitude cyclones, which are represented in ensemble models most commonly with mean sea level pressure (SLP), impact the east coast of the United States frequently in meteorological winter, causing significant economic disruption (Goodwin 2002). Considering these factors and the desire for the weather enterprise to transition to a probabilistic system of disseminating forecast information (Hirschberg et al. 2011), this thesis will focus on answering a few key

questions:

1. How does ensemble spread and mean absolute error of sea level pressure relate for cyclones in the eastern United States?

2. What effect does constraining the spread climatology to similar events have on the spread-error relationship?

3. Does the GEFS Reforecast and the operational GEFS have a similar spread-error relationship?

4. Are there distinct synoptic patterns which correspond to patterns in the spread-error relationship?

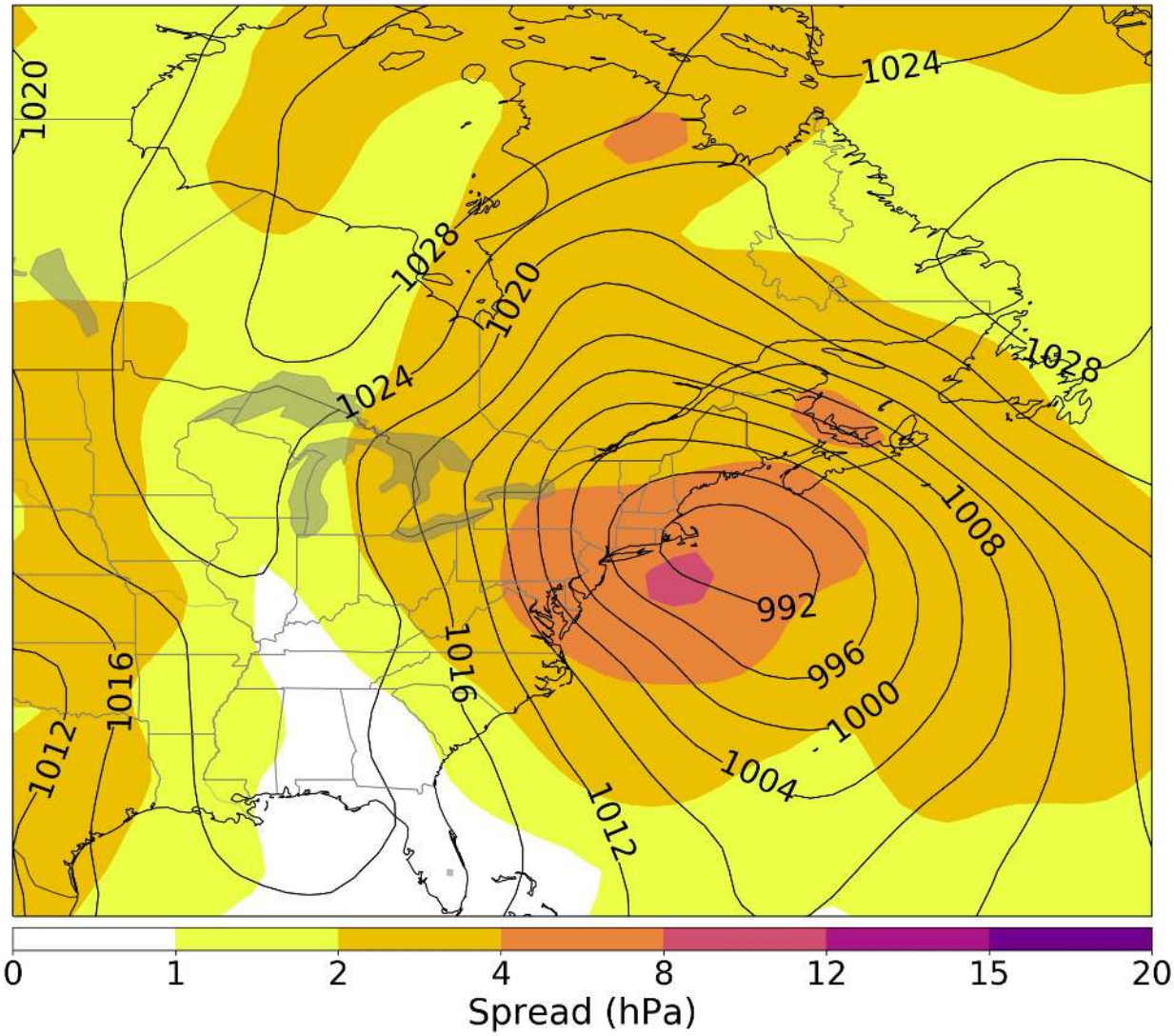


Figure 1.1: GEFS ensemble mean and spread forecast at hour 72, valid 26 Feb 2010 00z.

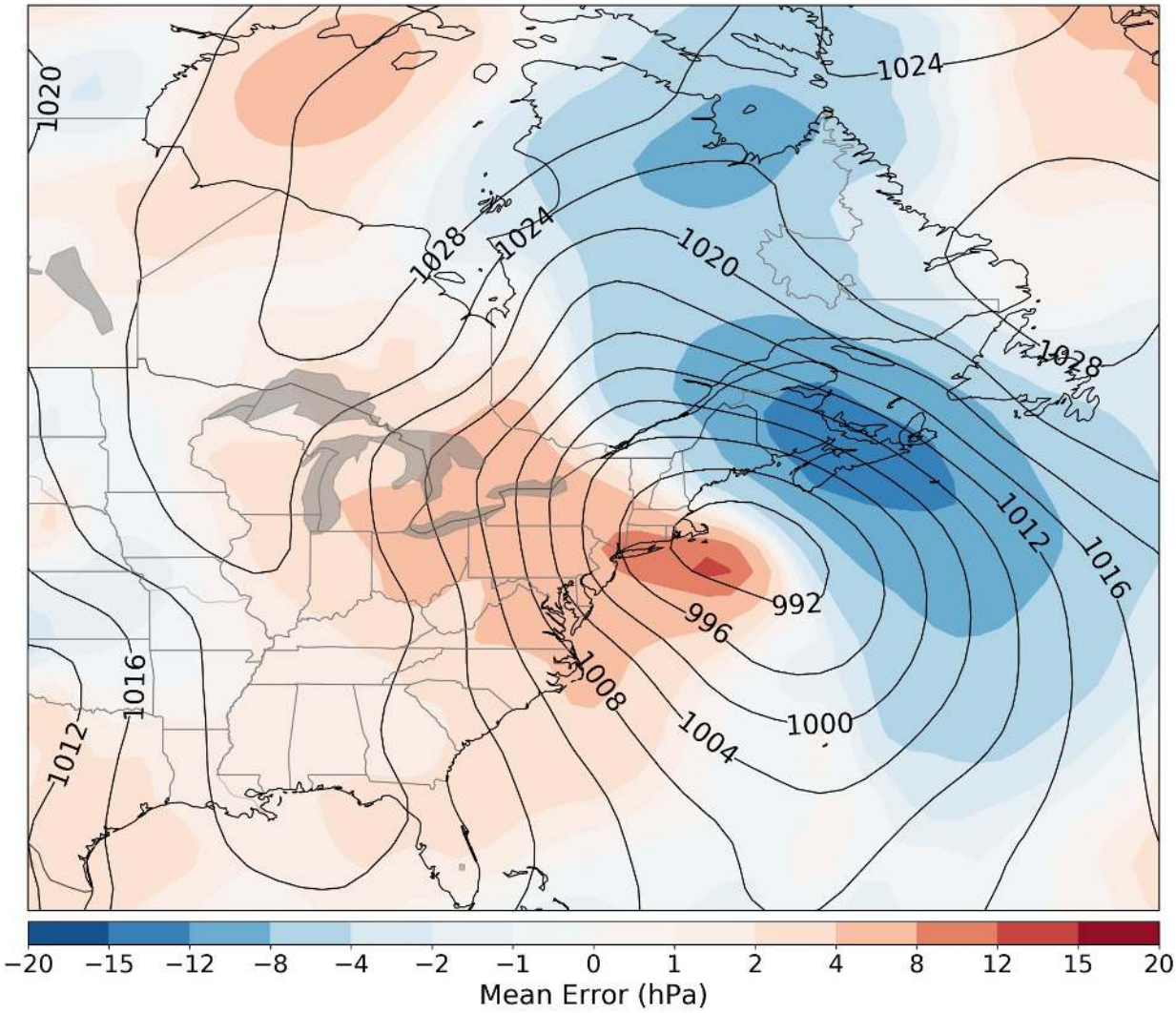


Figure 1.2: Similar to Fig. 1.1 but the GEFS ensemble mean error forecast. Positive values indicate a lower analyzed SLP, negative values indicate higher analyzed SLP.

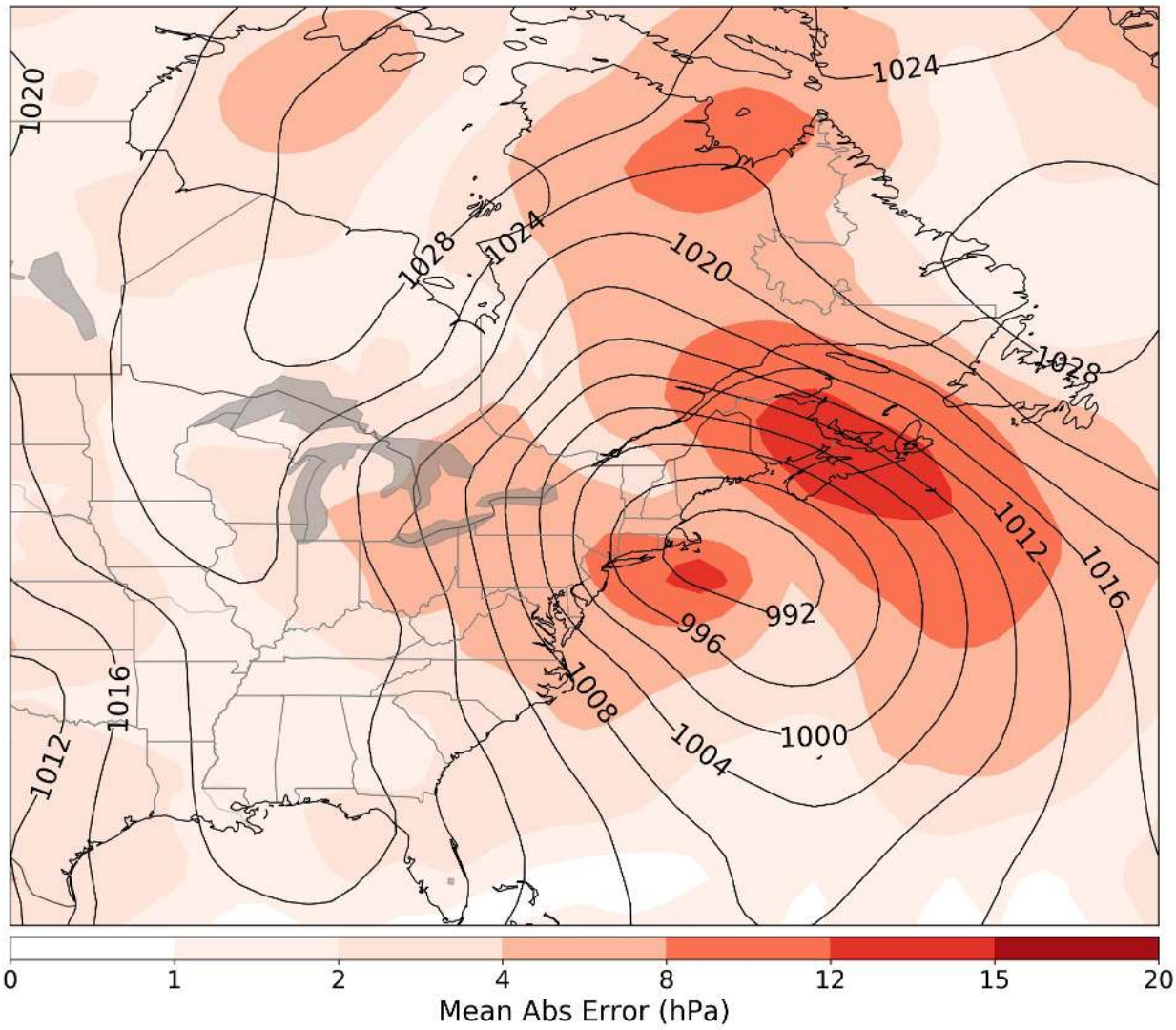


Figure 1.3: Similar to Fig. 1.1 but the GEFS ensemble mean absolute error forecast.

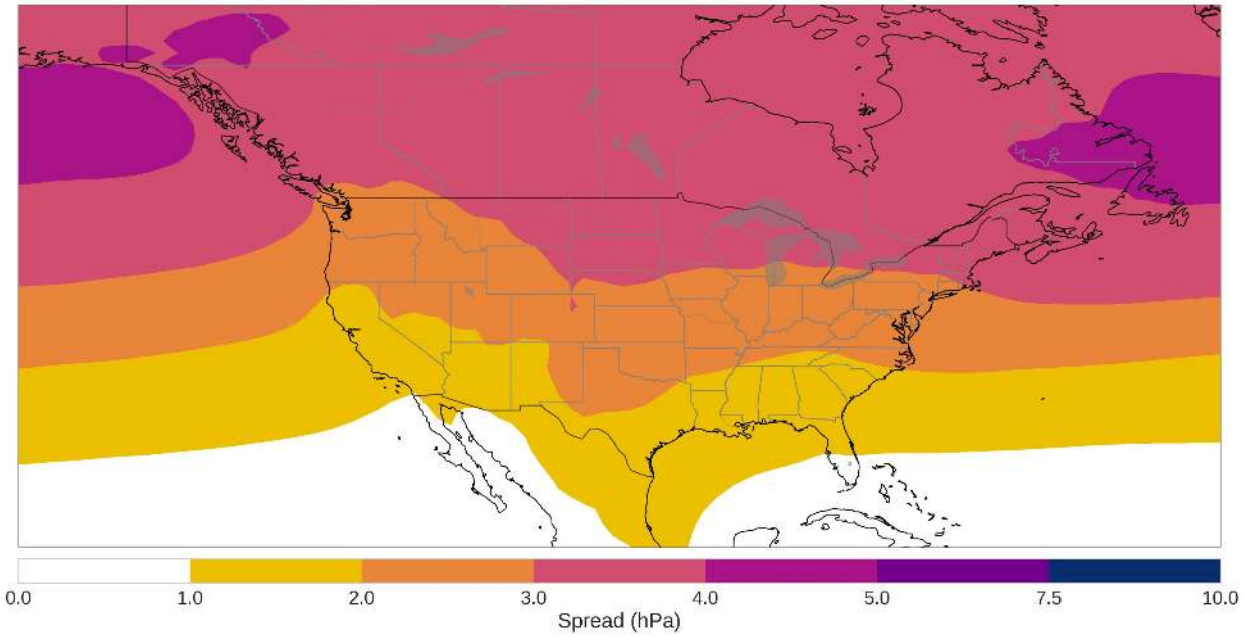


Figure 1.4: December-January-February average GEFS Reforecast ensemble spread in North America, 1985-2017. Darker colors indicate larger values.

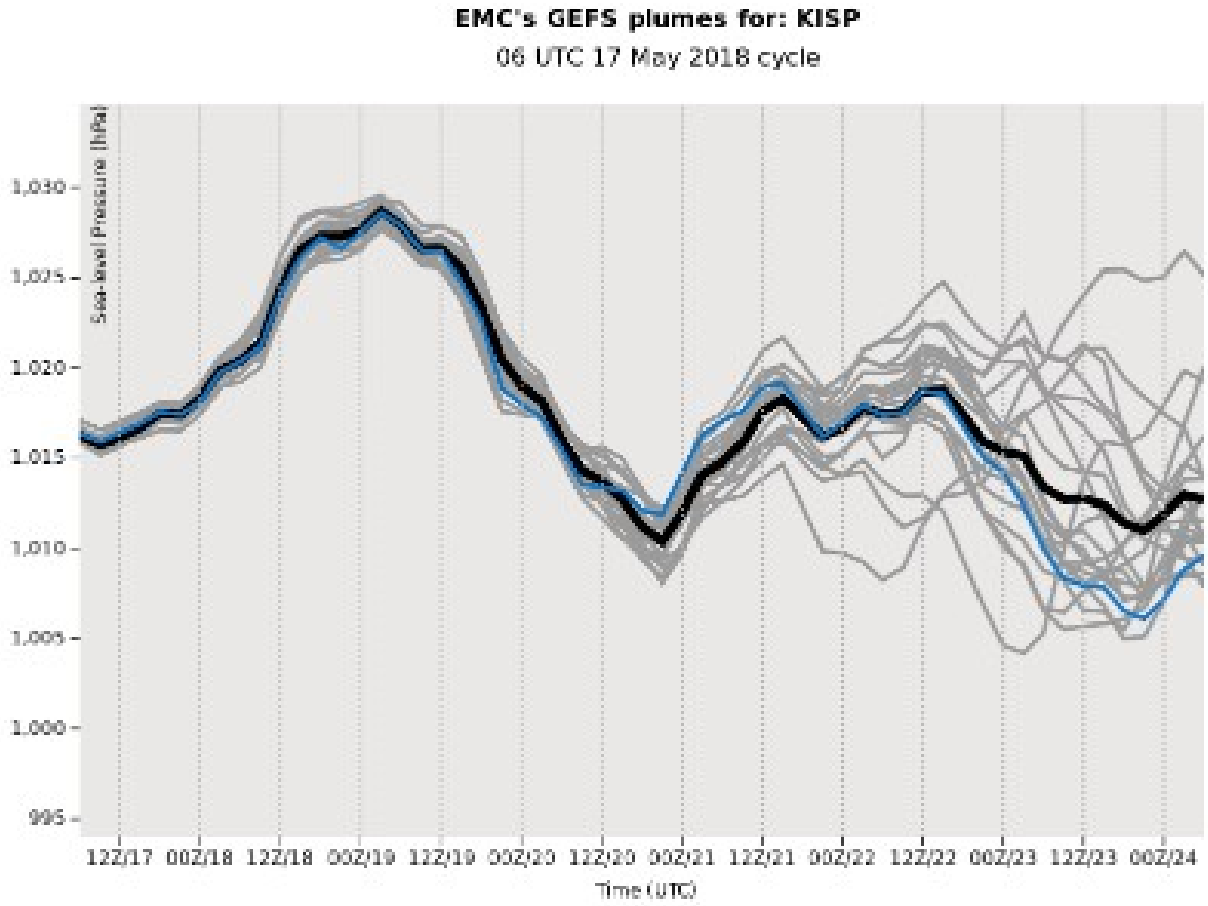


Figure 1.5: An example of a way to visualize ensemble spread, in this instance the mean sea level pressure plume diagram for KISP for the 17 May 2018 06z GEFS forecast. The ensemble mean and the control member are denoted by the black line and blue line, respectively.

2 Data & Methods

2.1 Description of Data

There are two main sections to this study: the relationship between ensemble spread and ensemble mean absolute error, and the construction of a utility to analyze subset spread anomaly (and the benefits and limitations of the procedure). Data were calculated from a blend of model reforecast, operational model, and verification/analysis datasets. Unprocessed ECMWF Reanalysis (ERA-40, ERA-Interim) data (Uppala et al. 2005, Berrisford et al. 2009) was downloaded from The Observing System Research and Predictability Experiment (THORPEX) Interactive Grand Global Ensemble (TIGGE) project (Bougeault et al. 2010) for 32 years from 1985 to 2017. Sea level pressure (SLP) and 500 hPa heights were both downloaded for the purposes of analysis data to calculate mean absolute errors and to observe and compare patterns in synoptic scale flow.

GEFS-R data were taken for the December, January, February period (DJF), beginning 21 November and ending 10 March for the purposes of utilizing 21-day centered climatologies (Grumm and Hart 2001) at 24 hour increments, beginning 21 November 1985 at 00z. The range of dates was chosen to utilize the largest possible sample space when partitioning the spread climatologies as well as the availability of the GEFS-R data, which begins December 1984. The data were downloaded previously interpolated to a 1° x 1° grid and utilized at the same resolution. The ERA-I data were also downloaded at 1° x 1° resolution. The ERA-I data was used in place of GFS analysis data to address any potential biases present in comparing an ensemble prediction system (EPS) with its own reanalysis product. Previous studies have shown that the ERA-I performs slightly better than other reanalysis products, though there is evidence of a warm bias over land (Mooney et al. 2011).

GEFS-R data were utilized from 1985 to 2017 to construct a 32-year climatology for SLP, 850 hPa temperatures, 500 hPa heights, precipitable water (PWAT), and 6-hour quantitative precipitation forecasts (QPF). All variables other than SLP have been analyzed but will not be focused on for the purposes of this research. Data were downloaded at a 1° x 1° grid resolution for hours 6 through 168, as the GEFS data was also interpolated to the same resolution. Data analyzing the spread and error relationship will focus exclusively on day-to-day forecasts, between hour 24 to 168, or days 1 through 7. The GEFS-R portal provides a useful feature in downloading ensemble mean and spread data in place of downloading all 11 members and calculating both manually. In the interest of data constraints as well as the goals for this research, only mean and spread data were downloaded for the aforementioned variables.

GEFS data for cyclone identification and analysis was utilized in correspondence with Nathan Korfe for 2007-2014 (Korfe and Colle 2018). Although the data were continuous, the GEFS resolution was improved during the time period of cyclones in this study from T190 to T254, which has consequences regarding the change in model skill (Korfe and Colle 2018). 90 cyclones were identified in the analysis in a boxed region from 33° N, 88° W to 48° N, 68° W (Figs. 2.1, 2.2). Forecasts were available every 24 hours at the 00z cycle, and as such the identified cyclones were only selected if they met the criteria at a valid time of 00z for purposes of maintaining the same lead times throughout the study for each cyclone. ECMWF and CMC data were also utilized and processed using the same methodology as the GEFS data. The CMC data contains 89 instead of 90 cyclones due to missing data for the December 26, 2008 cyclone in the dataset used.

Cyclones for the years of interest were selected based on a set of other criteria that utilized Korfe's matched analysis track dataset. The cyclones also had to have a central low pressure of at least 998 hPa, and were identified as uniquely identified cyclones (Korfe and Colle 2018). A

dataset was generated with latitude and longitude locations, central pressure, valid time, and cyclone ID. Valid times were corrected due to a discrepancy between MATLAB and python ordinal time by 366 days and 12 hours. Cyclones were then identified within the ERA-I analysis using this dataset by identifying minimum pressures for each cyclone case in the predefined bounding box. A secondary procedure utilized a selection criteria with varying thresholds to maintain a minimum of one centroid identified to add another method to address location accuracy, though the similarity in location made it advantageous and less computationally intense to use the minimum SLP identification. The Hodges tracking algorithm in part utilizes a similar procedure of identifying features using image filters on various datasets (Hodges 1994).

2.2 Methods

2.2.1 Cyclone Relative Procedure

To assess the relationship of DJF mid-latitude cyclones in the climatology, compositing a geographical domain to identify patterns would "wash out" information between cyclones in different locations. In order to account for this in the study, it was advantageous to utilize a different spatial framework to compare cyclones. The use of a cyclone relative approach centers all analysis products, ensemble mean, spread, and error about a central point. Two types of cyclone relative frameworks are utilized: centered on the ERA-I analysis cyclone minimum SLP, and centered on the mean position of the ensemble members' minimum SLP positions. A caveat to this approach is if there are two cyclones within the domain box, such as two cyclones below 998 hPa. The analysis cyclone relative (ACR) approach produces one 11°x11° box for each cyclone. The forecast cyclone relative (FCR) approach produces an 11°x11° box for each forecast hour for each cyclone,

depending on the ensemble members' latitude and longitude centroids, or:

$$\frac{1}{n}(\sum x_n, \sum y_n) \tag{1}$$

Such that the cyclone center is identified not by the minimum SLP on the ensemble mean field, but the unweighted position of the minimums of the individual ensemble members (Fig. 2.3). The 11°x11° box size was chosen as larger boxes would extend beyond the range of the available data. The ensemble MAE for the GEFS, CMC, and ECMWF was tested to identify if the average about the box size was significantly different between 5°, 7°, 9°, and 11° box sizes (Figs. 2.4, 2.5, 2.6). The boxes were not significantly different and tended to converge as the size increased. The position of the FCR box shifts with each lead time and must be accounted for when considering the model's relationship between spread and error. This method also addresses both the issue of weighting the center based on the magnitude of the low pressure systems in each ensemble, as well as the issue of an ensemble mean SLP field not producing a local or global minima to process a cyclone center from. All cyclone relative data were taken via the array indices identified, where:

$$latitude_c \pm 5^\circ, longitude_c \pm 5^\circ \tag{2}$$

such that each cyclone relative array centered on the SLP centroid has 5 gridpoints adjacent to each side, leading to an 11°x11° array for 7 days of forecast hours (days 1-7), corresponding to around 1100 km x 1100 km.

2.2.2 Ensemble Mean and Spread

The most common application for ensemble forecasting is ensemble mean and spread, which condenses many individual perturbed members of an ensemble into two 2-dimensional datasets which can be viewed in tandem with each other. The decision to explicitly focus on these two procedures is largely due to the frequency that the methods are used in the operational forecasting and decision making process. The ensemble mean is also considered the "best guess" forecast (WL98), which can then be assessed in situations of extremes, such as during cyclogenesis off the East Coast. The ensemble mean is defined as:

$$\frac{1}{n}\sum_{i=1}^{n} F_i \tag{3}$$

Where n is the number of ensemble members and F_i is ensemble member i.

The ensemble spread for the purposes of this research is considered only in the context of standard deviation. Other metrics, such as variance, range, and log-normalized residuals have been utilized in study (VSV16, Toth et al. 2001, Hopson 2012). The spread of a forecast denotes the probability density function's (PDF) variance, which provides a first-order quantitative metric to assess how "uncertain" or "confident" a forecast is. Spread is defined as:

$$\sqrt{\frac{\sum_{i=1}^{n}(F_i - \overline{F})^2}{n}} \tag{4}$$

Where $\overline{F}$ is ensemble mean, F_i is the ensemble member i, and n is number of ensemble members.

2.2.3 Ensemble Error

Ensemble error can be quantified in multiple metrics depending on the intent of the data. Typically ensemble mean error (ME) is represented by:

$$\frac{1}{n}\sum_{i=1}^{n} F_i - A \tag{5}$$

Where i is the ensemble number, n is the total number of ensemble members, F_i is the ensemble forecast, and A is the analysis. ME is often utilized to assess biases in model performance. The ensemble ME takes into account both positive and negative values in the field being analyzed. The benefit of this measure is an easy metric to gauge if the performance of a model variable is consistently positive or negative. Spatially, ensemble ME can provide context on an explored variable's positional bias or magnitude bias. Positional biases in ME manifest in spatial dipole (positive and negative tendencies adjacent to each other) patterns. Magnitude biases present as areas with either a positive or negative mean error tendency. A well calibrated forecast (with respect to the dimensionality being assessed) is likely to have a value of zero for ensemble ME. There is also the potential for large errors to be cancelled out due to the nature of the ME calculation, so careful attention must be paid to the variability of the sample space. Mean absolute error (MAE) simplifies the ME by changing the sign for all differences below zero of each ensemble member prior to averaging, which is represented by:

$$\frac{1}{n}\sum_{i=1}^{n} |F_i - A| \tag{6}$$

Such that the resulting values, plotted on a spatial grid, are non-negative. The MAE can be

compared to spread due to the similarity in both expansion of each variable as well as their magnitudes. Often, ensemble spread is compared to root mean squared error (RMSE), but for the purposes of this study, RMSE is not as similar to spread as MAE is (Fig. 2.9). The goal of comparing error and spread is best represented by MAE due to the similarities in expansion and magnitude to spread, though RMSE, when calculated alongside spread using the method explained by Fortin et al. (2017), is similar to MAE. For the GEFS Reforecast, the MAE utilized in this study is:

$$\frac{1}{n}\left|\overline{F} - A\right| \tag{7}$$

Where $\overline{F}$ is the ensemble mean due to the dataset used in this study only containing the ensemble mean and spread of the GEFS Reforecast.

2.2.4 Ensemble Spread/Error Quartile Calculations

Forecasters often look to spread to present a probabilistic measure of where the error is likely to occur in a model forecast. There have been few, if any, studies which address the colocationality of spread and error in global models. There is value in assessing the relationship of how well the spread acts as a "predictor" of error, henceforth represented by MAE. Two methodologies are introduced in this study to gauge the performance of the GEFS, ECMWF, and CMC spread and error relationship from the FCR perspective. Both approaches utilize the upper quartile of FCR gridded data for spread and MAE, where:

$$\begin{cases} n : n \geq p_{0.75} \\ 0 : n \leq p_{0.75} \end{cases} \tag{8}$$

Such that the top 30 values from each forecasted cyclone are preserved and the other 75% of data is removed. This method prevents weighting of less significant spread and MAE data. The first approach, upper quartile error and spread centroids (error and spread centroids henceforth), utilize a mass weighted centroid approach to find the center for each metric (Figs. 2.8, 2.9) The approach provides a latitude and longitude of the centroids and the distance between the centroids calculated using a haversine formula:

$$2rsin^{-1}(\sqrt{sin^2(\frac{lat_2 - lat_1}{2}) + cos(lat_1)cos(lat_2)sin^2(\frac{lon_2 - lon_1}{2})} \tag{9}$$

Where r is Earth's radius and lat and lon are the latitudes and longitudes of the centroid points.

The second approach, upper quartile grid point spread/error overlap (percent overlap here on), utilizes the indices of the upper quartile of spread and MAE to calculate the percentage of the values which overlap, such that:

$$O = \frac{100}{m}\sum_{i=1}^{m}\begin{cases} 1 : \sigma_{x,y} \wedge MAE_{x,y} \neq 0 \\ 0 : \sigma_{x,y} \vee MAE_{x,y} = 0 \end{cases} \tag{10}$$

Where σ is the upper quartile spread values at point (x,y), MAE is the upper quartile MAE at point (x,y), and m is the number of points in the upper quartile (in the 11x11 array, that number is $11 * 11 * 0.25$, or 30).

2.2.5 Standardized Spread Anomaly

The original purpose of this research was to develop a metric that utilizes M-Climate to compare the spread (uncertainty) of a forecast to historical events of similar magnitude anomaly. As

a result, the standardized spread anomaly approach was constructed based on the standardized anomaly, where:

$$\frac{F_{x,y} - \overline{F_{x,y}}}{\sigma_{x,y}} \tag{11}$$

Where $F_{x,y}$ is the forecast at a gridpoint, $\overline{F_{x,y}}$ is the climatology, and $\sigma_{x,y}$ is the standard deviation of that climatology.

Different approaches were considered for the subsetting procedure for the M-Climate, but it was settled to have a similar approach to Toth et al. 2001 where the forecast was identified and 5% of the historical spread forecasts above and below the mean standardized anomaly value were taken from the M-Climate, or

$$C_{\sigma s} = \begin{cases} C_{\sigma P_{\sigma_{x,y} \pm 0.05}} : P_{0.10} \leq P_{\sigma_{x,y}} \leq P_{0.90} \\ C_{\sigma P_{0.90-1.00}} : P_{\sigma_{x,y}} \geq P_{0.90} \\ C_{\sigma P_{0.00-0.10}} : P_{\sigma_{x,y}} \leq P_{0.10} \end{cases} \tag{12}$$

Where $\sigma_{x,y}$ is the forecast spread at point (x, y), C is the spread M-Climate at (x, y), P_i denotes percentile of the spread M-Climate at percent i, and $C_{\sigma s}$ is the subset spread M-Climate

The standard deviation was taken from the newly constructed $C_{\sigma s}$. The forecast and the M-Climate spread values are utilized to calculate standardized anomalies. This calculation is repeated for each gridpoint for each forecast. In order to maintain seasonality and due the extent of the GEFS Reforecast's temporal coverage, each subset spread M-Climate is a smaller distribution (n=40-67) of the total dataset of GEFS Reforecast values (n=3528-3598).

Ensemble spread is not represented by a Gaussian distribution and the subset spread M-Climate is not as well. As a result, utilizing standardized anomalies to provide context upon the uncertainty

of an event is inherently problematic. To address this, a simple normalizing approach which transforms $C_{\sigma s}$, along with the forecast spread included with the data, to a range of -0.99 to 0.99 is performed, where:

$$(0.99 - (-0.99))\frac{C_{x,y} - min(C_{x,y})}{max(C_{x,y}) - min(C_{x,y})} + -0.99 \tag{13}$$

The data are then transformed to a normally distributed PDF by a Fisher transformation, or:

$$C_{F_{\sigma s}} = arctanh(C_{\sigma s}) \tag{14}$$

The result restores a statistical validity to the standardized anomalies from the resulting calculations, where the traditional values of $\pm$ 1, 2, and 3 σ represent $\pm$ 68, 95, and, 99.7% percent of values, respectively. The resulting plots are sensitive to changes in $C_{F_{\sigma s}}$, and as such the data is smoothed with a Gaussian filter. The approach provides a mesoscale to synoptic scale historical context of uncertainty.

When comparing SSA to error, it is important to note the scale which spread is transformed into as well as the context of the subset M-Climate. As a result, it is also prudent to create a product from MAE which can relate to SSA, where:

$$(0.99 - (-0.99))\frac{MAE_{x,y} - min(MAE_{x,y})}{max(MAE_{x,y}) - min(MAE_{x,y})} + -0.99 \tag{15}$$

$$MAE_{x,y} = arctanh(MAE_{x,y}) \tag{16}$$

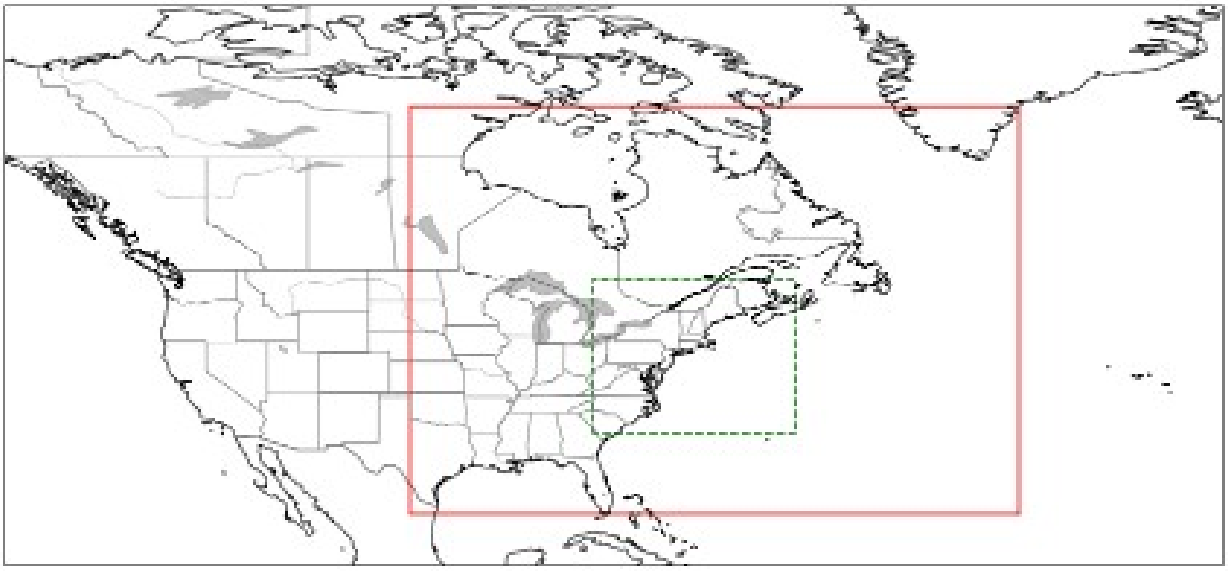


Figure 2.1: The domains utilized in this study. The solid red box indicates the extent of model data, the green hatched box represents the location which cyclones were identified.

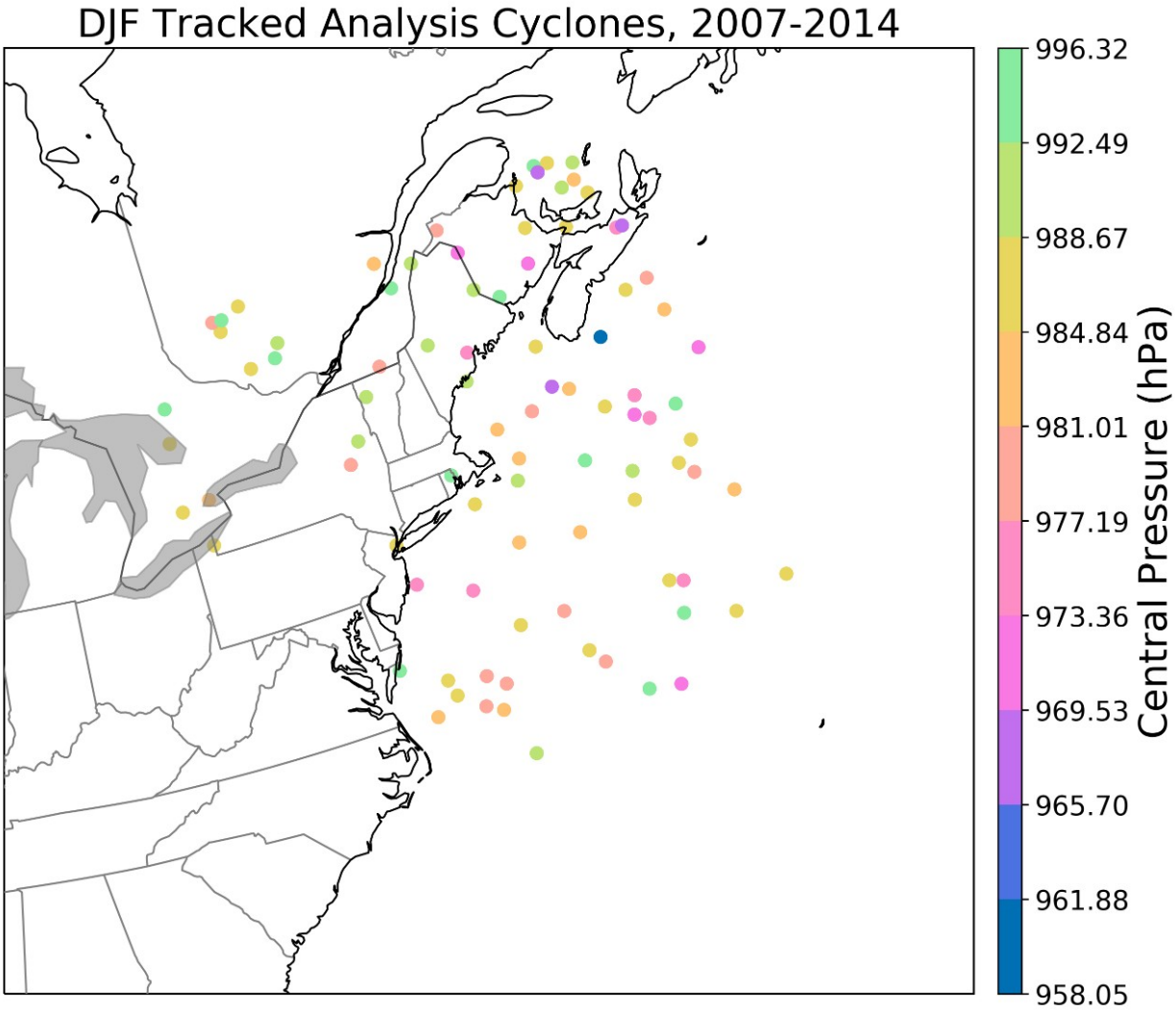


Figure 2.2: Locations and magnitudes of all tracked cyclones from Korfe and Colle's (2018) dataset and restrictions set in this study.

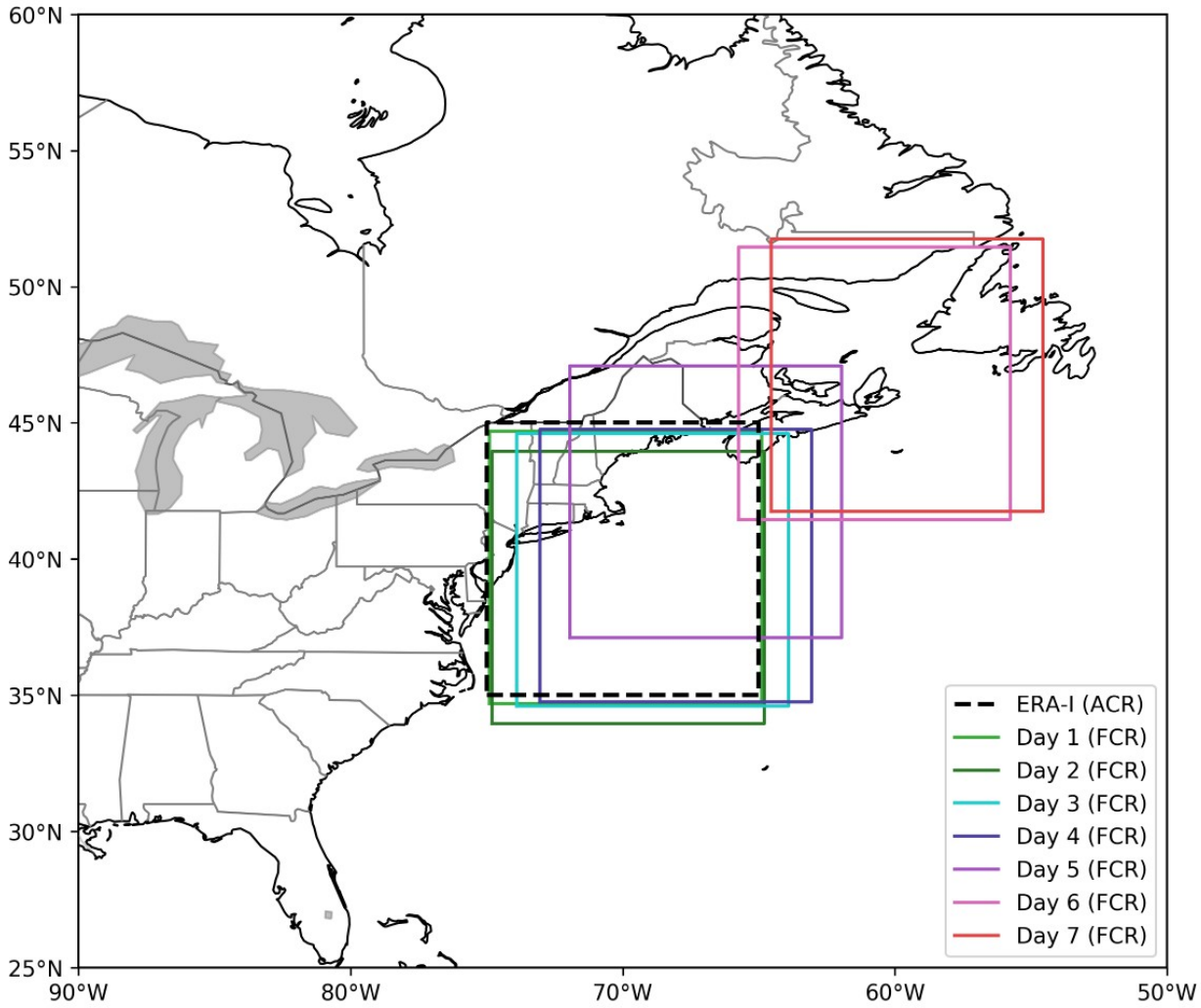


Figure 2.3: Locations of the analysis cyclone relative (ACR) box (black hatched line) and the forecast cyclone relative (FCR) boxes (colored lines).

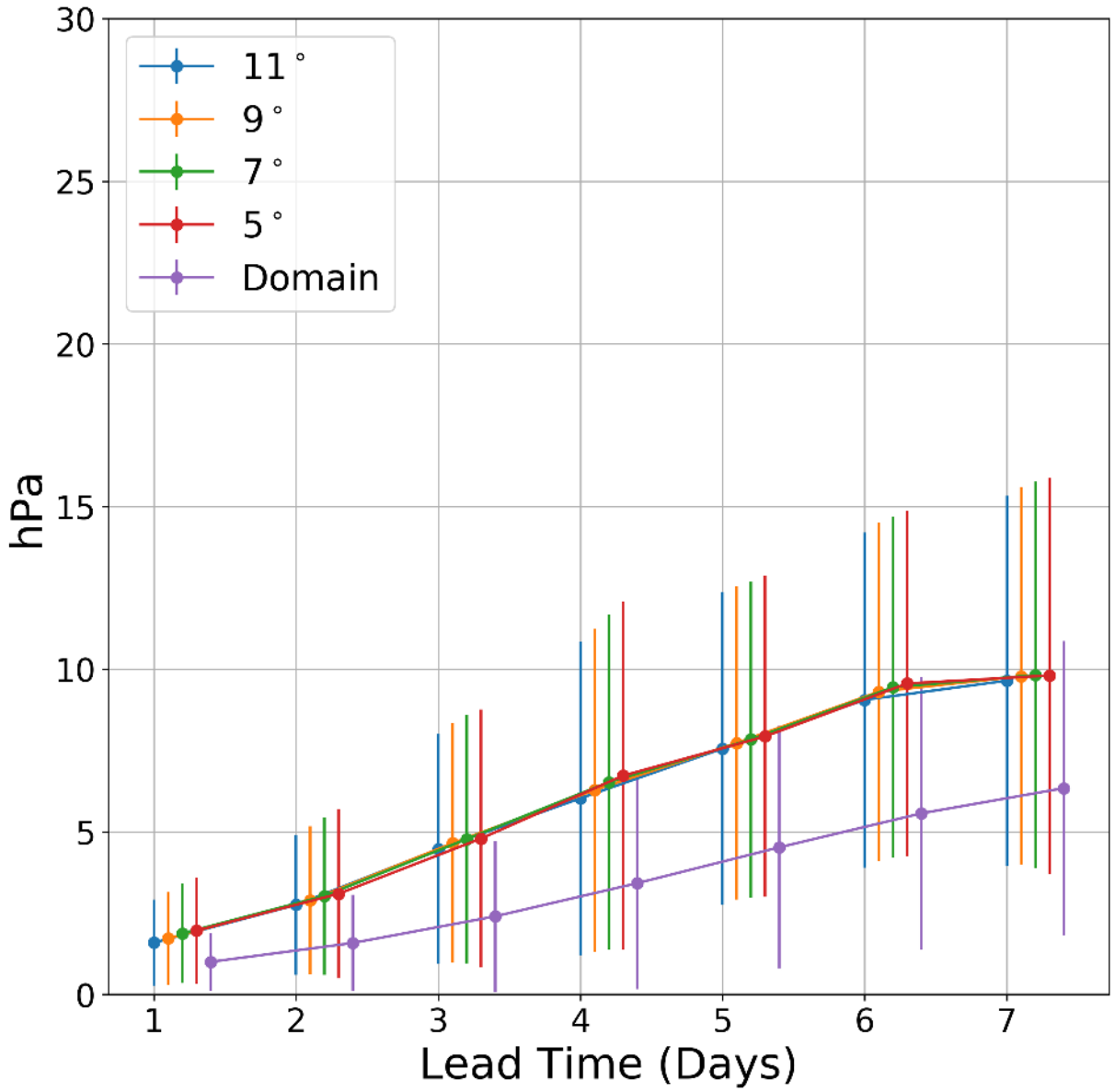


Figure 2.4: Comparison of box averaged MAE between different cyclone relative box sizes for the GEFS. The averaged GEFS domain MAE from Korfe and Colle's (2018) dataset is also plotted.

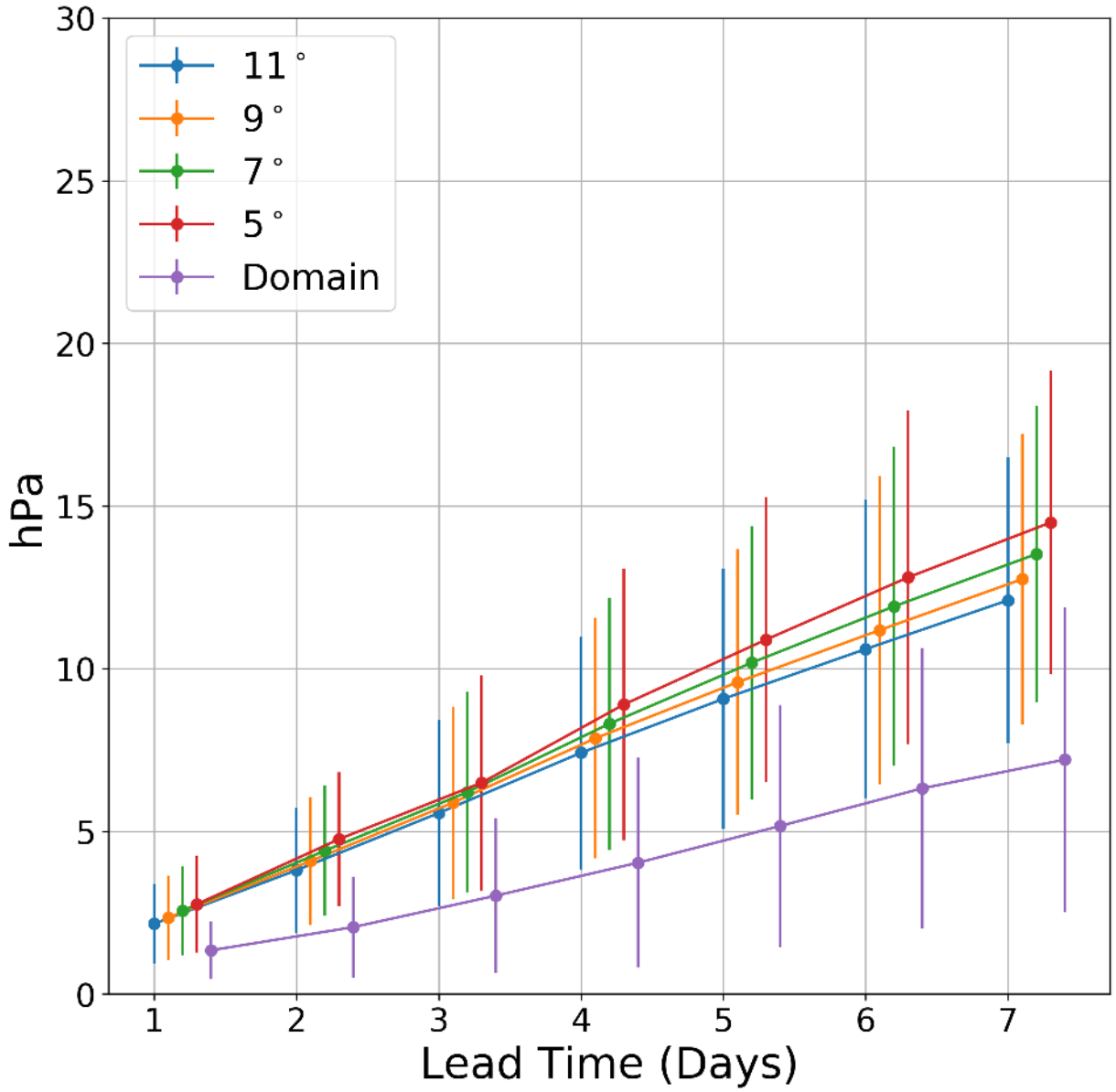


Figure 2.5: Same as Fig. 2.4 but for the CMC.

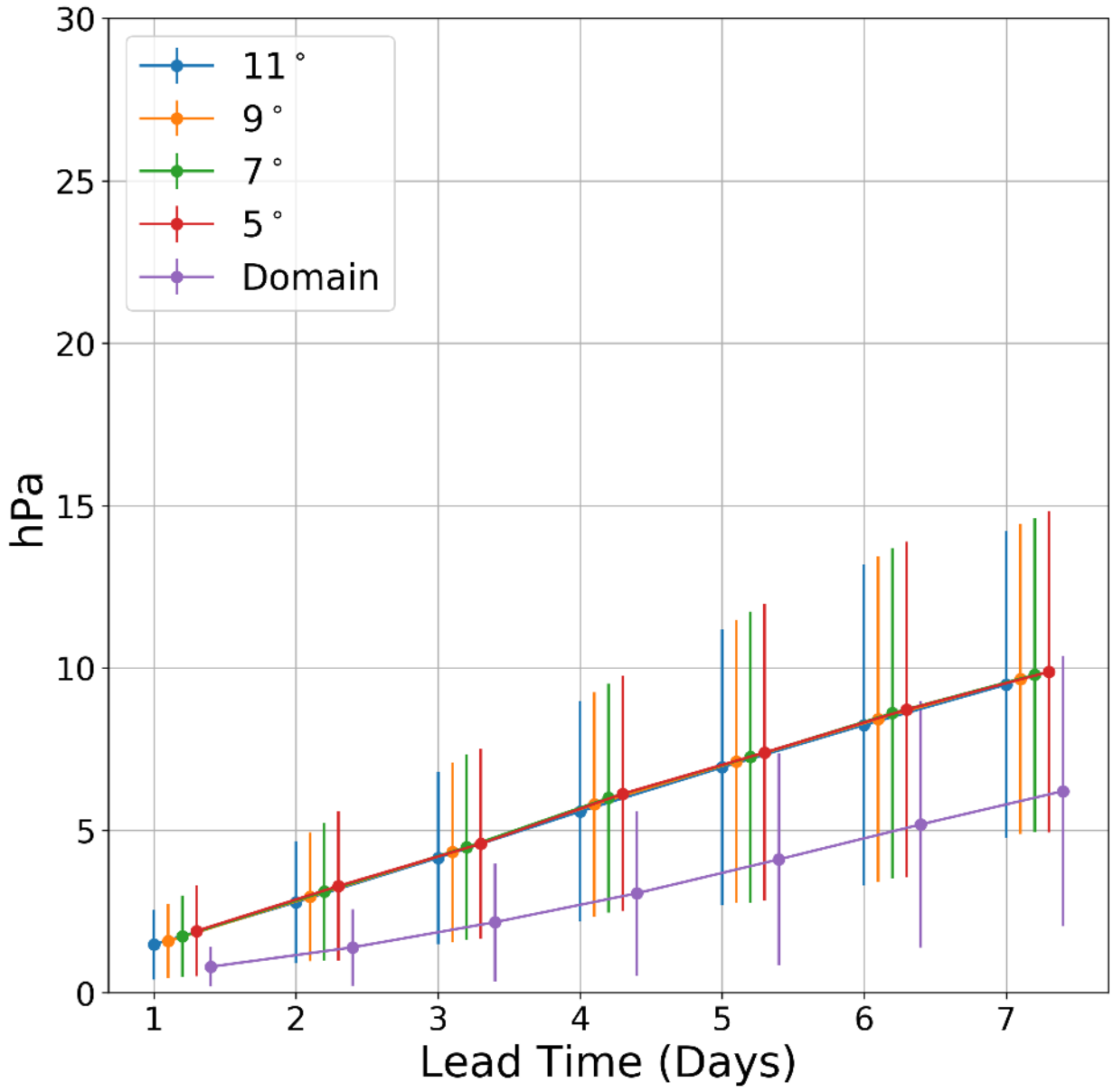


Figure 2.6: Same as Fig. 2.4 but for the ECMWF.

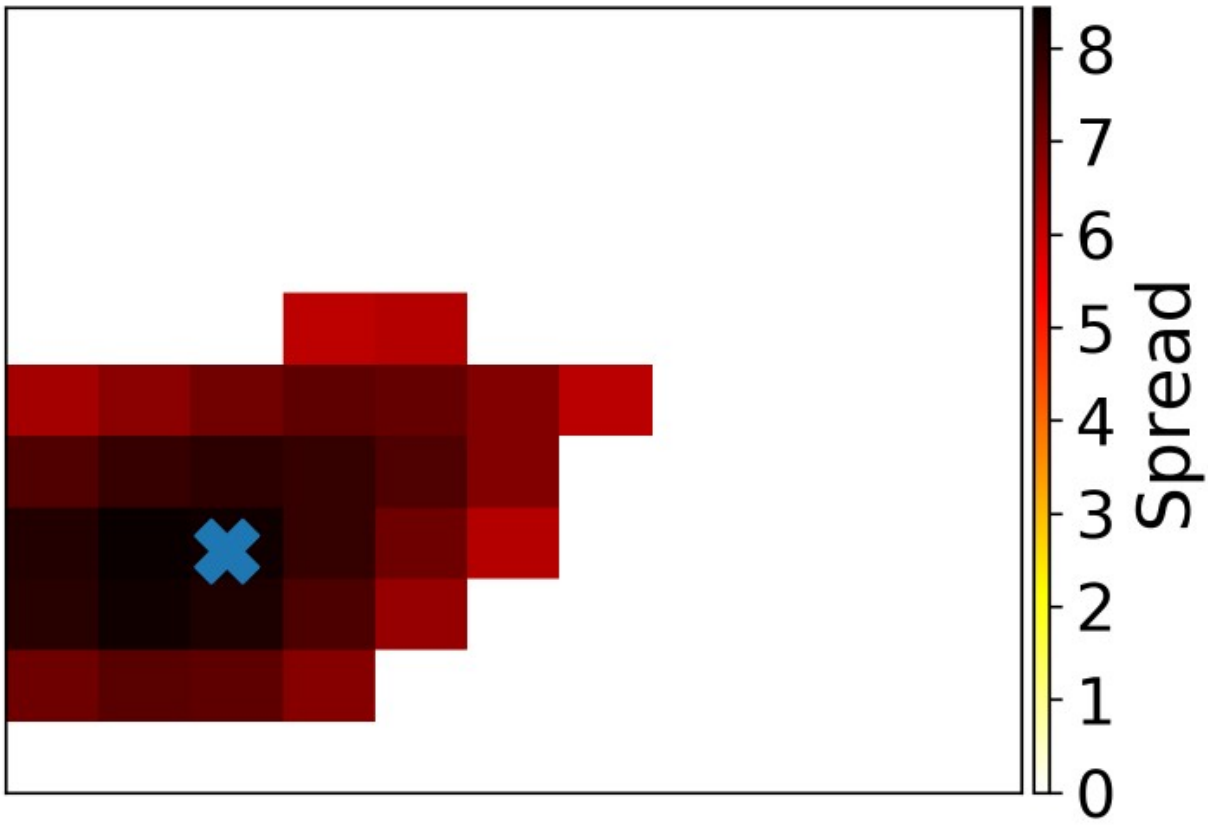


Figure 2.7: Mean absolute error (MAE) of the 26 February 2010 cyclone at hour 72 lead time for the GEFS. White colors are zero, red colors denote the gridpoints within the upper quartile of the 11x11 array. The blue x icon indicates the MAE ”center of mass” (centroid).

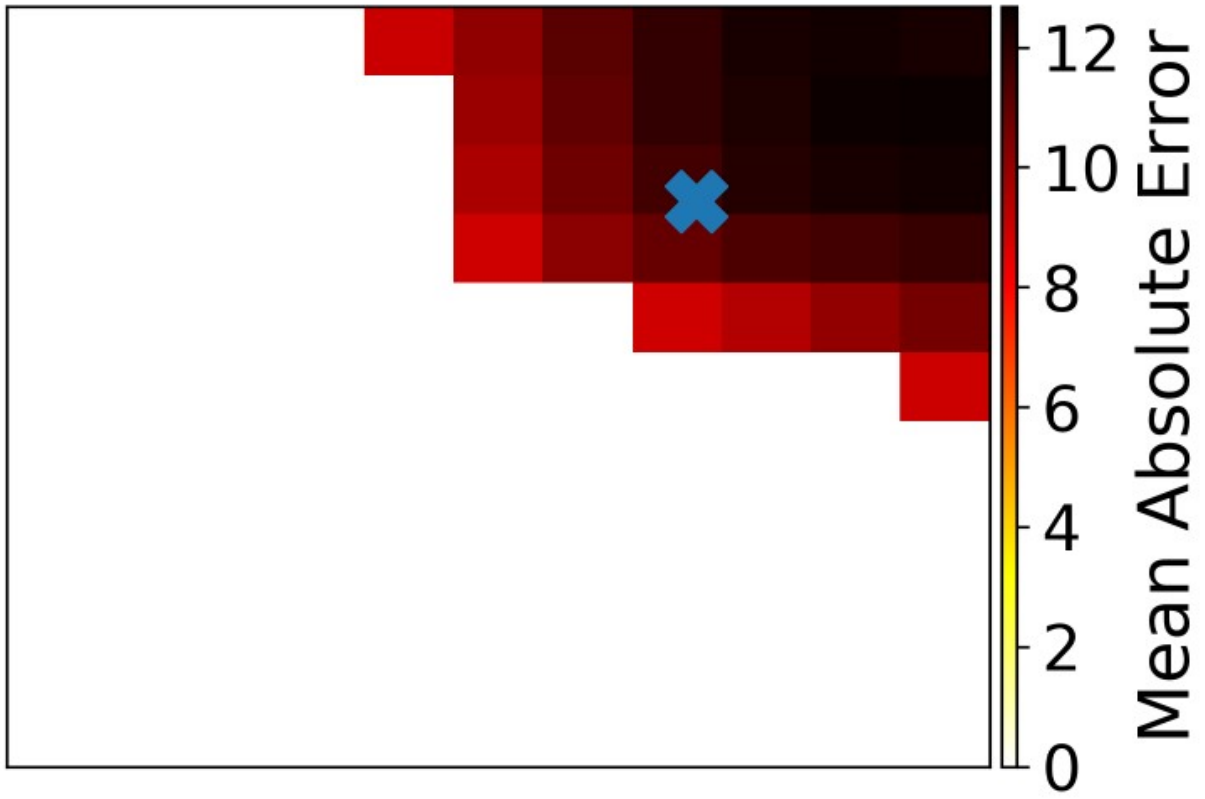


Figure 2.8: Ensemble spread of the 26 February 2010 cyclone at hour 72 lead time for the GEFS. White colors are zero, red colors denote the gridpoints within the upper quartile of the 11x11 array. The blue x icon indicates the spread ”center of mass” (centroid).

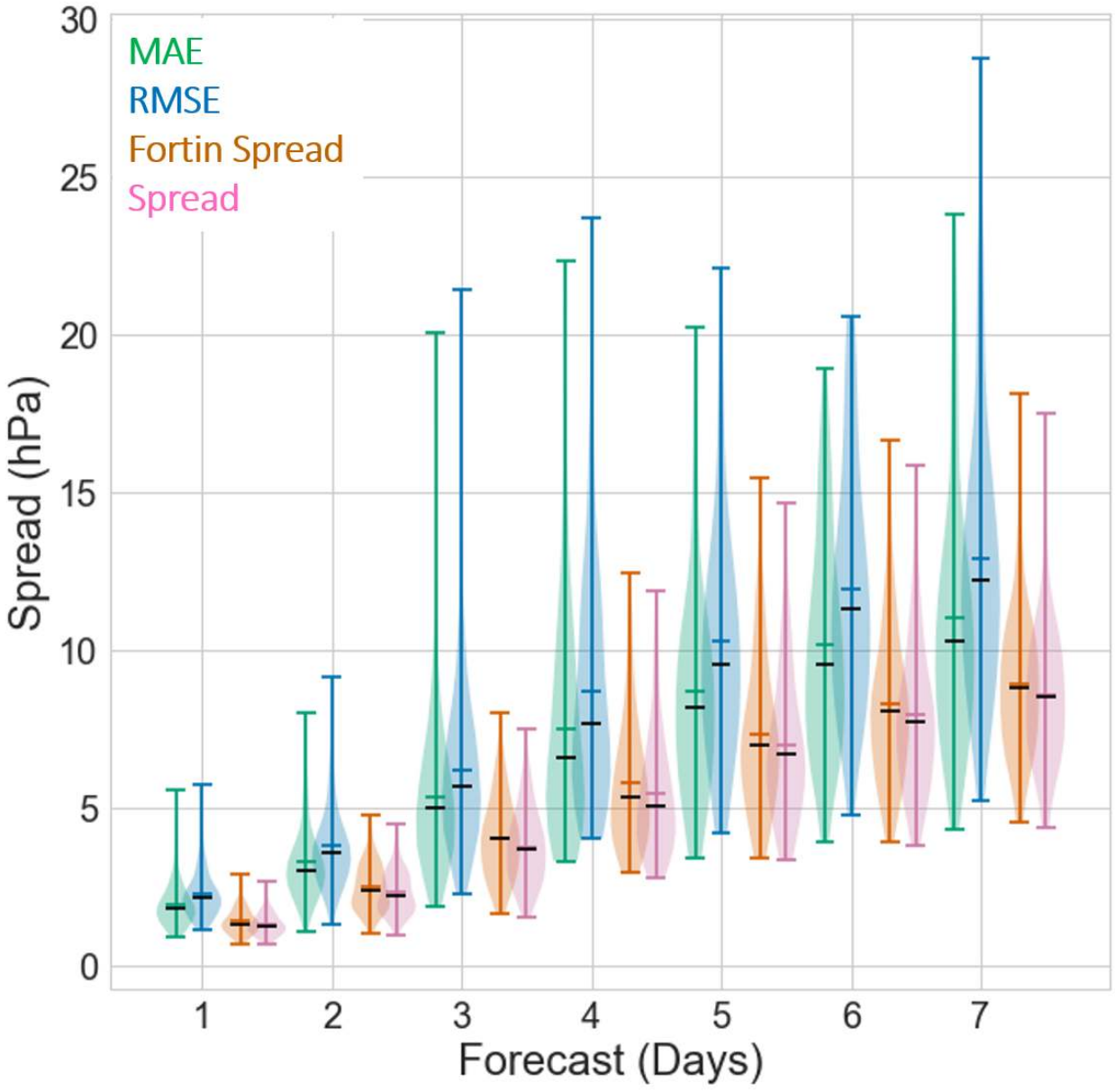


Figure 2.9: Violin plots for RMSE, MAE, spread, and spread as calculated in Fortin et al. (2017) for all 90 cyclones for the GEFS.

3 Error & Spread

3.1 Introduction

This chapter investigates the relationship between ensemble spread and error, represented as mean absolute error (MAE, Eq. 6). The purpose of ensemble spread is to identify areas of largest forecast uncertainty which should predict location and/or magnitude of errors in an ensemble. All 90 U.S. East Coast cyclones were compared for the ECMWF, GEFS and CMC, while the CMC had 89 storms due to missing data for 2 December 2008. For each cyclone, SLP spread, MAE, upper quartile spread/error centroid distances and spread/error gridpoint overlap, and comparisons therein were analyzed to assess differences between the three models as well as how the statistics changed with lead time.

3.2 Magnitude and Expansion of Spread and Mean Absolute Error

Ensemble spread is often compared to RMSE to assess how well dispersed a model is relative to its error. Ensemble spread is not only consistently smaller in magnitude than RMSE, but also MAE. RMSE is consistently larger than MAE for data in this study, therefore the error metric for the purposes of this study is MAE. The values of spread extend from 0.3 hPa to 11.9 hPa whereas the MAE range from $\sim$0.4 hPa to 34.4 hPa for the day 1-3 forecast histograms (Figs. 3.1a, b); a maximum value 288% larger than the spread. The ECMWF and CMC also show a discrepancy between the magnitudes of spread and MAE, with a difference of 183% and 156%, respectively, between the spread and MAE (not pictured). This discrepancy is also evident when looking at the frequency of cases in which the MAE is larger than spread, 6.2% of all values of MAE are larger

in magnitude than the maximum spread value within days 1-3. For the ECMWF, 644 gridpoints (2.0%), and the CMC, 676 (2.1%) gridpoints are above the maximum value of spread for days 1-3.

The long range (days 5-7) forecast spread and resulting MAE for the GEFS also shows a magnitude difference (Figs. 3.1c, d). The GEFS performs slightly better than the other models when comparing the magnitudes between both metrics. The maximum value of the GEFS spread and MAE maintains a 160% difference between the largest values (23.5 hPa spread, 37.5 hPa MAE). The difference between the maximum values of spread and MAE for the ECMWF is 206%; the CMC, 167%. 9.4% of MAE values for the GEFS are above the maximum spread value (3068 points), 18.3% for the ECMWF (5964), and 30.5% for the CMC (9856). All 3 ensemble models do not maintain the same magnitude of spread as error for either range.

Two-dimensional probability density functions (PDFs) provide insight into the overall correlation between spread and error. All values for all forecast hours are compared (Fig. 3.2); the resulting two-dimensional PDFs show a relatively strong relationship that has a moderate linear relationship, with Pearson correlation coefficients of 0.71, 0.75, and 0.74 for (a), (b), and (c), respectively. The CMC MAE distribution has an overall lower magnitude compared to the ECMWF or the GEFS. None of the models follow a clear 1 to 1 ratio between MAE and spread. When separating the PDFs into short range (days 1-3) and long range (days 5-7), the linear pattern is more clear in the short range, but it is less clear in the long range, where correlation coefficients decrease to 0.37 for the CMC, 0.44 for the ECMWF, 0.47 for the GEFS. There is evidence of a linear relationship when looking at the short range and days 1-7 but not the long range.

To analyze the differences between the increase with lead time of error and spread between methods of viewing forecasts, the ACR spread and MAE were compared with the FCR spread and MAE. The four metrics were plotted, the main lines representing the medians, showing a

divergence between the GEFS FCR and ACR (Fig. 3.5). The FCR MAE and spread both have more variability in comparison to the ACR but are not statistically significantly different at any lead time. The rate of change between lead times increases in both FCR metrics after day five; median values of the ACR approach begin to level off. Both spread and MAE do not diverge when looking at the entire suite of cyclones. The rate of increase of spread versus MAE with lead time for the EC is similar, as opposed to the CMC or the GEFS which diverge more in comparison.

The maximum values of spread and MAE through day 7 were plotted for FCR and ACR approaches (Figs. 3.8, 3.9, 3.10). Both show statistically significant differences between their respective spread and MAE, further illustrating that the GEFS does not represent the potential magnitude or expansion of error when viewing ensemble spread. The spread values between the ACR and FCR methods are statistically similar, but the ACR and FCR MAE maintain statistically significant differences. Overall, the extreme values for the MAE are statistically different than the largest values for the spread for all three models. The ACR spread increases at a greater rate than the FCR when looking towards longer lead times until day 7; there is also statistical significance between FCR and ACR spread for day 1 and 2 lead times (Fig. 3.9). A similar rate of increase in errors with lead time occurs for the ECMWF (Fig. 3.10), although the average max CMC FCR MAE is lower by $\sim$9 and $\sim$10 hPa for days 5 and 6 lead times, respectively.

3.3 Centroid Distance Between Ensemble Spread and MAE

Utilizing the upper quartile approach as described in Chapter 2, the distance between the centroids of the spread and MAE on the FCR domain were calculated and compared between lead time and model. The distance varies significantly, where the GEFS distances between centroids

range from 3.5 km (at day 3) to 914 km (at day 7), the CMC from 5.8 km (at day 4) to 888 km (at day 5), and the EC from 3.4 km (at day 4) to 807 km (at day 7). The distribution of these values shift further from 0 with lead time, but as shown with each minimum and maximum, the smallest values are not necessarily constrained to day 1 and day 7 lead times, respectively.

Histograms of the distance between the spread and error centroids are separated by forecast range and model (Fig. 3.11). Figure 3.11a shows the histogram for days 1-3 lead time for the GEFS (light green), CMC (dark green), and ECMWF (light blue). The ECMWF has significantly more cases 100 km or below for the short range as shown as compared to either the GEFS or CMC. The GEFS has not only a smaller number of points within 100 km, but also has consistently higher numbers at each larger bin with $\sim$5 to 10 cyclone cases more in the 200, 300, and 400 km bins. The ECMWF has consistently less frequency of cyclones for all bins greater than 100 km. There are $\sim$140 cases throughout all models where, even at the 24 hour range, the ensemble spread and MAE centroids are well over 300 km apart, which has implications on any model's inherent ability to consistently and accurately predict its own error with the ensemble spread.

The long range (day 5-7) forecasts for East Coast cyclones show a more dispersed histogram (Fig. 3.11b). The ECMWF has significantly more cyclones (17 compared to GEFS, 31 compared to CMC) with an ensemble spread and MAE centroid distance under 100 km. Significantly less cyclones in the CMC had centroid distances below 100 km compared to the other models. Above 100 km distance, no model is clearly better; however, the CMC has significantly more cyclones with greater centroid distances between 400 and 500 km.

The centroid distances were also separated by day to asses daily variability in model performance (Fig. 3.12). The CMC (Fig. 3.12a) performs significantly better at day 1 than days 2 or 3, with a general trend towards increasing centroid distance with days - the maximum count for

day 2 and 3 is located between 100-200 km, with a significant maximum of values at 300-400 km for day 3. The ECMWF (Fig. 3.12b) performs well all 3 days, where between 37-40 of the 90 cyclones have centroid distances within 0-100 km. The GEFS (Fig. 3.12c) does not perform well at short range, with cyclone counts for 0-100 km centroid distances between 17-25 for days 1-3. The ECMWF perform significantly better for all 3 days in the short range in comparison to the GEFS. For the day 5-7 range, all models perform expectedly worse than the short range, with a more dispersed histogram showing larger centroid distances for the cyclones in comparison. The CMC (Fig. 3.12d) 0-100 km bins for days 5 and 7 have a lower count than the 100-500 km bins, and near equal for day 6 in the same range. The ECMWF (Fig. 3.12e) performs similarly, though there is significantly more 0-100 km centroid distances for days 5 and 7. Slightly more cyclones in the GEFS (Fig. 3.12f) were identified than the CMC within the same 0-100 km bin, but is similar in terms of the number of forecasts which are in bins greater than 400 km.

3.4 Percent Overlap Between Upper Quartile of Ensemble Spread and MAE

The number of grid points in the FCR approach that overlap between the upper quartile of ensemble spread and MAE were calculated using the approach described in Chapter 2. The range of percentage varies between 0 to 100 for both the short and long range forecasts, thus indicating how well the upper quartile of spread and MAE co-located. The higher the percentage, the better the co-location for the model. The short range lead time percent overlap (Fig. 3.13a) shows that the ECMWF, in light blue, has significantly more cyclone cases in the 90-100% category, with $\sim$35 out of 90. The CMC, in dark green, lags behind with $\sim$15 and the GEFS, in light green, has $\sim$5. The ECMWF also performs slightly better in the 70-80% and 80-90% bins, but it is not

statistically significant. The GEFS is statistically worse than the CMC or ECMWF in the 80-90% bin range. In the middle percentage ranges, the models are similar, with the ECMWF slowly decreasing in cyclone count as percentage bins decrease. The GEFS has a significant number of cyclones within the 0-10% bin and 30-40% bin. Similar to the percentage overlap, the ECMWF and CMC are significantly better than the GEFS for lead times of 24-72 hours when considering the spatial similarity between ensemble spread and MAE.

Overall, the 5-7 day lead time range shows a weak relationship between the percentage overlap. The long range (Fig. 3.13b) percent overlap between ensemble spread and MAE show a much more dispersive histogram, with the ECMWF showing better overlap for 80-90%, and slightly better overlap compared to GEFS and CMC for 90-100%. All of the models' percentage overlap do not show statistical significance amongst each other except for the CMC, in purple, at 20-30%. The GEFS, in dark blue, shows slightly more 0-10% values than the other models but is not statistically significant.

The short range shows the overlap for the CMC (Fig. 3.14a), ECMWF (Fig. 3.14b), and GEFS (Fig. 3.14c) are relatively similar. The GEFS shows statistically significant overlap at day 3 lead time for 70-80% for around 20 cyclones. The ECMWF, for all 3 lead times, has between 7-15 cyclones in the 90-100% range, compared to the GEFS with 0-3. There is a statistically high number of cases in the 60-70% bin for the ECMWF, with $\sim$23 cases. The majority of cases for all 3 models in the short range lie between 50-60% and 80-90%, which details a moderate relationship day-to-day when forecasting in the short range without too much variability between lead times. The long range forecasts (Figs. 3.14d, e, f) show a similar pattern as Fig. 3.13b, where each model is similar throughout the 3 long range lead times. The values being dispersive for days 5-7 lead times show that the relationship tends to break down between ensemble spread and MAE percent

overlap when forecasting for medium- to long-range.

3.5 Comparing Colocation Statistics Based on Cyclone Minimum SLP

The weakest and deepest 50% of cyclones were identified and sorted to identify differences in the approaches with respect to cyclone stage and MSLP magnitude for both centroid distance and percent overlap. The variations between the weaker and deeper cyclones are affected by synoptic flow, diabatic processes, feeder lows, and mid- to upper-level divergence, all of which are handled through different variations in each model's dynamical cores and parameterizations. The weakest cyclones range from 995 hPa to 972 hPa and the deepest cyclones range from 972 hPa to 951 hPa, based on the ERA-I data. It is useful to compare both the short range (Fig. 3.15a) and long range (Fig. 3.15b) along with their counterpart in 3.16 to provide context on how each model performs relative to the strength and maturity of cyclones.

The ECMWF performs significantly better than the CMC or GEFS, with the GEFS performing the worst in comparison (Fig. 3.15a). The ECMWF does not produce as extreme of values in centroid distances. The long range (Fig. 3.15b) shows that the ECMWF performs significantly better than either two models for the 0-100 km bin, though the trend for all three models are similar for the tails of their respective distributions, i.e. the models all struggle similarly at longer ranges. The ECMWF is not only significantly better, once again depicted by the 0-100 km bin, but also the three models perform better compared to weaker cyclones (the tails of all three taper off in less extreme bins than weaker cyclones, Fig. 3.16a). For longer ranges (Fig. 3.16b), there is no discernable trend between any of the models, which leads to a similar conclusion as weaker cyclones (models struggling to capture the centroids spatially close to each other).

Percent overlap results show the relationship of spread and MAE is more robust for deeper cyclones in comparison to the weaker in the shorter range. The ECMWF performs better in the short range (Fig. 3.17a), as evidenced by having significantly more cyclones in the 90-100% bin. The GEFS performs worse, with significantly more cyclones (60) in the bins below 50% compared to the other two models (31 for ECMWF, 48 for CMC). Both sets of cyclones have a widely distributed histogram in the long range without a distinct pattern, though the ECMWF does have significantly more cyclones in the 80-90% bin for the weakest cyclones (Fig. 3.17b). The GEFS and CMC perform slightly worse for the deepest cyclones but there is no statistically significant model at any bin (Fig. 3.18b).

3.6 Variability in Synoptic Regimes Associated with Spread/Error Colocation

A method to assess differences in the flow patterns that were associated with the cyclones in this study is to compare 500 hPa geopotential heights composites based on sorting of variables. Identifying differences in flow patterns can provide context on the patterns which contribute to larger spatial differences between the spread and MAE. The lowest (highest) 50% of centroid distance (percent overlap) was subtracted from the highest (lowest) 50% of centroid distance (percent overlap) in order to identify differences in forecasts with better colocation of spread and MAE. The lower the centroid distance, the better the colocation; the higher the percentage overlap, the better the colocation.

The GEFS, CMC, and ECMWF centroid distances were sorted individually and the data were separated into the top and bottom 50%. The ranked index was applied to the 500 hPa geopotential

heights ERA-Interim analysis data. The GEFS shows a generally weaker wave pattern associated with smaller centroid distances (Fig. 3.19a), where a weaker trough in the eastern US is paired with a weaker ridge in the western half of the US, both of which are significant using the difference of means test for statistical significance. There is also an area of 80-120 m geopotential height ridging south of Alaska in the eastern Pacific that is significant. A smaller area of lower heights near Newfoundland indicates a slightly weaker ridge (statistically significant) that could be associated with less latent heating amplifying the downstream ridge. The 540 dm line is less amplified in the lower 50% of centroid distances than the upper 50% for the GEFS. The ECMWF shows a much weaker ridge to the northeast of Newfoundland, which also could be the result of less ridge amplification (Fig. 3.19b). The higher height signals present in the GEFS are not present in the ECMWF. A similar, but weaker, signal exists in west central Canada of a slightly weaker ridge that is statistically significant. There is also more troughing around the Bering Strait in the ECMWF related to the shorter centroid distances. The CMC shows a pattern of much stronger ridging north of Hudson Bay, without any clear troughing signals (Fig. 3.19c). There is slight ridging in the eastern Pacific, but it is not statistically significant. The difference in the analysis cases being indexed for the sorted cyclones of each model could be the result of varying model microphysical schemes, data assimilation procedures, and parameterizations, which bias the individual models towards certain solutions.

For the percent overlap sorting, the model values were individually sorted from high to low (the opposite of centroid distance to represent the "better" colocation subtracted from the "worse") and separated between the top and bottom 50% values. The GEFS (Fig. 3.20a) produces a similar solution to the centroid distance approach, which helps to convey the consistency of both when used in tandem. The signal with the percent overlap is slightly weaker but identifies a weaker pattern that

is associated with better performing cyclones, with statistical significance near Newfoundland, the Midwest, and the West Coast. There is a slight ridging (or, less troughing) near the north central Pacific, but it is not statistically significant. The ECMWF (Fig. 3.20b) produces a similar signal. There is still a significant weakening of the trough to the north and east of Newfoundland as well as a dampening of the ridge near the West Coast. There is a weaker trough pattern which is also evident in the GEFS in the Midwest and into Ontario and northeast Manitoba. The CMC (Fig. 3.20c) is slightly different from the GEFS and ECMWF patterns, with a stronger ridge into British Columbia and Alberta, a slightly stronger trough into northern Maine and Quebec, and a trough over the Aleutian Islands and the south of the Bering Strait. Similar to the centroid distance disparities, the differences in the 500 hPa geopotential height composites from model to model are likely the result of model differences in physics and data assimilation.

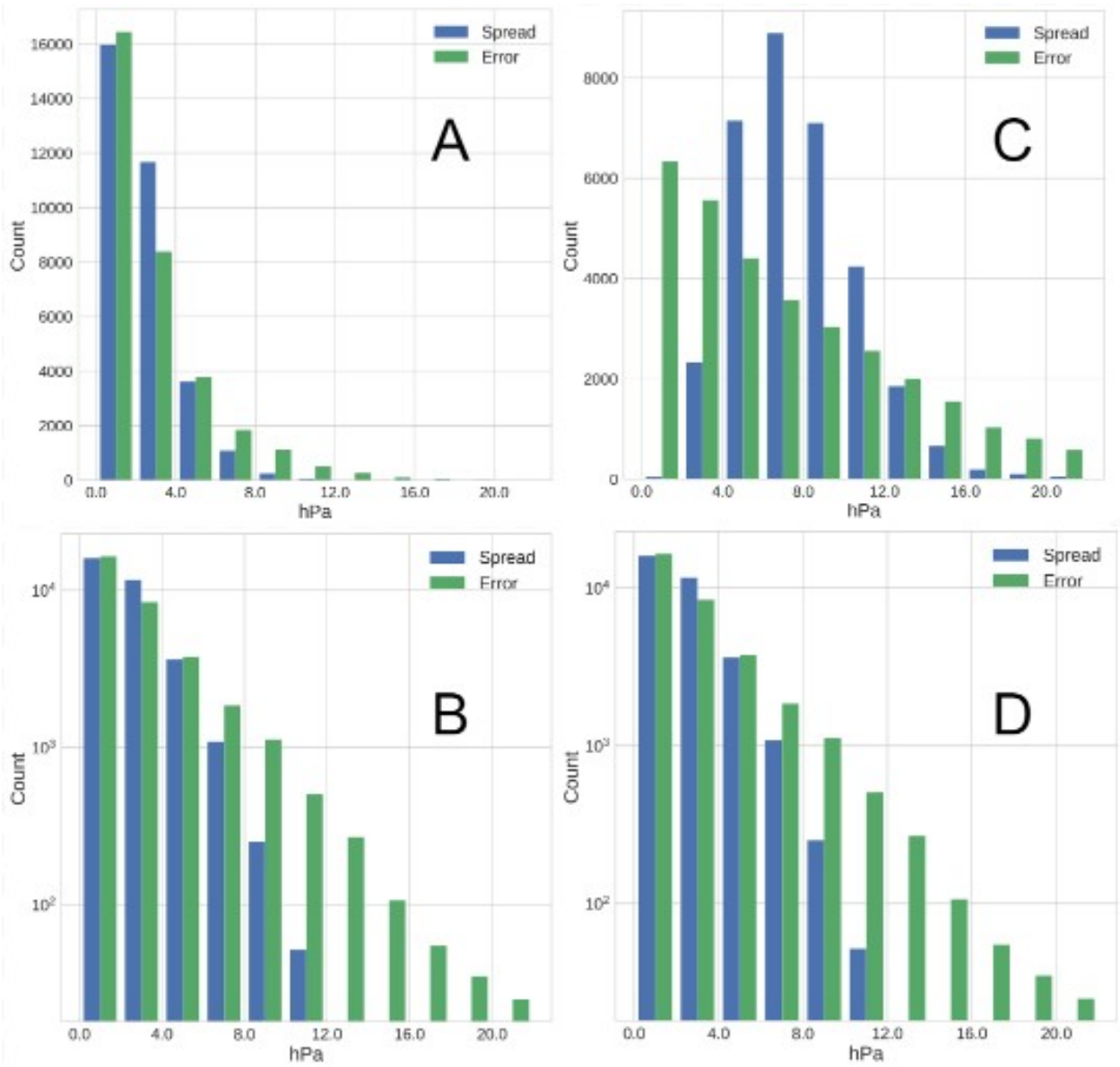


Figure 3.1: Distribution of GEFS spread and mean absolute error for short range (days 1-3, a, b) and long range (days 5-7, c, d), GEFS mean cyclone relative. (b) and (d) are the logarithmic histograms of (a) and (c), respectively.

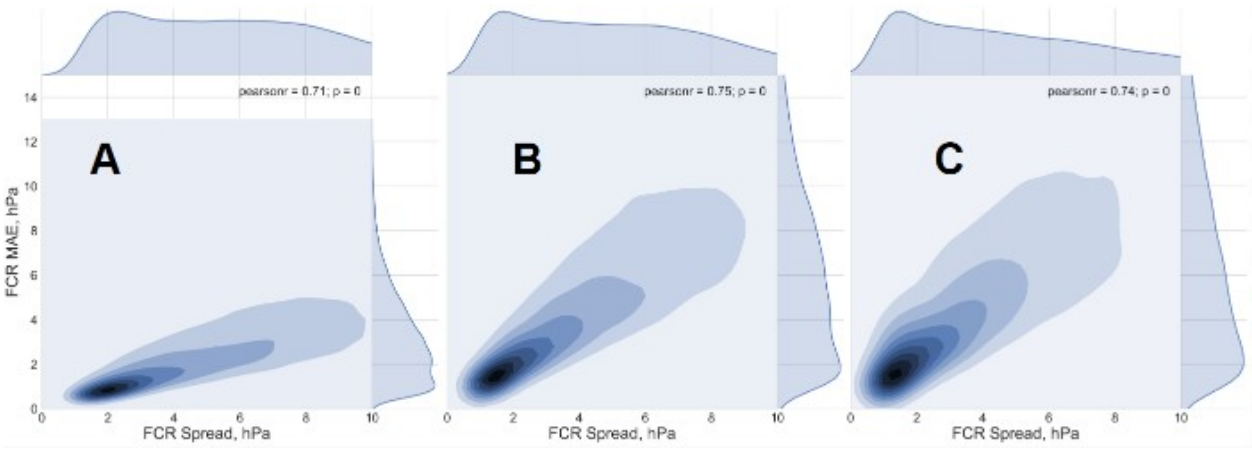


Figure 3.2: Forecast cyclone relative MAE and Spread 2D PDFs for the CMC (a), ECMWF (b), and GEFS (c) at all forecast hours. The Pearson correlation coefficient values for the distributions are 0.71, 0.75, and 0.74, respectively.

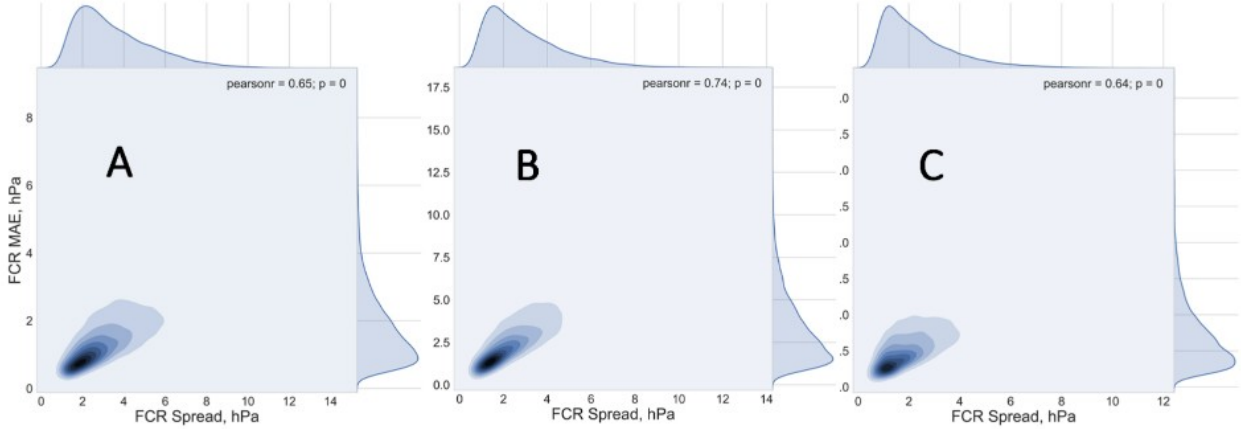


Figure 3.3: Same as Fig. 2.2, but for forecast days 1-3.

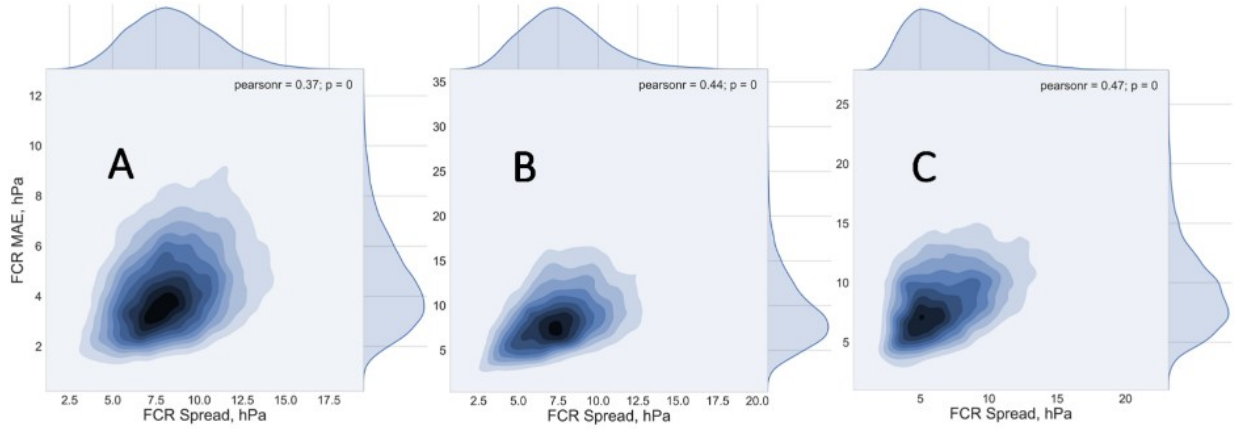


Figure 3.4: Same as Fig. 2.2, but for forecast days 5-7.

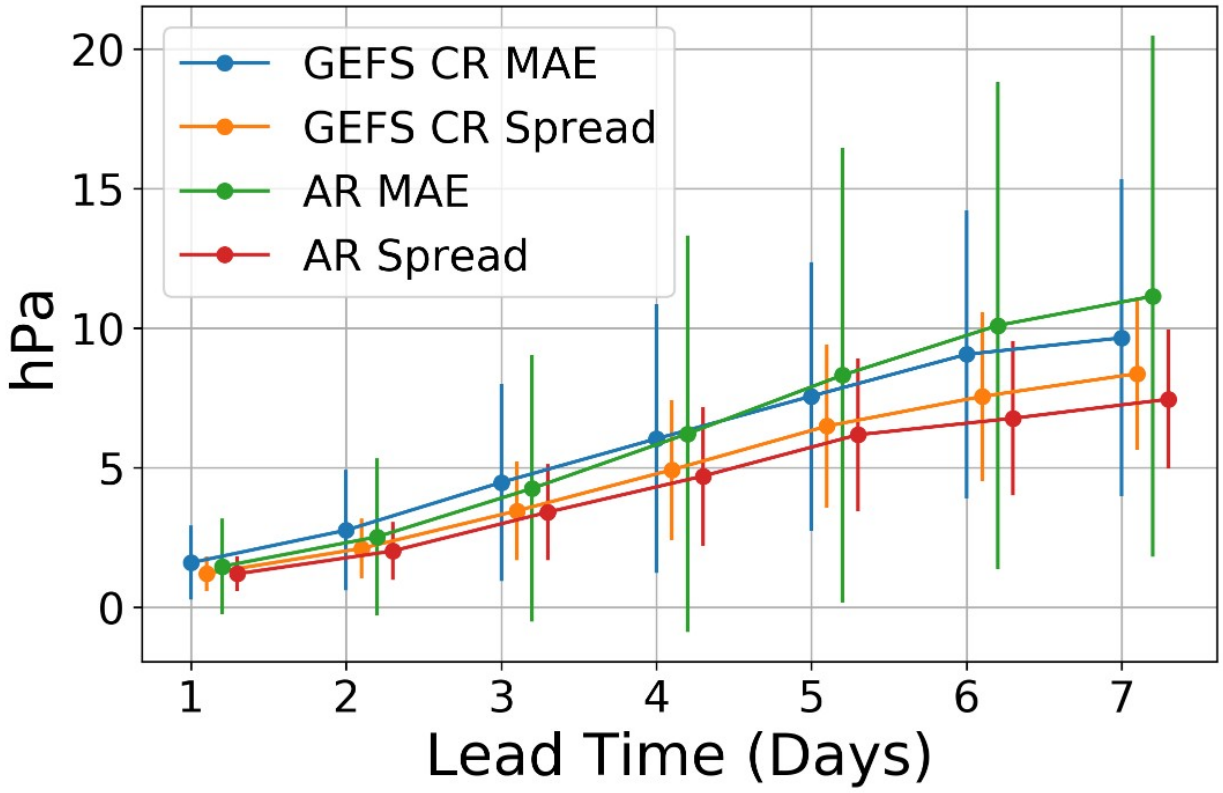


Figure 3.5: Expansion of average values of mean absolute error and spread over Days 1-7. GR denotes cyclone relative data centered on the forecast mean minimum SLP (FCR). AR denotes cyclone relative data centered on the ERA-Interim minimum SLP (ACR).

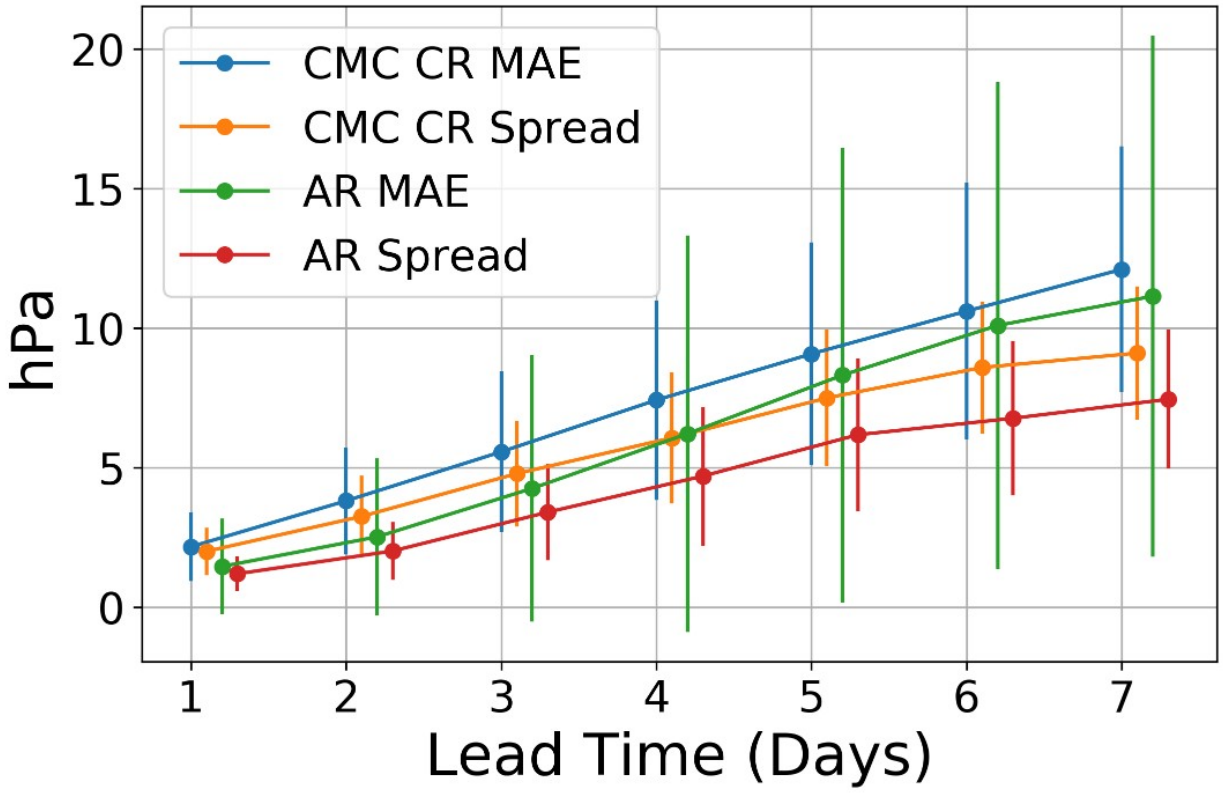


Figure 3.6: Same as Fig. 3.5, but for the CMC.

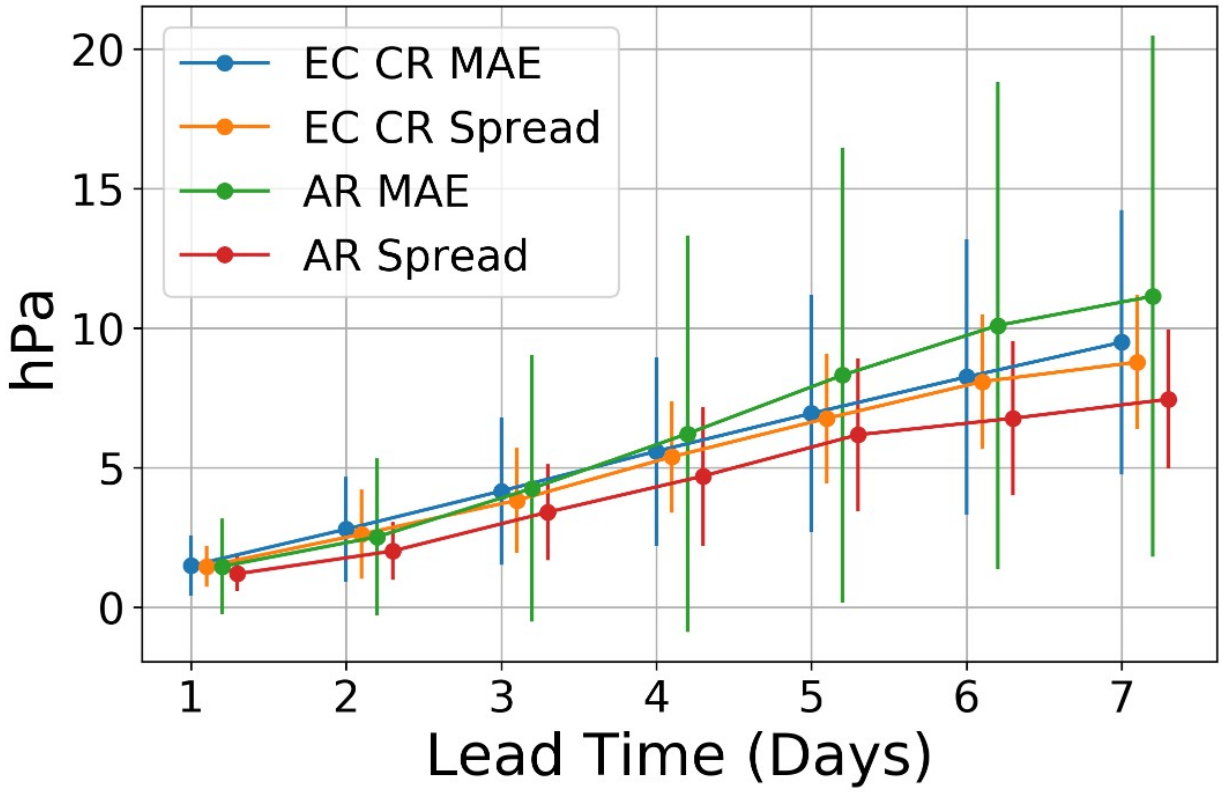


Figure 3.7: Same as Fig. 3.5, but for the ECMWF.

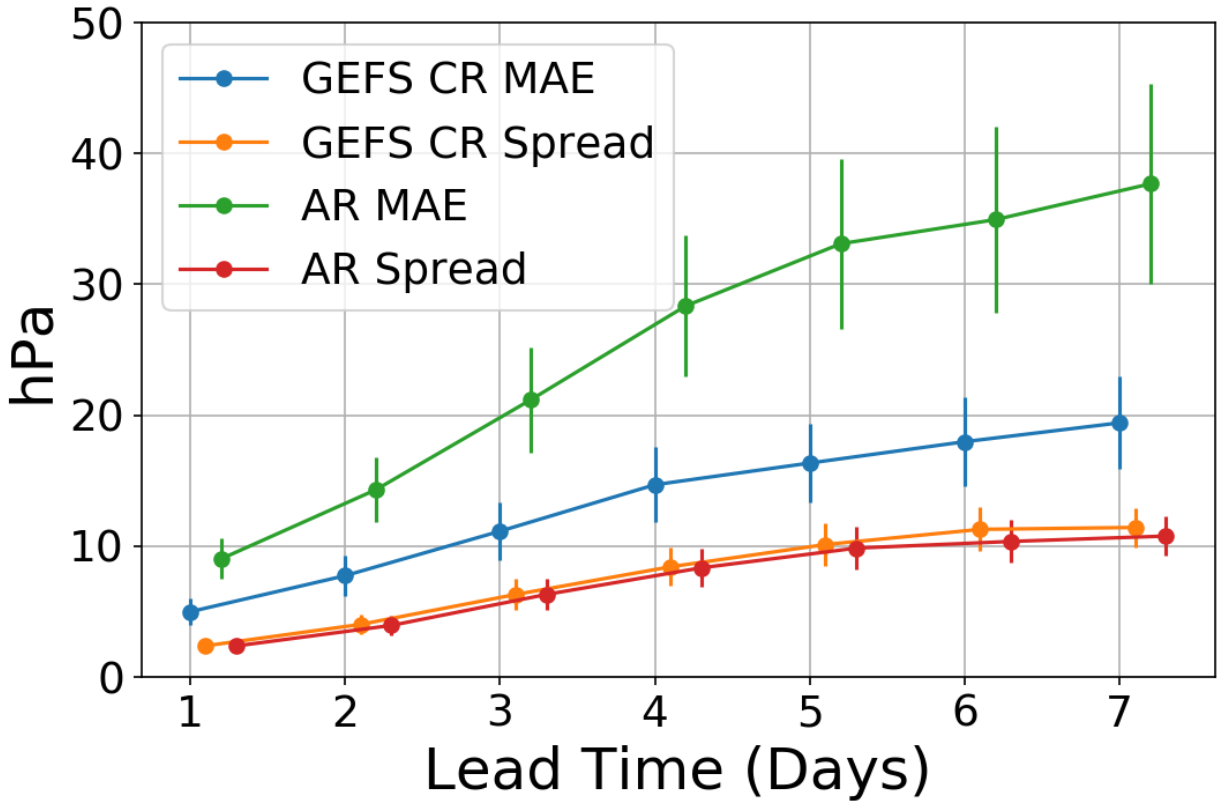


Figure 3.8: Same as Fig. 3.5, but expansion of maximum values of mean absolute error and spread for GEFS relative (FCR) and ERA-Interim relative analysis (ACR).

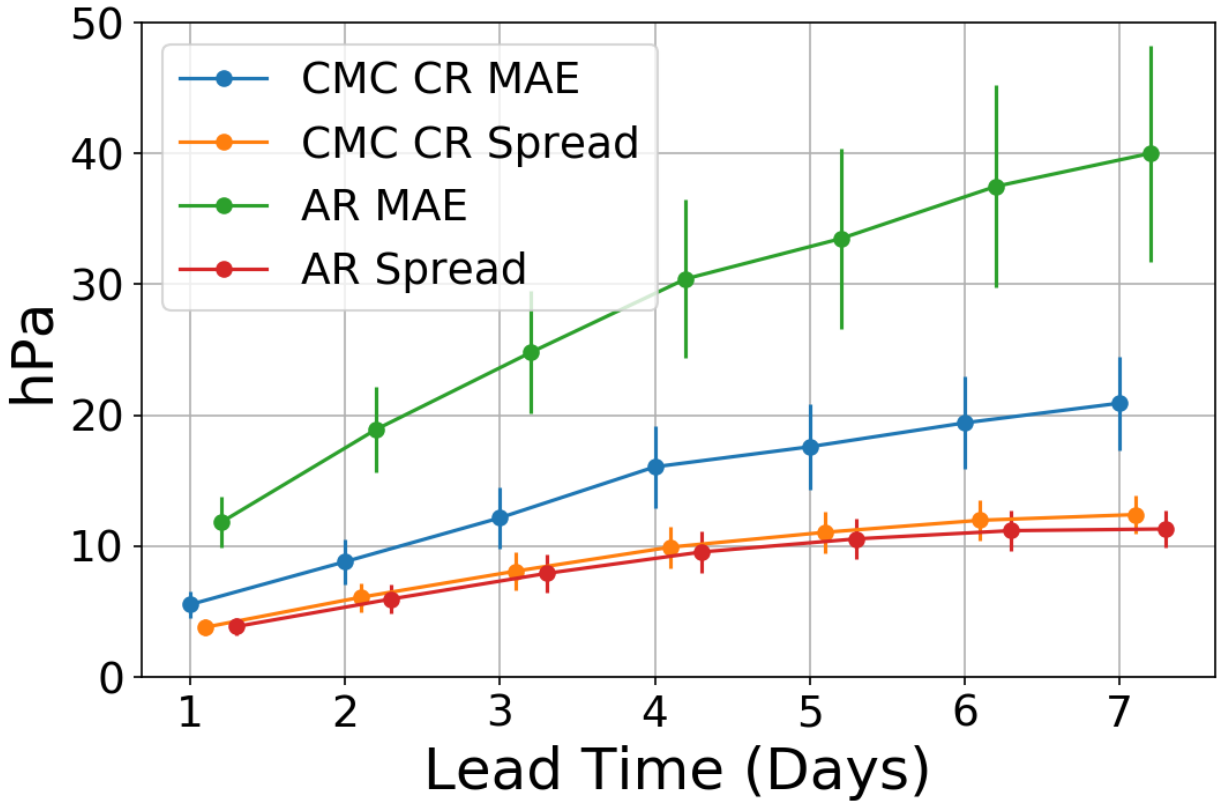


Figure 3.9: Same as Fig. 3.8, but for the CMC.

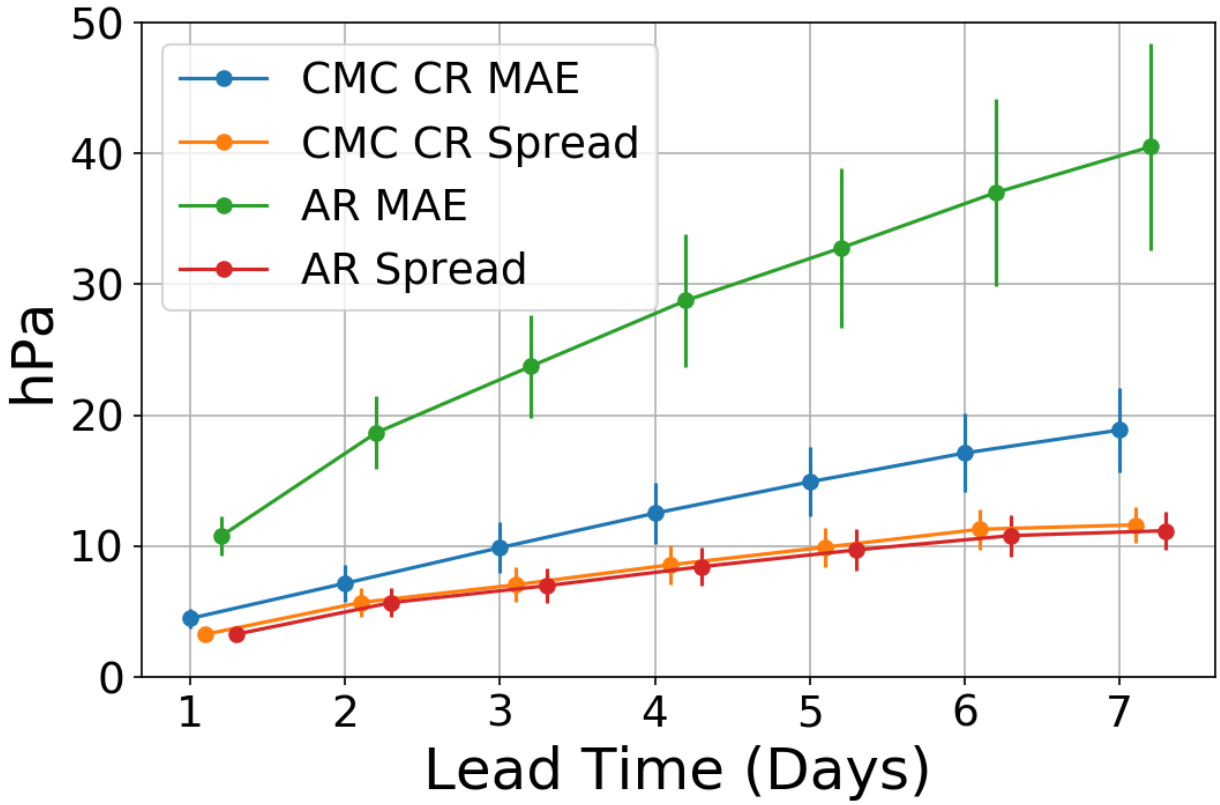


Figure 3.10: Same as Fig. 3.8, but for the ECMWF.

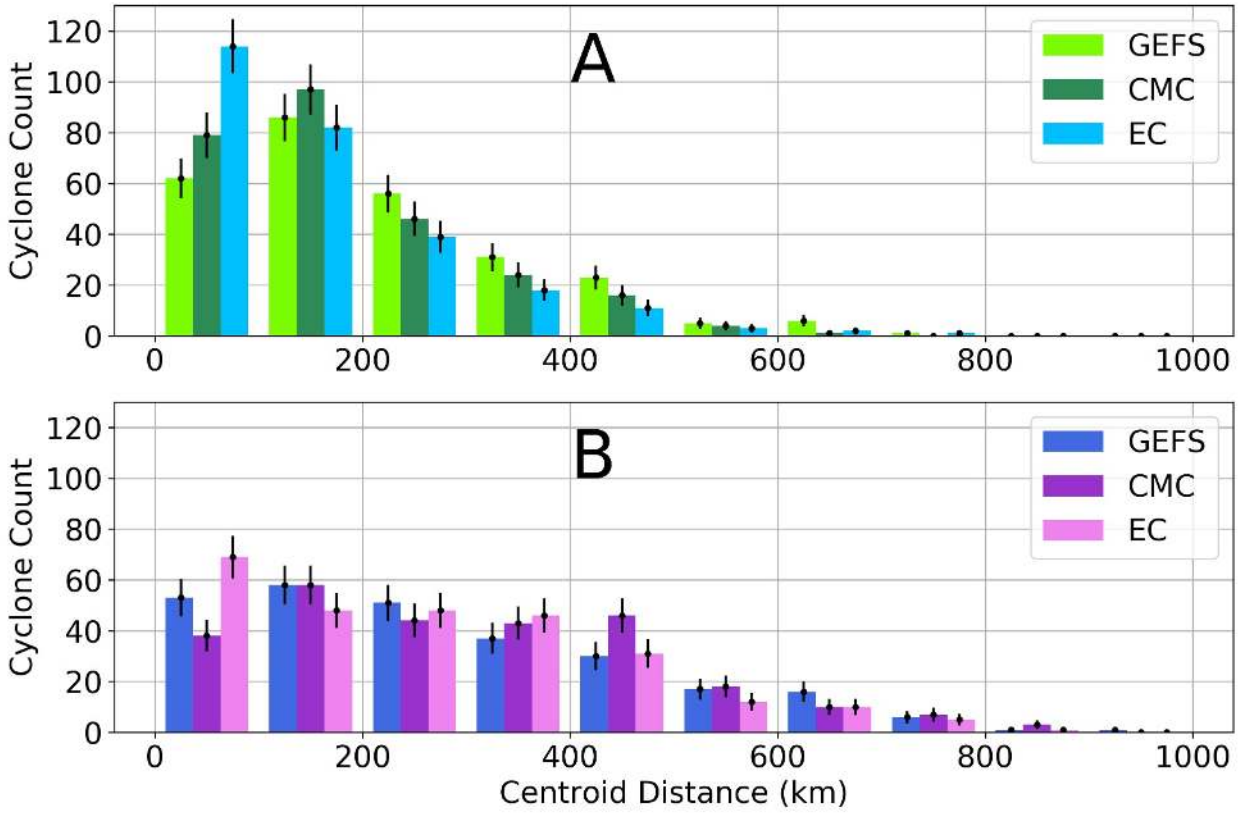


Figure 3.11: Centroid distance broken down by model and forecast range. (a) represents the short range, (b) represents the long range.

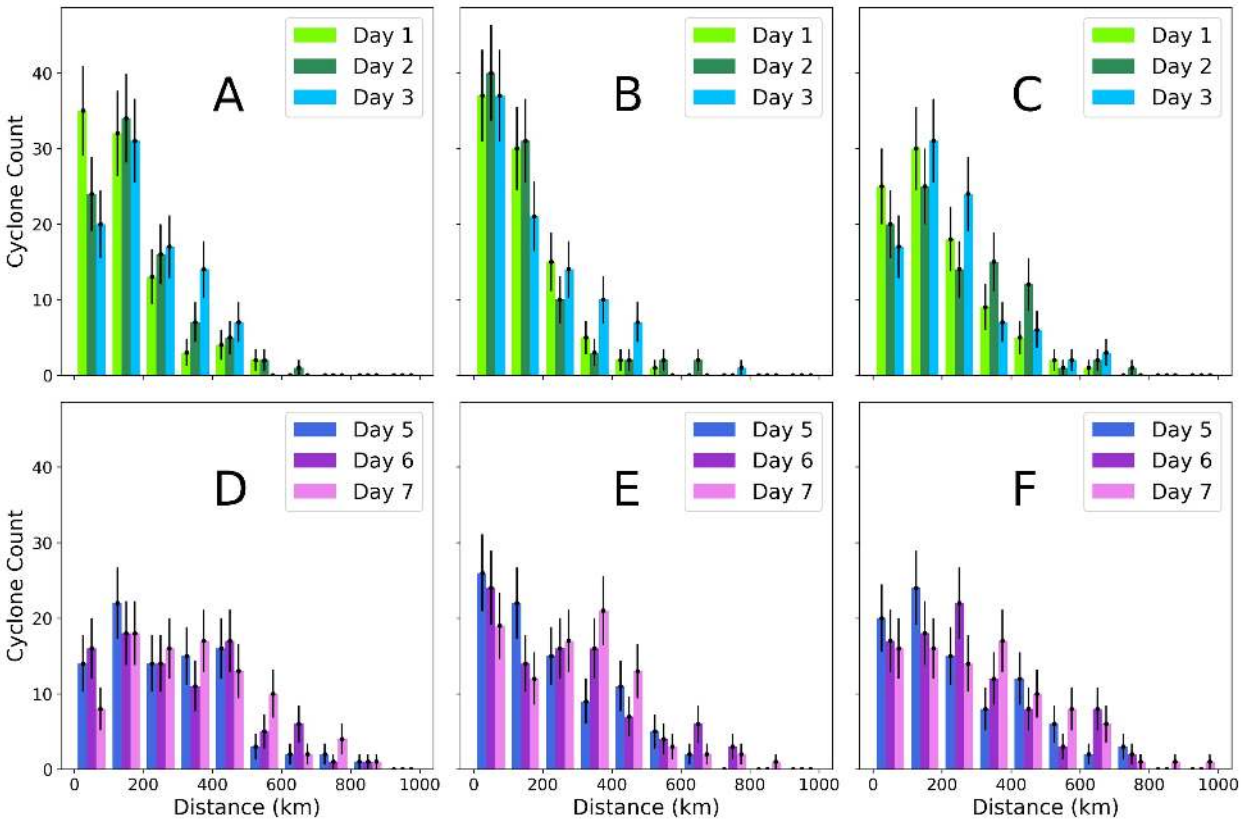


Figure 3.12: The distribution of centroid distance, error bars are the square root of the bin length, binned every 100 km. (a, b, c) represent the short range and (d, e, f) represent the long range. The left column is the CMC, the middle column is the ECMWF, and the right column is the GEFS.

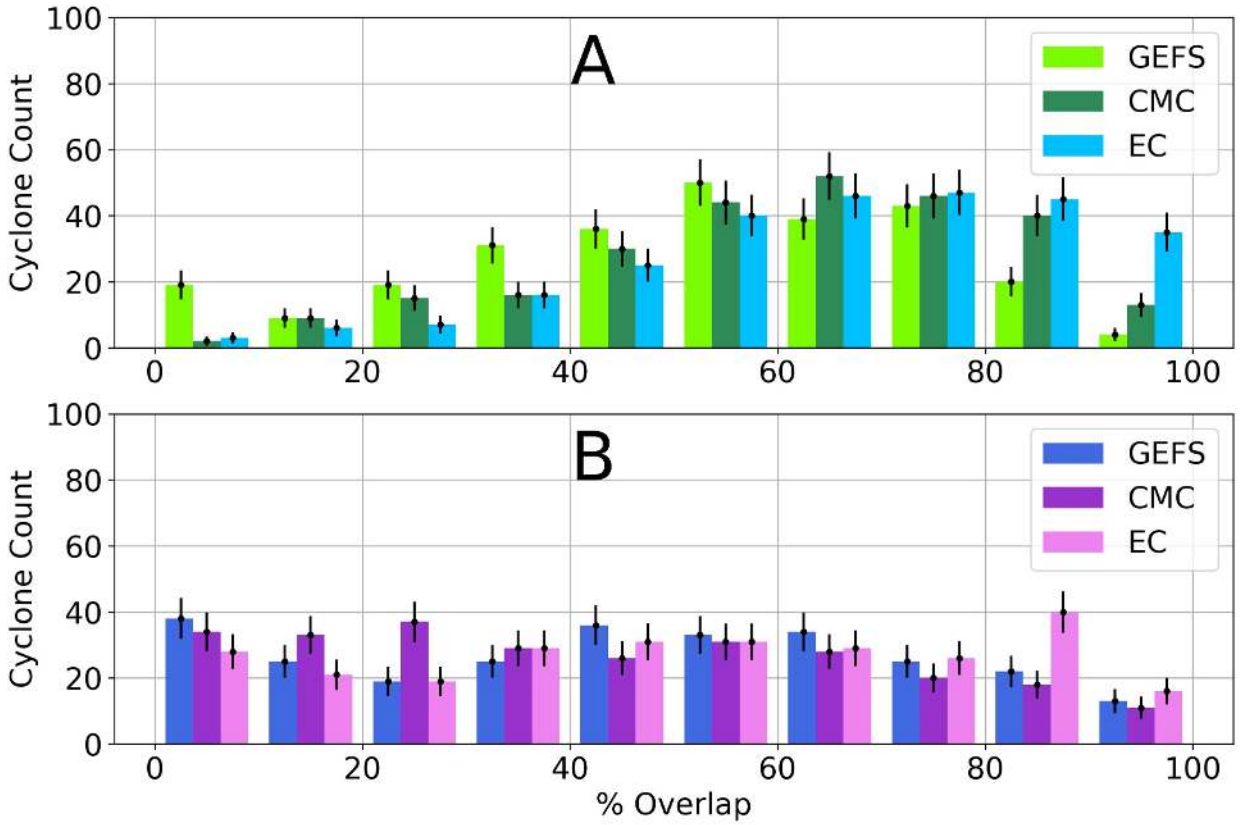


Figure 3.13: Percent overlap separated by forecast lead time and model. (a) represents the short range, (b) represents the long range.

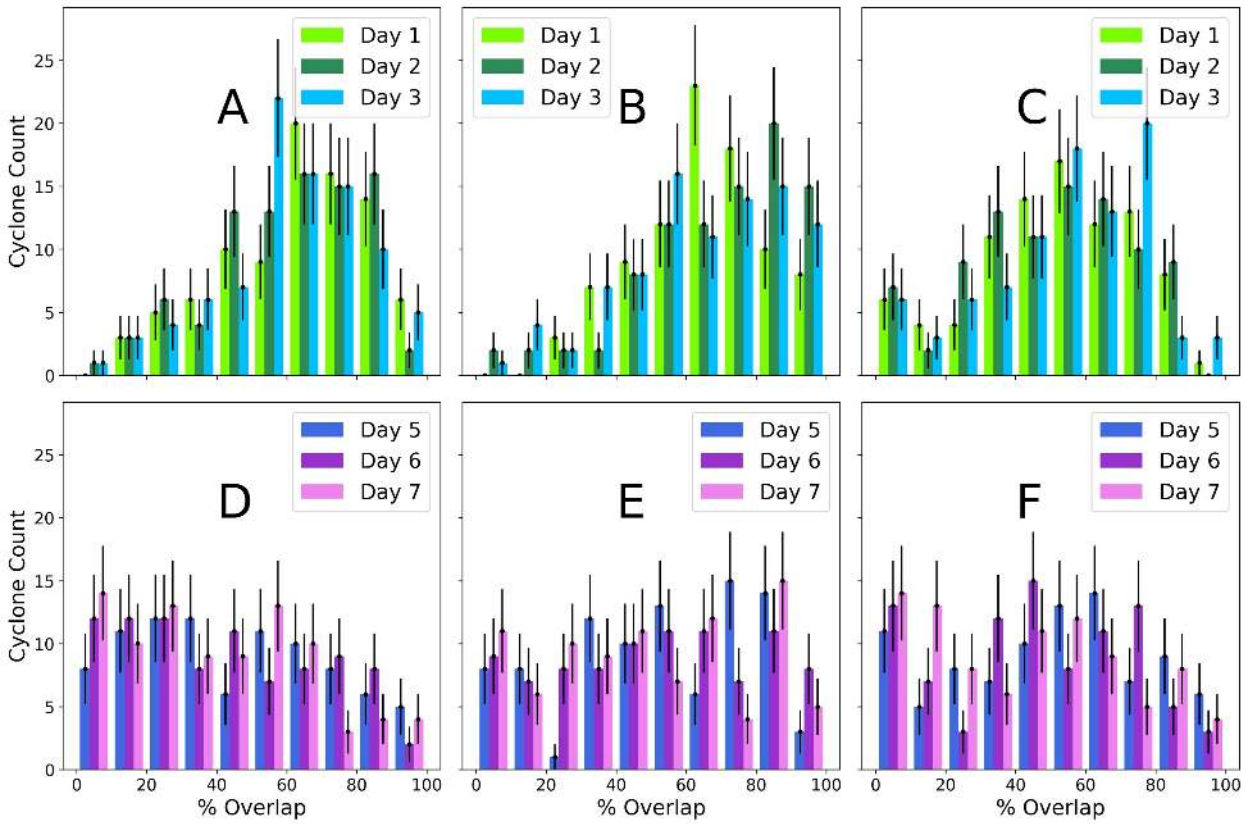


Figure 3.14: The percent overlap values, separated by model and each individual lead time. (a, b, c) represent the short range and (d, e, f) represent the long range. The left column is the CMC, the middle column is the ECMWF, and the right column is the GEFS.

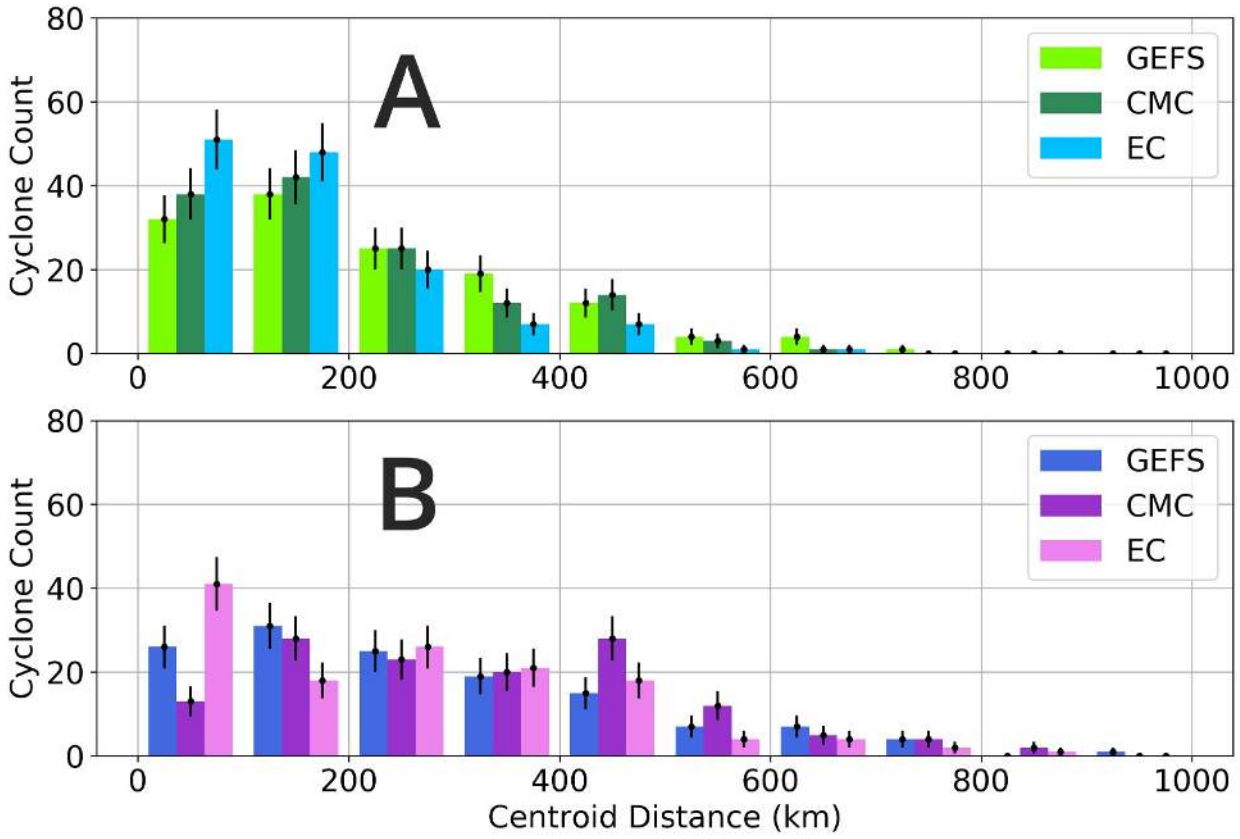


Figure 3.15: Same as Fig. 3.11, but the centroid distances for the weakest 50% of cyclones (n=135, n=132 for CMC). (a) represents the short range, (b) represents the long range.

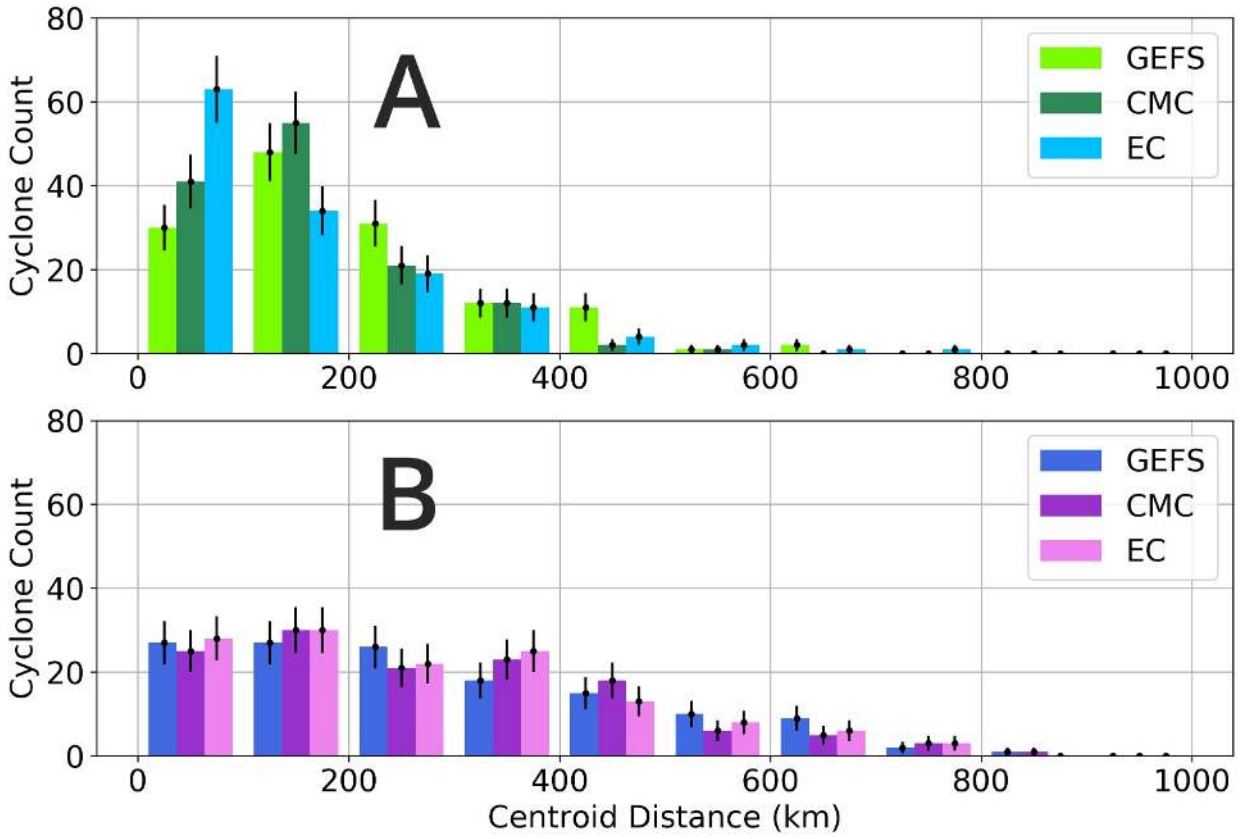


Figure 3.16: Same as Fig. 3.15, but for the deepest 50% of cyclones. (a) represents the short range, (b) represents the long range.

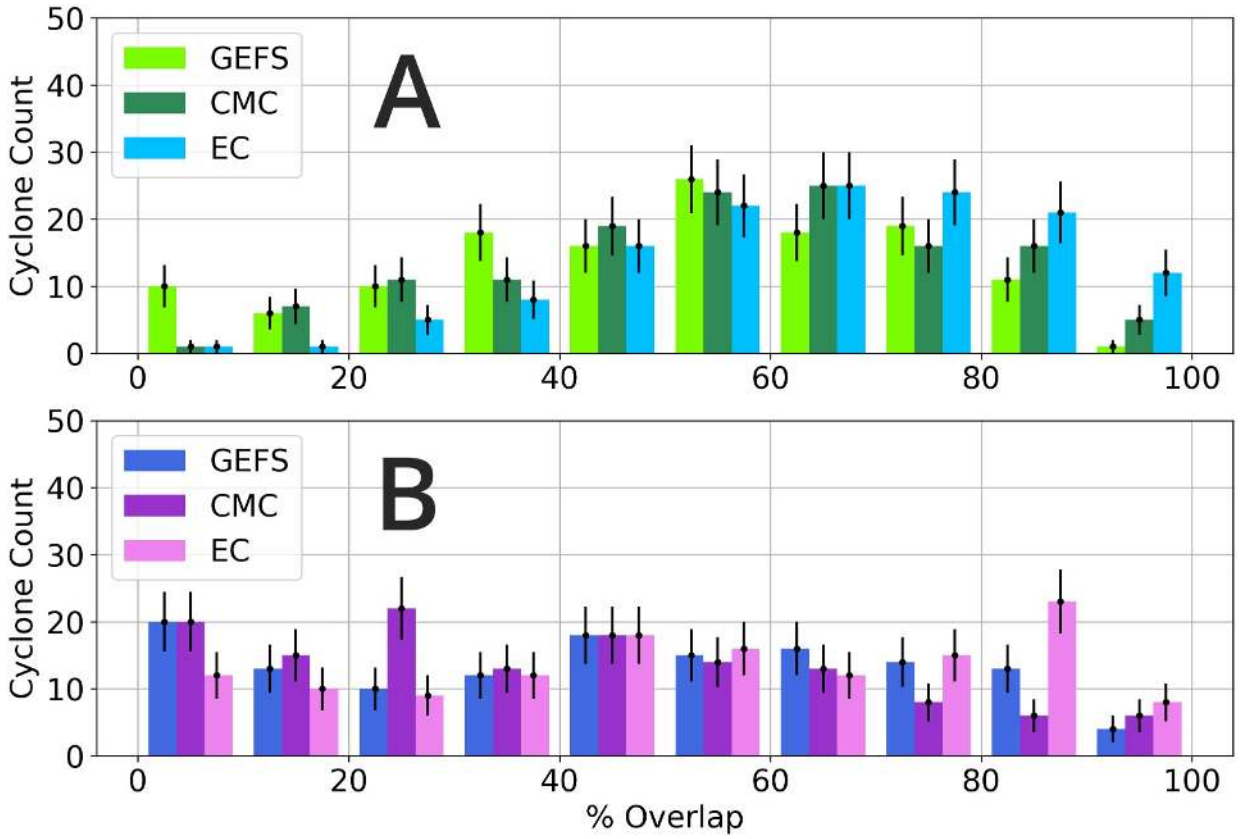


Figure 3.17: Same as Fig. 3.13, but percent overlap for the weakest 50% of cyclones

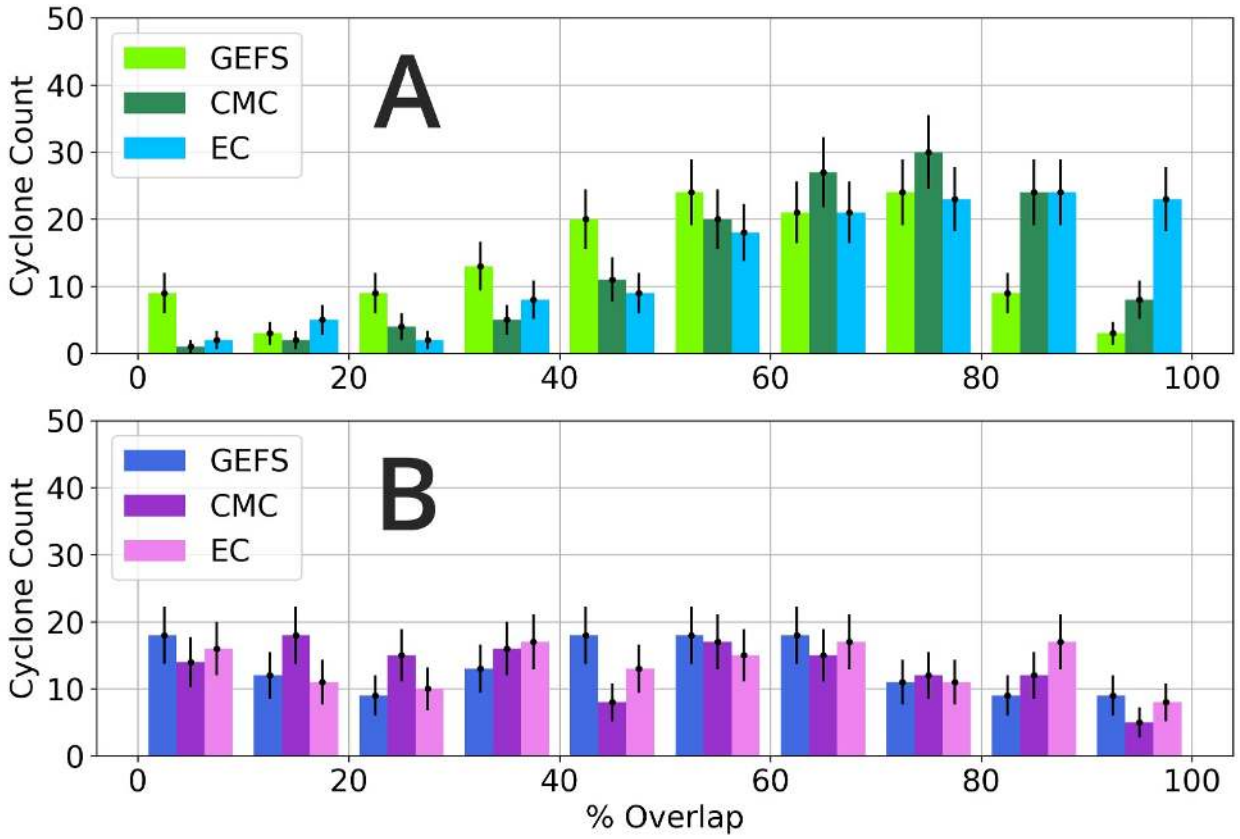


Figure 3.18: Same as Fig. 3.17 but for the deepest 50% of cyclones. (a) represents the short range, (b) represents the long range.

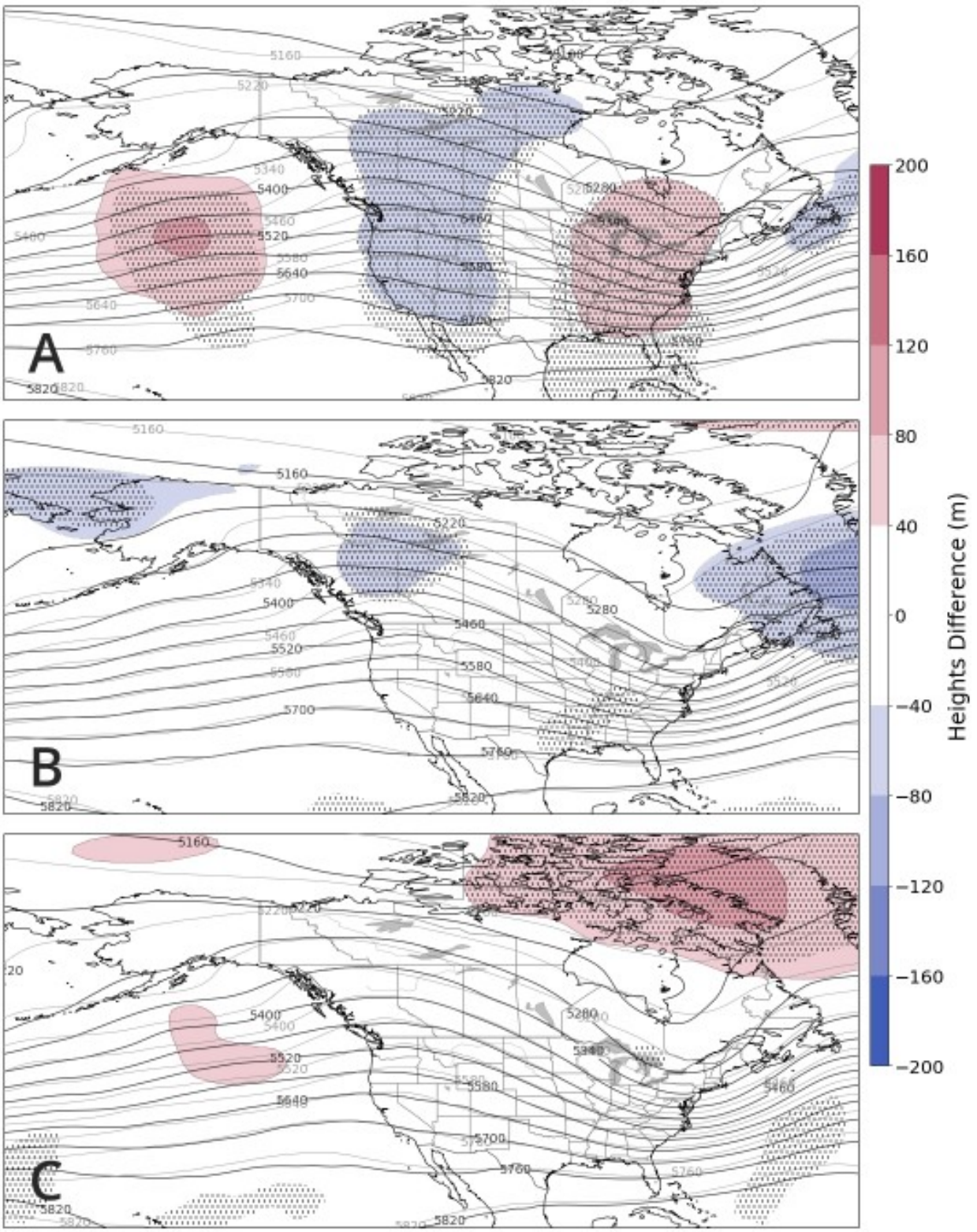


Figure 3.19: Differences between the GEFS (a), ECMWF (b), and CMC (b) flow patterns for the lowest 50%-highest 50% centroid distances in the short range (days 1-3). The hatched pattern represents 95% confidence interval via difference of means test for significance.

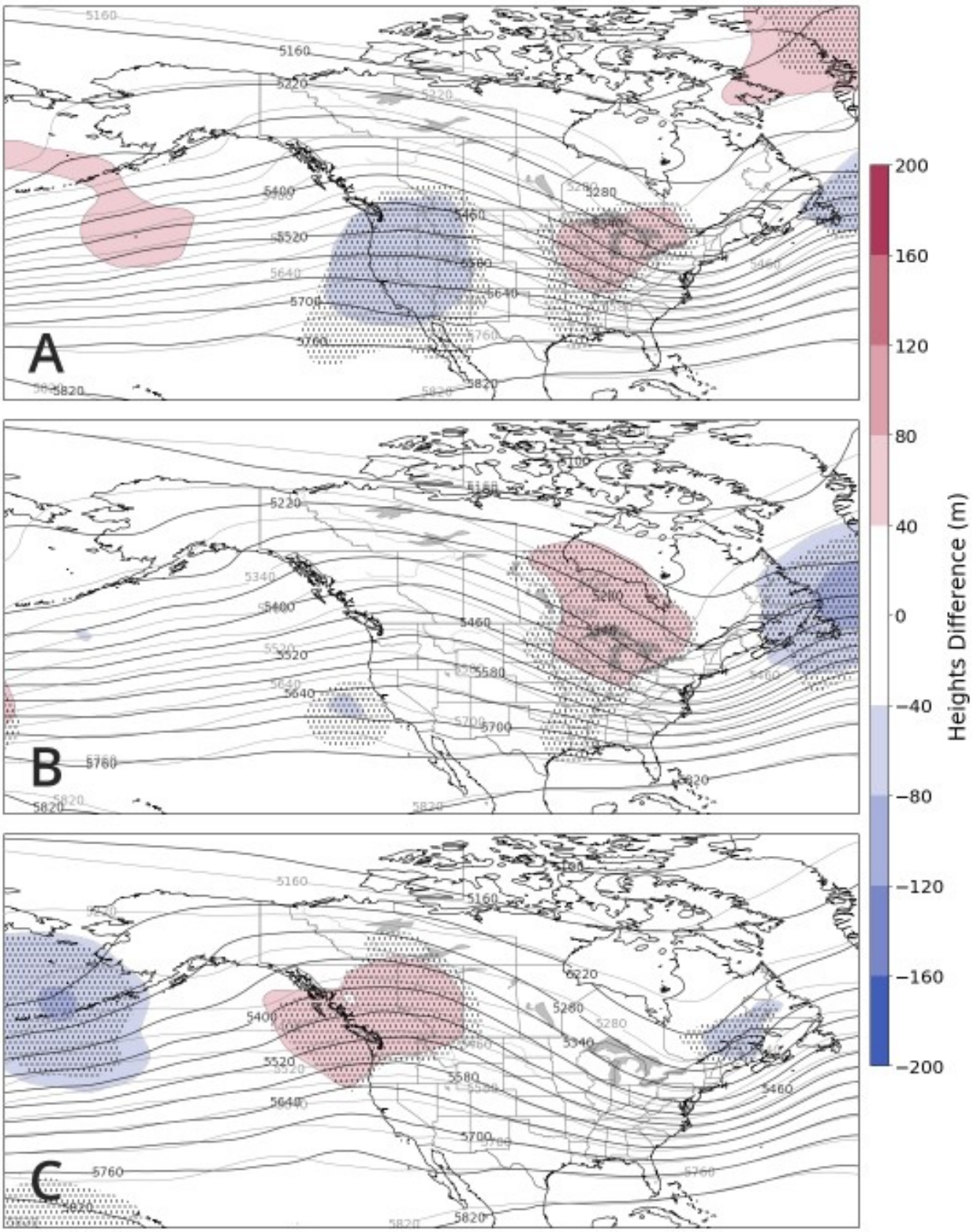


Figure 3.20: Same as Fig. 3.19 but sorted by overlap 50% highest percentage - 50% lowest percentage for GEFS (a), ECMWF (b), and CMC (c)

4 An Analog Standardized Spread Anomaly Tool

4.1 Introduction

The Ensemble Situational Awareness Table (ESAT) provides context on whether an ensemble is predicting an unusual or significant event. The ESAT analyzes forecasts from the GEFS as well as the GEFS combined with the CMC, or North American Ensemble Forecast System (NAEFS). The tool utilizes reanalysis climatology (R-Climate), which generates anomalies based on the Climate Forecast System Reanalysis (CFSR, Saha et al. 2010) 30 year (1979-2009) climatology. There is also a model climatology (M-Climate) portion, where GEFS forecasts are compared with the GEFS Reforecast (Hamill et al. 2013) for 27 years (1985-2012) of data. The goal of the M-Climate is to place a forecast in context of a climatology of forecasts given similar seasonal conditions and lead time. The M-Climate is generated by subsetting the Reforecast to similar seasonal conditions (21 days, centered about the forecast). It is further subset with the same lead time, e.g. only 72 hour forecasts are considered for the current hour 72 forecast calculation.

Taking the ESAT a step further, the ensemble spread of the GEFS can be treated as the variable to be compared to understand the historical variability of the forecast at each gridpoint. As explained in Chapter 1, the variability of forecasts increase with latitude in the Northern Hemisphere (Fig. 1.4); thus, it is important to compare each gridpoint individually to avoid consistently biasing data with increasing latitude. The spread of the forecast can then be compared to the GEFS Reforecast spread in a similar seasonal subset and lead time, outputting a standardized spread anomaly (SSA, Eq. 10). The restricted SSA tool contextualizes the spread further, to only assess similar forecasted events in the M-Climate. Utilizing the methods and calculations explained in Chapter 2, the tool was used to generate SSA data for the 90 cyclones examined in this study.

There are a few key benefits to using this tool over ensemble spread which benefit the forecasting enterprise. An immediate benefit is that, unlike ensemble spread, the magnitude of SSA is not inherently dependent on forecast lead time. It provides context of the forecast uncertainty explicitly related to the forecast hour being examined. The SSA tool also explicitly references forecasts of $\sim$10% similarity to their mean values (Eq. 11), which helps to provide context on how the variability of a certain ensemble mean value compares to similar historical forecasts. The current calculation normalizes the data to a more Gaussian distribution (Eqs. 12, 13), which provides a statistical reasoning for the values of the standardized anomalies. In other standardized anomaly outputs, such as on the ESAT, it is possible to reach values of 8 σ or greater. Values that extreme provide a clear signal to end users but is limited in that it is bound to Chebyshev's Theorem only - the return interval of the output is not intuitive to end users. The ESAT produces percentiles to address this limitation.

There are also limitations to the SSA tool which must be considered. The tool does not take into account variations in synoptic flow or tendency with time or space of that variable. It is also limited by the model itself. If the GEFS ensemble spread is poor at diagnosing colocational discrepancies as the ensemble spread, the tool will also be similarly poor. Biases and limitations in the spread-error relationship will likely manifest in the SSA procedure. This limitation motivates the creation of an M-Climate for other ensemble models, such that biases which are present in each model can provide a more robust solution for end users.

4.2 GEFS and GEFS Reforecast

As the M-Climate being used to compare the operational GEFS is a slightly different ensemble, it is important to compare the output of both to assess similarities and differences in ensemble mean, MAE, and spread values. The GEFS Reforecast v2 (Hamill et al. 2013) is utilized for the M-Climate. The reforecast uses the GEFS 2012 dynamical core but outputs only 10 perturbed members, where the GEFS outputs 20. Both models up to 2013 utilize the ensemble transform technique with rescaling (ETR) to initiate perturbations in members (Wei et al. 2008). To assess the efficacy of utilizing the GEFS Reforecast (GEFS-R) for the SSA tool, the model output was compared to the operational GEFS cyclone data. The ensemble mean of the GEFS-R MSLP for day 1 shows a similar distribution to the GEFS cyclone ensemble mean MSLP, with the mean (blue and green bars) for each model nearly identical and the median similar (Fig. 4.1). When comparing the distributions using the difference of means test for statistical significance (Wilks 2011), the GEFS-R FCR ensemble mean MSLP is statistically significantly deeper for days 3-7. Mean FCR MSLP values for GEFS-R are $\sim$1.8 hPa (day 3) to $\sim$3.8 hPa (day 7) lower. The variability amongst the FCR ensemble mean MSLP for the cyclones is larger for the GEFS than the GEFS-R which could be due to the difference in ensemble member space.

The ensemble MAE for the GEFS and GEFS-R are also similar, including the tails of the distributions (Fig. 4.2). Day 1 maximum MAE values for the GEFS-R and GEFS are $\sim$5.3 hPa and $\sim$5.6 hPa, respectively; the means are $\sim$1.7 hPa and $\sim$2 hPa. Day 2 presents a similar solution with the GEFS-R MAE maximum, mean, and median lower than the GEFS MAE. The model with the largest maximum values for days 3-7 fluctuate, but the median and mean values continue to show the GEFS-R MAE as lower than the GEFS. The mean values, when comparing the two

models, are ∼14% to ∼27% lower for each area averaged cyclone case. The minimum values for the GEFS-R also range from ∼32% to ∼61% lower. The MAE is significantly different at the 95% confidence level for all lead times. The ensemble spread values for the GEFS-R compared to the GEFS paints a similar picture as the ensemble MAE (Fig. 4.3). The spread of the GEFS-R cyclones is consistently lower in nearly all aspects except for maximum values. The mean of all spread values is ∼5% to ∼30% lower for the GEFS-R compared to the GEFS, where the minimum values are also ∼18% to ∼49% lower. The mean ensemble spread of the 90 cyclones is significantly different at the 95% confidence level for days 1-6 but is not for day 7. The GEFS-R values for the 90 cyclones is lower in magnitude compared to the operational GEFS, which is important to note when considering the SSA tool.

4.3 Cyclone Case Examples of SSA

To provide context of the tool's usage, the output data can be plotted alongside GEFS ensemble spread to identify its patterns and where a forecaster would seek to look when using it. The SSA is compared against the spread for three lead times, hours 120 (Fig. 4.1a), 72 (Fig. 4.1b), and 24 (Fig. 4.1c), for the 26 February 2010 cyclone event valid at 00z. The GEFS ensemble mean is plotted as black contours, the standardized MAE as dashed black contours. At hour 24, the cyclone center is east of Long Island with a tight MSLP gradient to the north and east over New England. The SSA for hour 24 shows a 1.5 - 2 σ standardized anomaly into northeastern Pennsylvania and southern New York, with a 0.5 and 1 σ anomaly spreading from Cape Cod to Washington, D.C. The spread maximum is located in a similar region into New Jersey and southern New York. The area of greatest uncertainty relative to similar historical forecasts would be located in the region

near Pennsylvania and southern New York, where the largest standardized MAE occurred over Long Island, New Jersey, Connecticut, and Southern New York.

At hour 72, the ensemble mean shows a weaker cyclone compared to hour 24 likely due to ensemble member dispersion. The center of the low pressure system is shifted to the east, off of Cape Cod. The SSA maximum is located off the Long Island and New Jersey coast, with values less than 0.5 σ covering the region. There is a region of SSA below -2 σ into Newfoundland and Labrador. The spread shows a very large region of greater than 6 hPa spread from offshore New Jersey to Cape Cod. That region of spread off the Long Island coast is, according to the SSA tool, the most uncertain location on the map compared to historical forecasts of similar magnitude. The verification based on the standardized MAE identifies an area of 1 σ offshore which overlaps well with the SSA.

At hour 120, the MSLP ensemble mean field is weaker in comparison, with a central low pressure well off the New England coast slightly below 1000 hPa. The SSA does not identify any regions greater than 0.5 σ anomaly, with the same region in Canada showing a low SSA of below -1.5 sigma. The spread maximum of slightly above 7.5 hPa is located in the center of the ensemble mean low pressure system. Although the spread may seem large, the SSA provides context showing that it is not abnormally uncertain relative to other forecasts in that MSLP range. The standardized MAE, though, shows that there was around a 3 σ anomaly offshore near the 40-70 benchmark.

A cyclone valid on 8 Jan 2009 at 00z occurred near Newfoundland (Fig. 4.2). The GEFS generated a cyclone at the valid time which shifted further east with decreasing lead time. At hour 120 (Fig. 4.2a), the GEFS mean placed a cyclone center south of Nova Scotia with a spread maximum to the northeast over Newfoundland, though, as seen in the SSA, the spread was not significantly

anomalous given the aforementioned parameters of SSA. At hour 72 (Fig. 4.2b), the spread was still large around Newfoundland, where in this instance it was significantly anomalous. The SSA at hour 72 identified areas of greater than 2 σ east and north of Newfoundland, where standardized MAE anomalies reached 3 and 4 σ in similar locations. The hour 24 forecast (Fig. 4.2c) produced spread values of between 3 and 6 hPa which was extremely anomalous when looking at SSA, with values above 2.5 σ - the maximum possible value of SSA in this formula is 2.64 - with an accompanying 4 σ standardized MAE anomaly.

A third case, valid 10 February 2013 at 00z, had an extremely deep central pressure of around 954 hPa in the ERA-Interim analysis dataset (Fig. 4.3). Variance in the GEFS was significant, with the hour 120 (Fig. 4.3a) maximum spread values ranging from 12.5 to $\sim$15.5 hPa accompanied by a central ensemble mean low pressure of $\sim$980 hPa. The SSA tool identified a $\sim$1.5 σ anomaly off the coast of Nova Scotia, where the standardized MAE produced a 3 σ anomaly over the island. Hour 72 (Fig. 4.3b) identified a low around 16 hPa deeper with the largest spread values around the western quadrants of the low of $\sim$6 to $\sim$9.5 hPa, with SSA values reaching slightly above 1 sigma, though the standardized MAE reached to 3 σ south of the GEFS mean low pressure center. A similar solution occurs at hour 24 (Fig. 4.3c), with 3 σ standardized MAE anomalies occur south of the GEFS mean low pressure center and SSA maximum values of $\sim$1 to 1.5 σ occurring west of the low center. A limitation in the tool is shown here, where there is significantly limited sample size due to the rarity of events such as ¡960 hPa lows in the GEFS Reforecast M-Climate. The SSA tool still was able to identify the potential for uncertainty although the ensemble spread was below $\sim$3 hPa about the cyclone for the valid time.

4.4 Comparison of SSA and Standardized MAE

It is optimal for the SSA to be similar to the GEFS ensemble spread in its relationship to MAE, such that any predictive "skill" of forecasting its own error would not be lost to the transformation. The calculations for the upcoming results are calculated from the FCR procedure. The mean values of all short range (days 1-3) cyclones' SSA values are averaged and plotted as the independent variable, where the standardized MAE is plotted as the dependent variable with the intention of the SSA being able to predict the error in some capacity (Fig. 4.7). The shaded contours around the regression line are representative of the 95% confidence interval, created with a bootstrap of $n = 1000$ resamples. The standardized MAE in this plot is different than the values shaded in Figs. 4.1 - 4.3. The calculation for the plotted data was the traditional standardized anomaly formula. In this calculation, Eqs. 14 and 15 are used to compare the cyclone averaged SSA to the cyclone averaged standardized MAE. The standardized MAE is mostly below 0 due to the majority of points being near zero and a PDF which is similar to a gamma distribution with shape parameter k of 1 and scale parameter theta of 2.

Although the regression is not 1 to 1, there is a clear linear relationship with increasing SSA corresponding with increasing standardized MAE. The Pearson correlation coefficient (r value) yields a result of 0.546. To test for incidence of extreme values affecting the result of the correlation, it is useful to look at the median as well as the mean of data. The medians of all the cyclone arrays for days 1-3 are also plotted (Fig. 4.8). The median as a statistical tool is a robust measure compared to mean; means are sensitive to extreme outliers whereas medians are not. The regression is very similar to the mean calculation with an r value of 0.537, similar to 0.546 for figure 4.4. Therefore, there is evidence that the relationship is not sensitive to extreme outliers in the data.

The relationship between SSA and standardized MAE is stronger for the long range than the short range, with an r value of 0.641 (Fig. 4.9). There is also a stronger relationship between median SSA and standardized MAE, which indicates that there is evidence that the SSA is a good predictor even in the medium to long forecasting range for the likelihood of errors in extreme events such as cyclones (Fig. 4.10).

To apply a verification technique to the data, the data were calculated similar to verification plots on the ESAT webpage ($http://satable.ncep.noaa.gov/verify/$), where the forecast M-Climate anomaly is compared with the analyzed CFSR anomaly. Similar to the ESAT graph, ten bins of SSA are sorted from low to high values, the index of the sorted values is retained, and the standardized MAE associated with the corresponding SSA values are indexed. The mean of each bin is calculated as well as the 10th and 90th percentiles. The short range (Fig. 4.11a) and long range (Fig. 4.11b) plots are calculated using all FCR gridpoints for the short range and long range for all cyclones. The large ranges of the x-axis and y-axis are due to the variance in maximum and minimum values of SSA and standardized MAE. Both lead time ranges show a similar pattern and variance between the 10th percentile, the mean, and the 90th percentile. The similarity shows the efficacy of the SSA tool to be able to predict the error with similar skill between both ranges. With that being said, these values are representative of the most extreme climatological cases of MSLP and would likely decrease in variance if the verification dataset and range was expanded to the same extent as the ESAT verification tables.

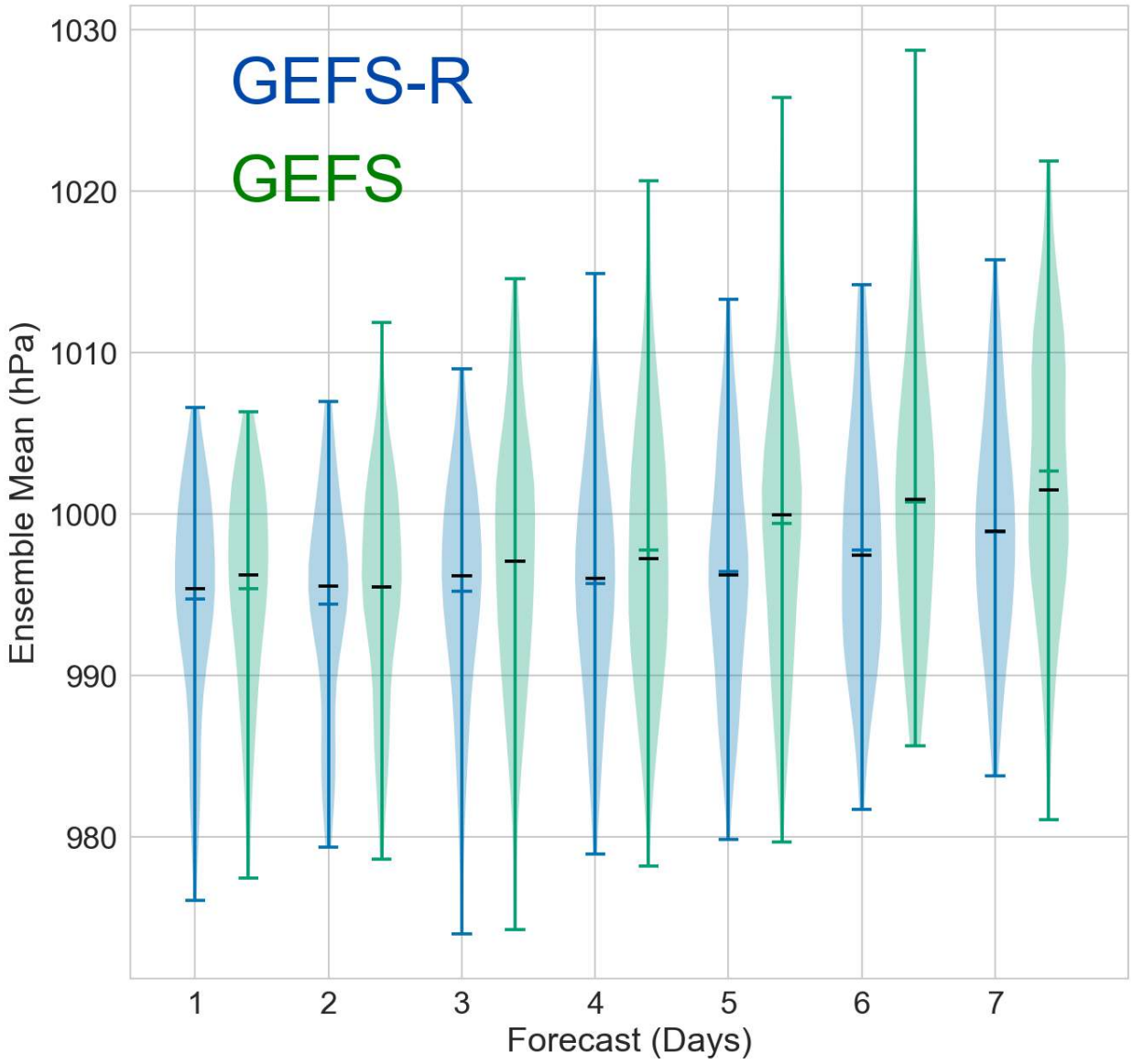


Figure 4.1: Violin plots for the GEFS-R (blue) and GEFS (green) ensemble mean for the FCR area averaged 90 cyclones for each lead time. Maximum and minimum values of each distribution are denoted by horizontal marks at the tails of each plot. Mean values are denoted by the blue and green marks within each plot and medians are denoted by black marks. The frequency of values in the distribution is depicted by the filled contours for each plot.

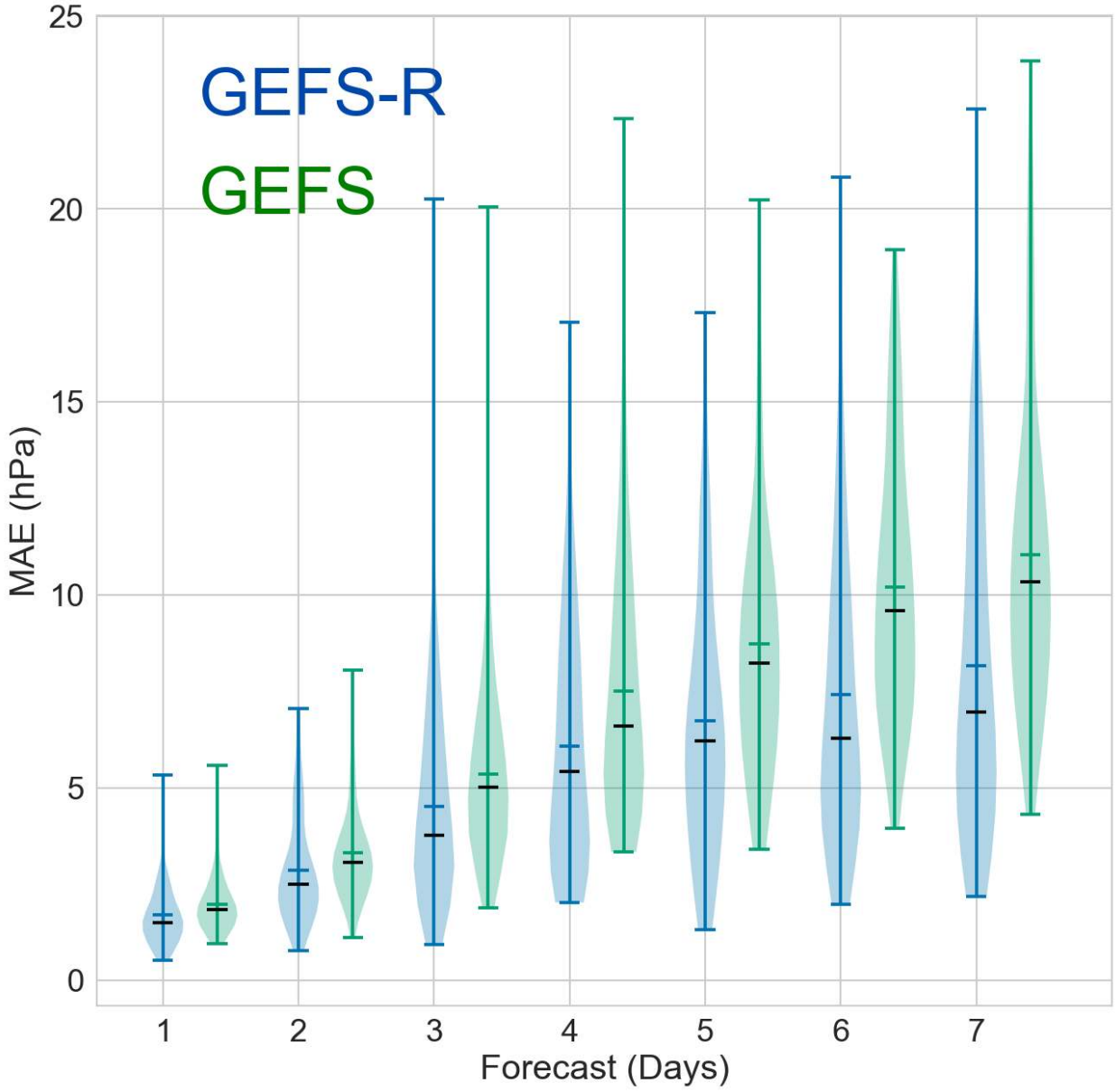


Figure 4.2: Same as Fig. 4.1, but the ensemble MAE for the GEFS-R and GEFS.

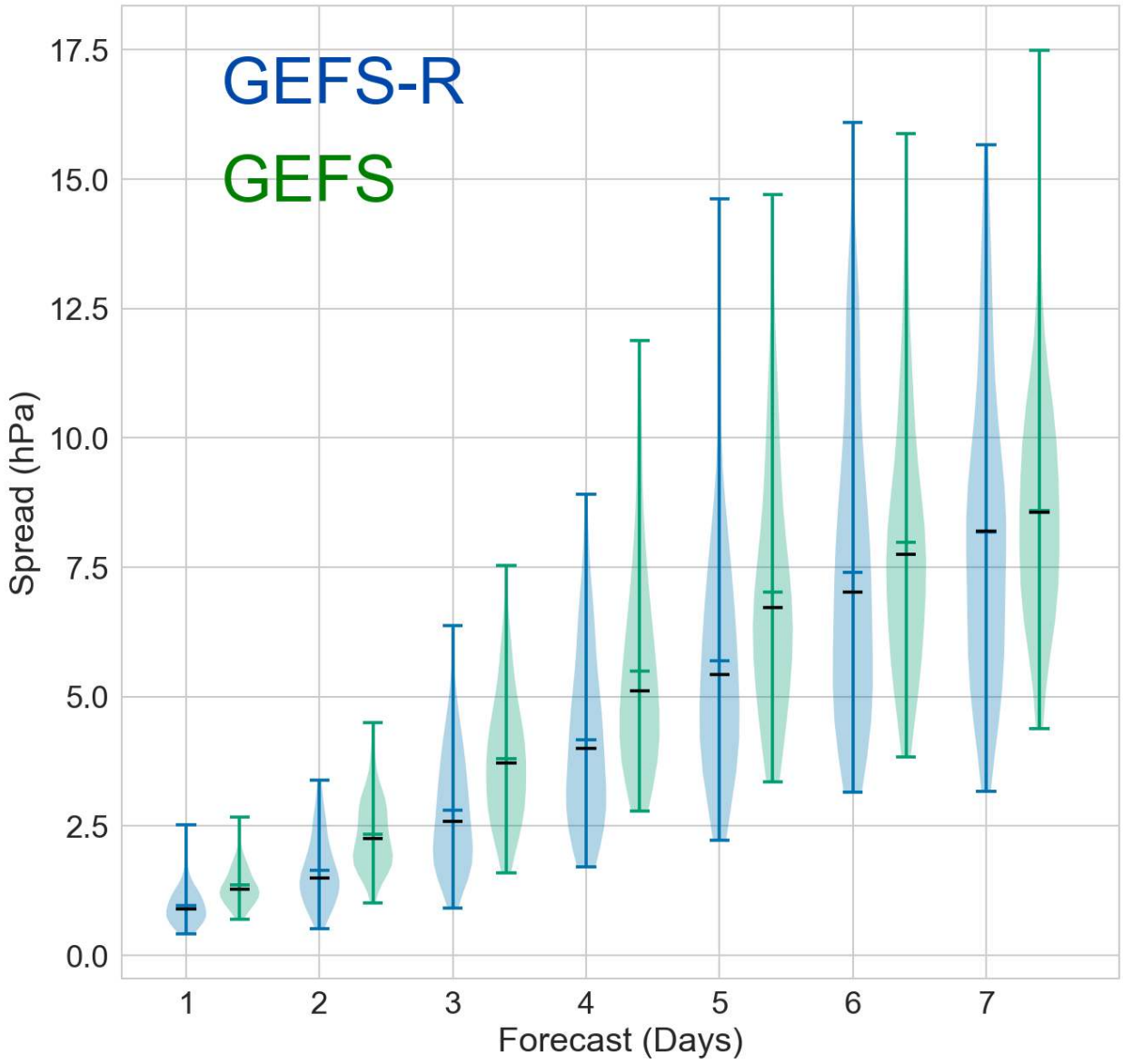


Figure 4.3: Same as Fig. 4.1, but the ensemble spread for the GEFS-R and GEFS.

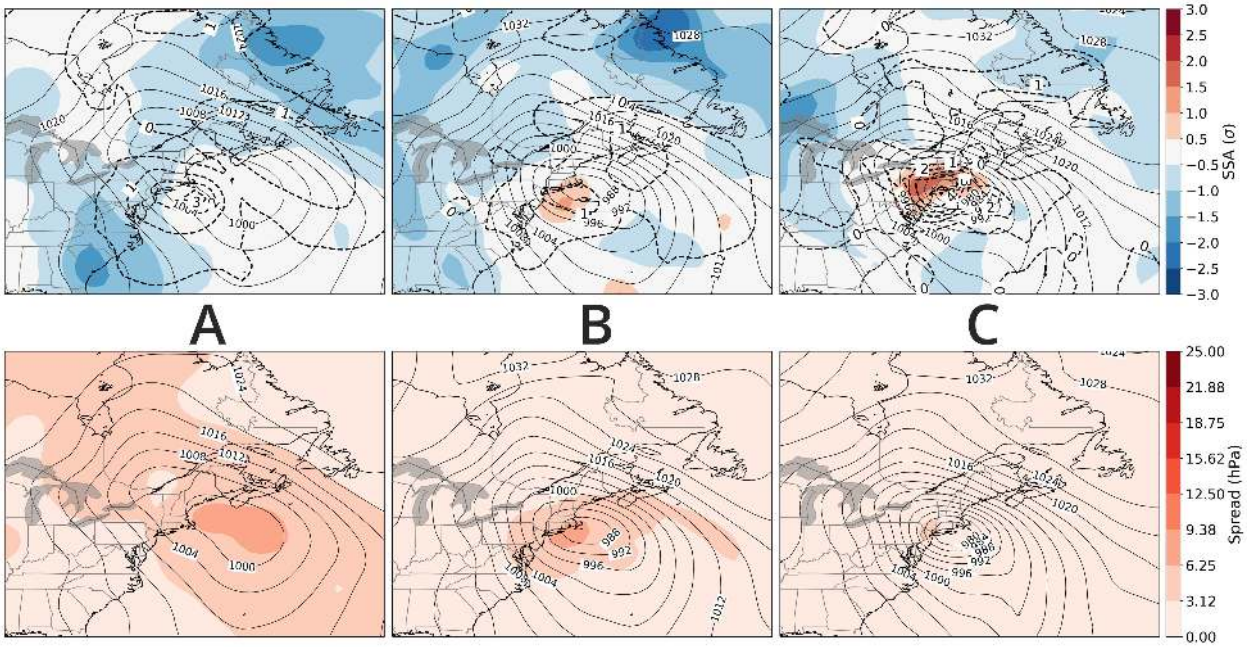


Figure 4.4: Standardized spread anomaly (SSA, top) and ensemble spread (bottom), for hours 120 (a), 72 (b), and 24 (c), valid 26 Feb 2010 at 00z. Standardized mean absolute error (MAE) is shown as hatched contours alongside SSA, GEFS ensemble mean sea level pressure (MSLP) is contoured on both SSA and spread plots.

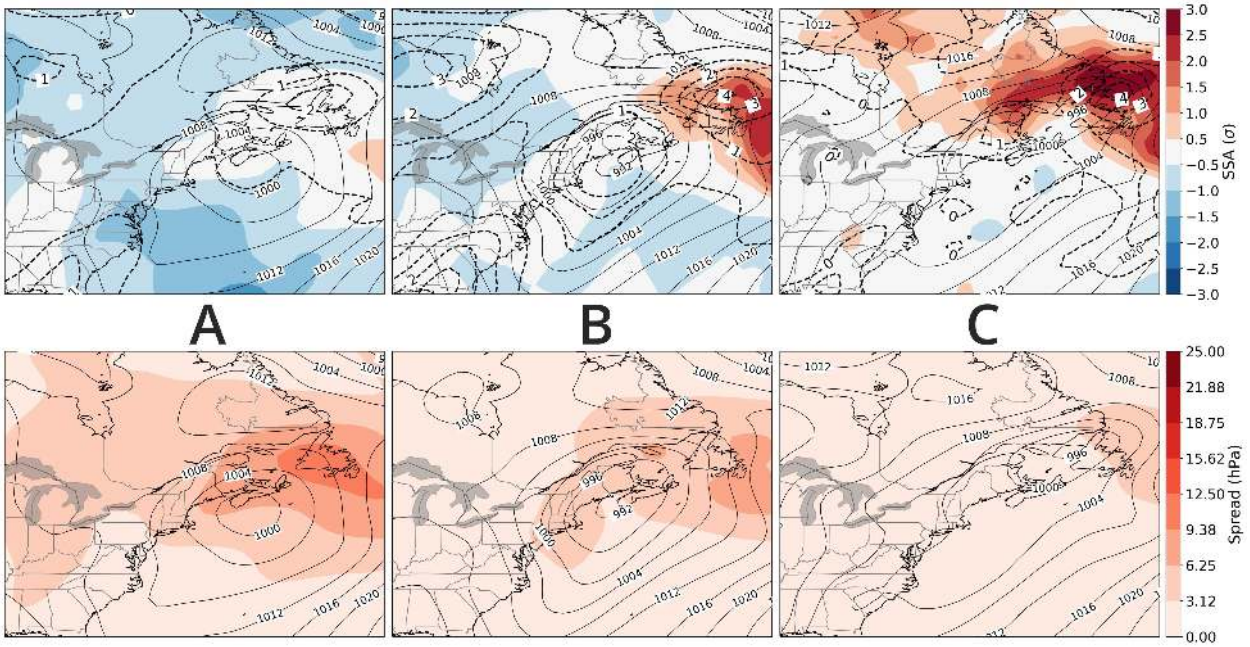


Figure 4.5: Same as Fig. 4.4, but valid 8 Jan 2009 at 00z.

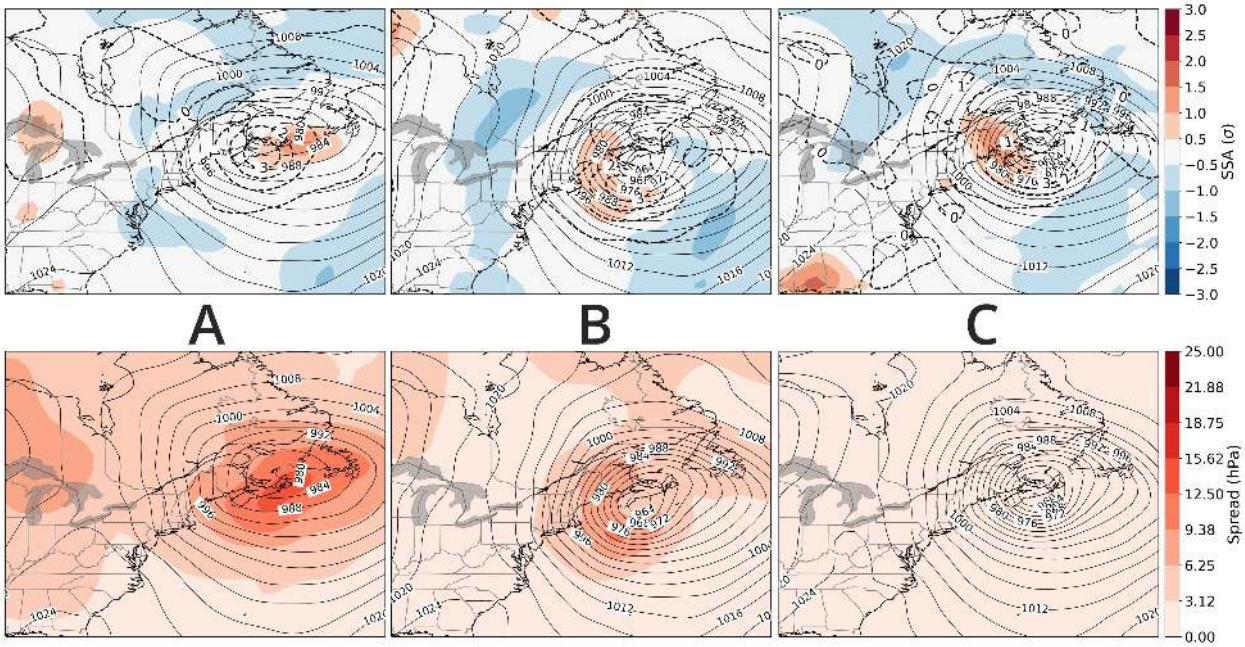


Figure 4.6: Same as Fig. 4.4, but valid 10 February 2013 at 00z.

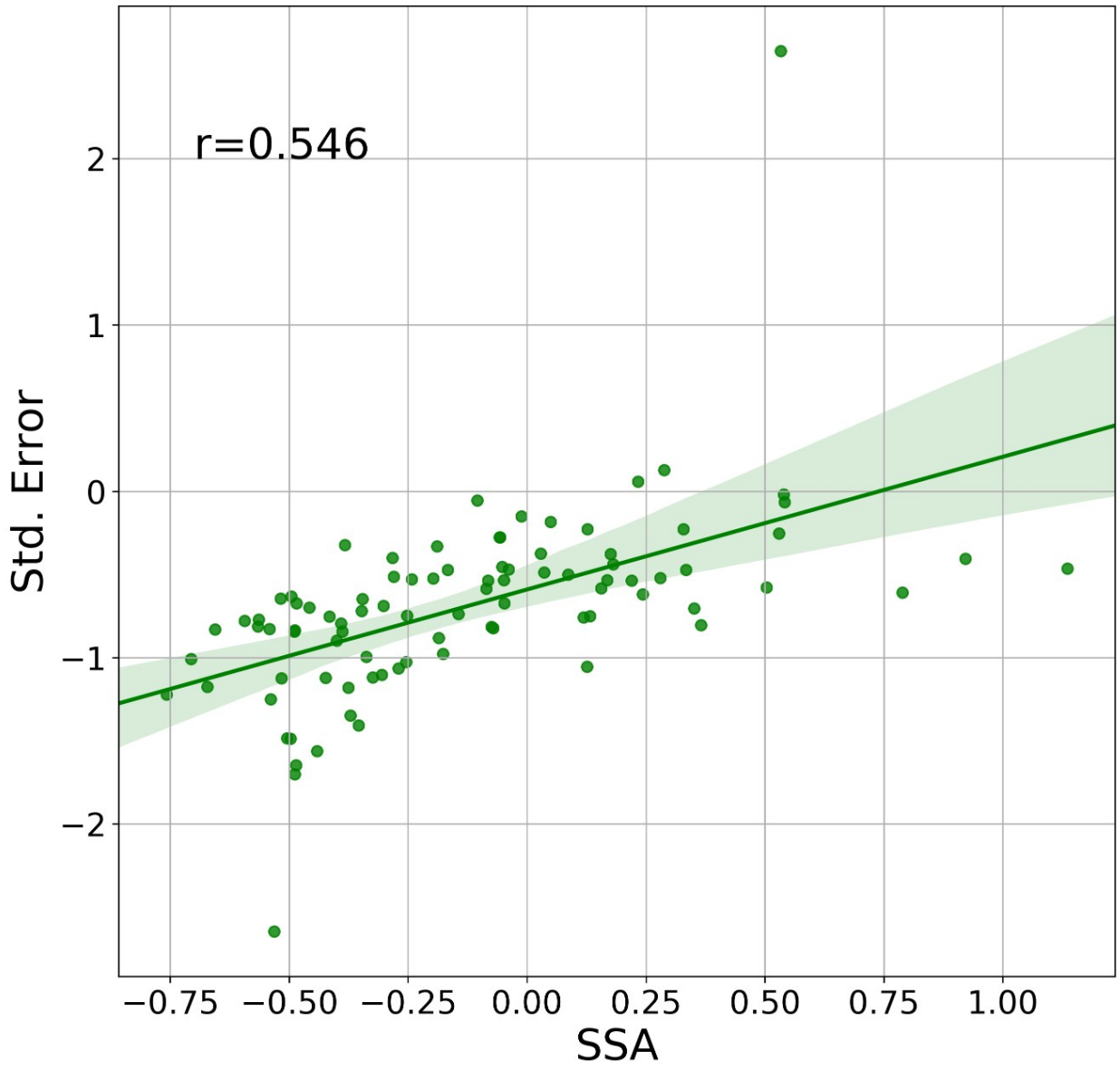


Figure 4.7: Mean SSA and standardized MAE FCR averaged short range (days 1-3) forecasts. The solid line represents the regression line, the shaded region represents an n=1000 95% confidence interval bootstrapping.

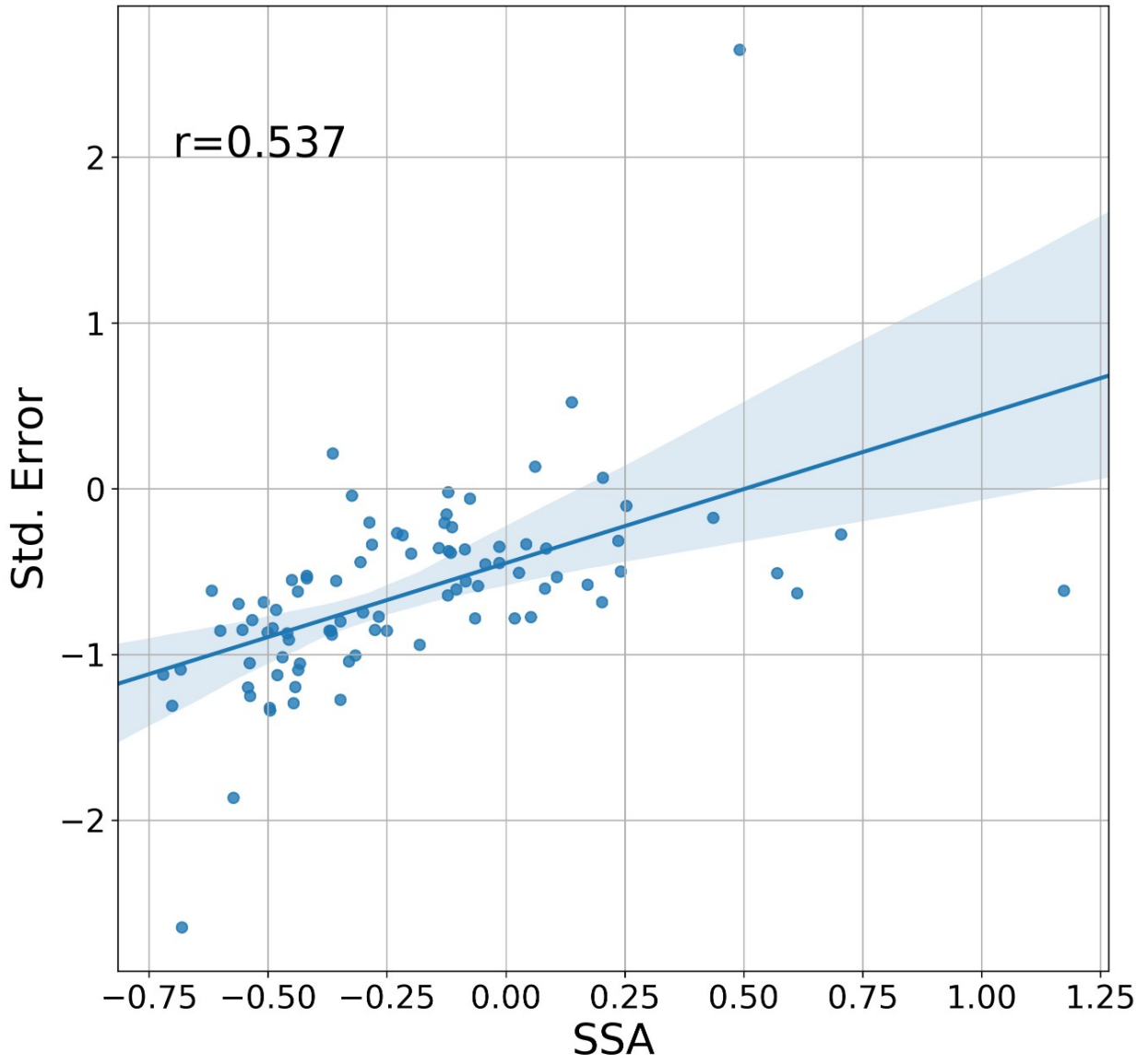


Figure 4.8: Same as Fig. 4.7, but for median in place of mean values.

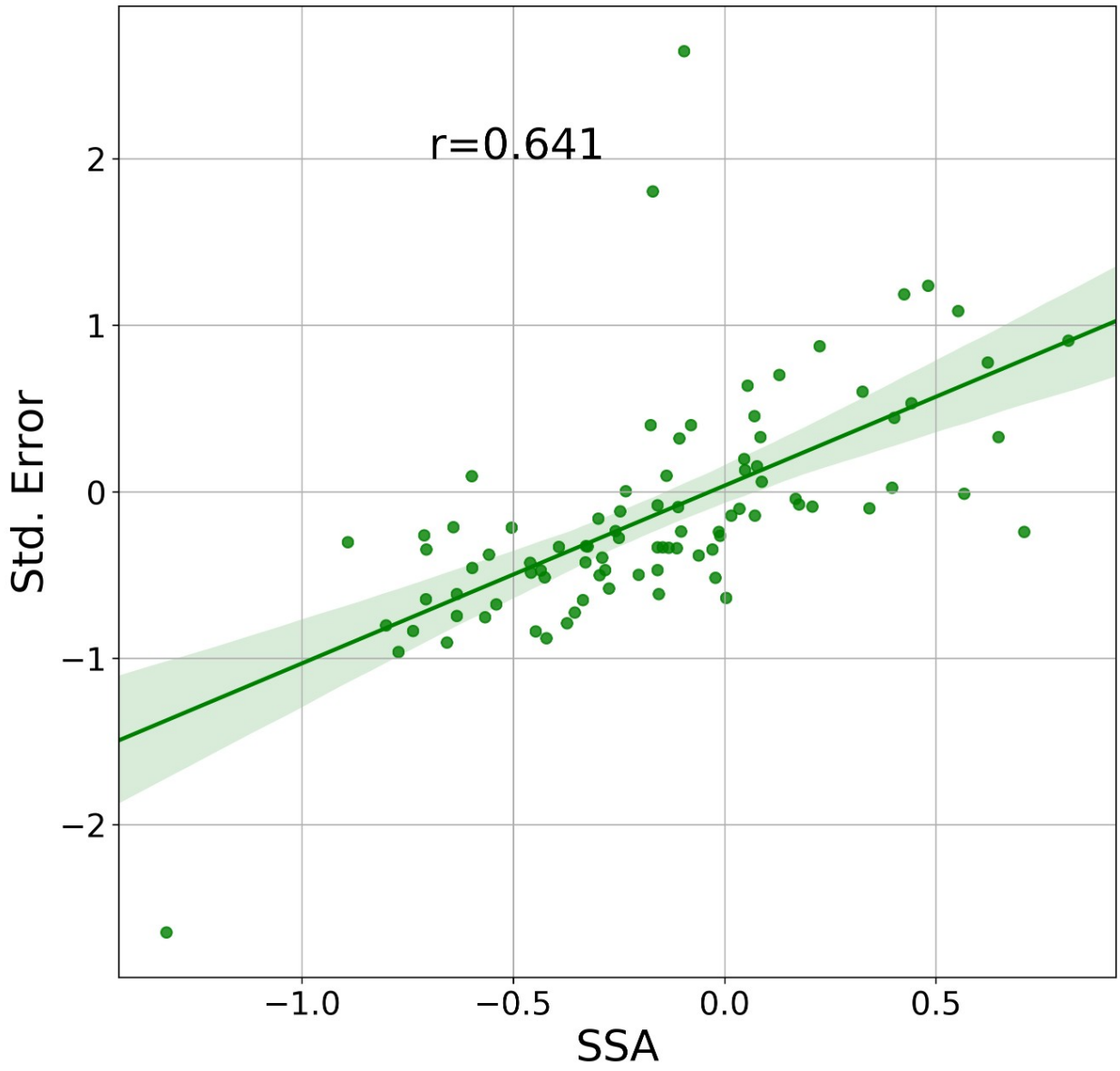


Figure 4.9: Mean SSA and standardized MAE FCR averaged long range (days 5-7) forecasts.

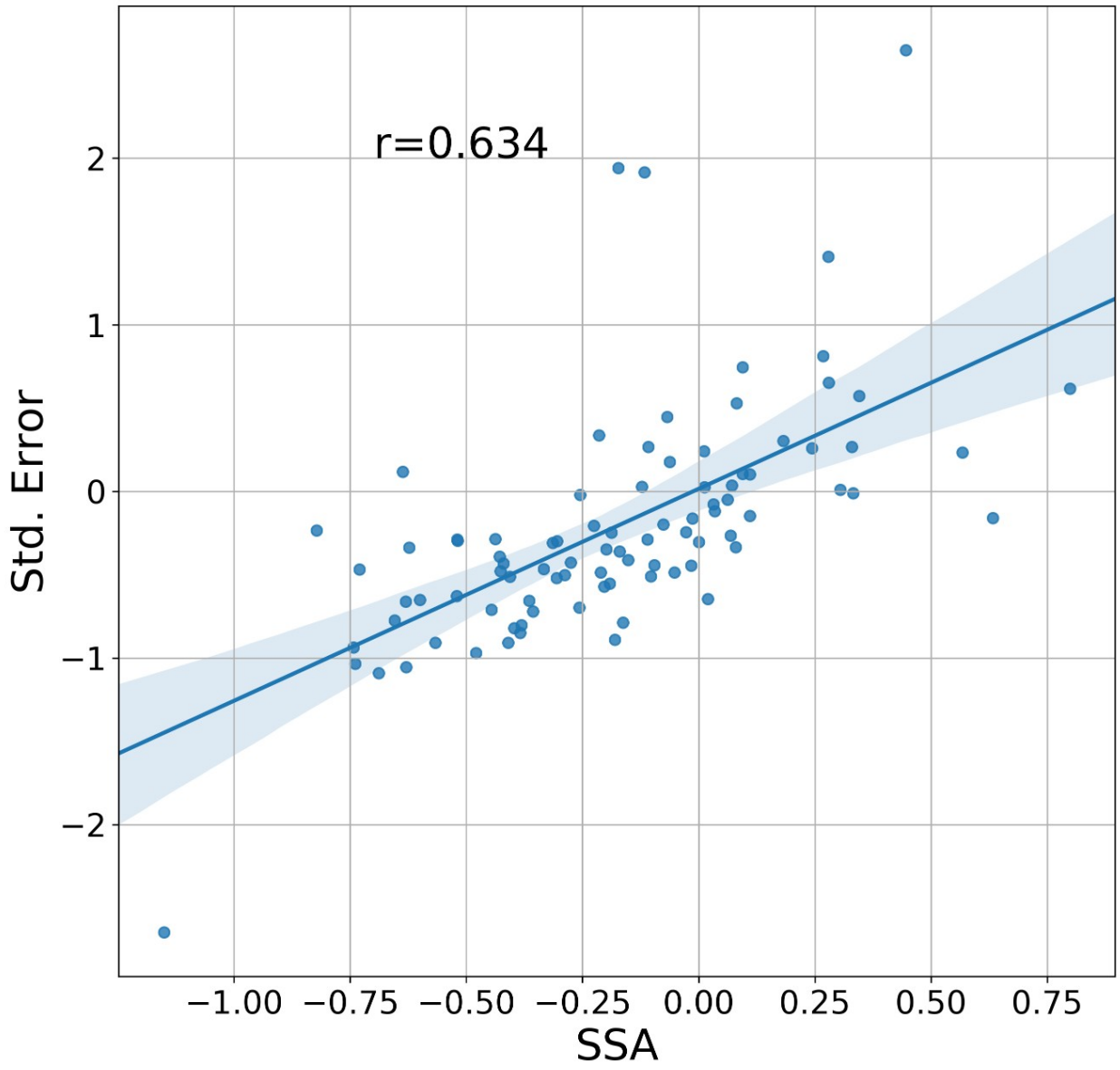


Figure 4.10: Same as Fig. 4.9, but for median in place of mean values.

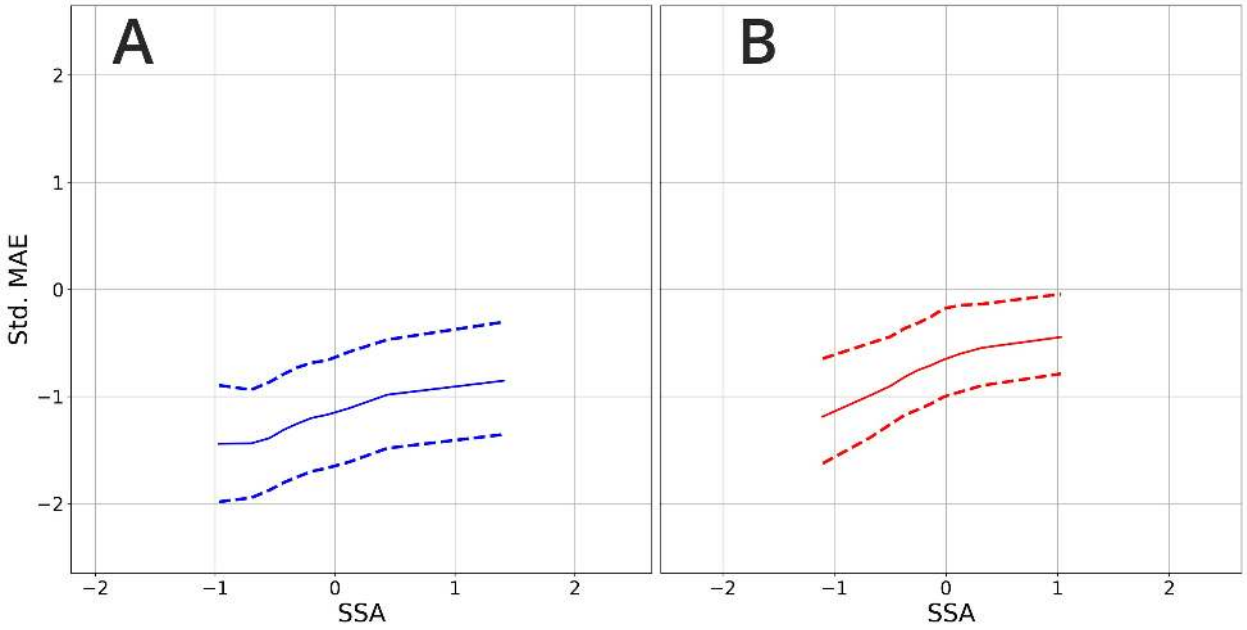


Figure 4.11: Mean values, 10th, and 90th percentiles of all short range (a) and long range (b) SSA forecast values compared against standardized MAE values for all FCR gridpoints for all 90 cyclones.

5 Conclusion

Mid-latitude cyclones along the east coast of the United States are particularly impactful with respect to extreme events, due to the economic and public health impacts that result. The density of population in the northeast United States provides further motivation to better understand the uncertainty in forecasts of mid-latitude cyclones. This thesis was focused on two components of ensemble forecasting that are desired by the forecasting enterprise. The first component, assessing the relationship between ensemble spread and ensemble error, utilized methods of spatial analysis to identify trends of spatial co-location amongst three global models. The second, developing and quantifying a tool to use for historical spread anomaly, provides a new methodology for the forecasting enterprise to both understand and contextualize the variability in an ensemble forecast.

5.1 Error and Spread Relationship

The colocation and magnitude of the error and spread of mid-latitude cyclones are an important aspect of the spread/error relationship to assess. If values are not related for significant events such as mid-latitude cyclones for MSLP, it does not provide a concrete estimation of the location and magnitude of potential errors to the forecasting enterprise. To assess this relationship, an analysis of the relationship between spread and error was performed. The ensemble spread is most often related to ensemble root-mean-squared error (RMSE) as the error metric. The mean absolute error (MAE) was utilized in this thesis due to the values being as close or closer to spread than the RMSE. 90 mid-latitude cyclones for the GEFS and ECMWF (89 for the CMC due to a missing datapoint) below 998 hPa, from 2007-2014 were identified within a bounding box from 33° N, 88° W, to 48° N, 68° W. The cyclones were subset and plotted from both an ”analysis” cyclone relative

(ACR) and ”forecast” cyclone relative (FCR) perspectives. The prescribed box for ACR uses the ERA-I analysis cyclone minimum pressure center, and the box for FCR moves with the average centroid of ensemble member minimum pressures for each forecast lead time for the GEFS, CMC, and ECMWF ensemble models.

The FCR ensemble spread and MAE for all three ensembles were compared using multiple metrics. The largest magnitudes of spread are consistently smaller than the largest magnitudes of MAE, indicating that the ensemble spread is limited in predicting the magnitude of error in the model solution. The resulting distributions of MAE and spread show patterns of correlation that degrade with lead time, with the overall patterns correlating with an Pearson correlation coefficient (r) value of 0.74, 0.71, and 0.75 for the GEFS, CMC, and ECMWF, respectively. The short range (days 1-3) r values for each model are 0.74, 0.65, and 0.64. The long range (days 5-7) r values are 0.44, 0.37, and 0.47. The GEFS and ECMWF overall perform better when comparing magnitudes of spread and MAE although without statistical significance. None of the means of the 3 models have a 1-to-1 ratio between spread and MAE.

FCR and ACR spread and MAE were compared to assess differences between both methods. Similar values would indicate either similar magnitudes with time or spatial characteristics, i.e. the FCR center remains close to the ACR center. Both cyclone relative methods were averaged for each cyclone and their means were plotted for each lead time. The mean FCR spread and MAE were slightly greater than mean ACR spread and MAE for the GEFS (until day 5) and CMC. The ECMWF FCR spread and MAE were similar to ACR spread and MAE, indicating that the ECMWF forecasted cyclones are slightly more spatially correlated to the analysis cyclone location. The variability amongst the 90 cyclones is large; this is likely due to varying mesoscale and synoptic conditions. When looking at the maximum values of FCR and ACR spread and MAE,

the maximum MAE is significantly higher for both cyclone relative approaches for all 3 models, with the ACR MAE consistently significantly larger than the FCR.

There have been few studies if any looking into the colocationality of operational spread and MAE. To address this, an approach producing centroids of the upper quartiles of FCR spread and MAE was derived for days 1-7. The FCR was chosen due to the interest in the relationship about the forecast cyclone as it pertains to the forecaster. The distance between latitude and longitude coordinates of the spread and MAE centroids of each cyclone was calculated for all lead times. The location of each gridpoint within the upper quartile was also compared between each other.

The ECMWF performed the best when assessing the forecast centroid distances for the short and long range. The distance between the spread and MAE centroid was below 100 km in $\sim$114 out of 270 cyclone cases in the short range and $\sim$69 cyclone cases in the long range. The GEFS performed the worst in the short range for centroids within 100 km ($\sim$61). The CMC performed the worst in the long range for centroids within 100 km ($\sim$38). For overlapping upper quartile gridpoints between spread and MAE, the ECMWF also performed the best with $\sim$75 out of 270 cyclone cases in the short range overlapping 80-100%. $\sim$52 cyclones had 80-100% overlapping gridpoints for the CMC; the GEFS performed the worst ($\sim$25). The long range overlapping gridpoints showed the ECMWF as significantly better with $\sim$56 out of 270 cyclones with 80-100% overlap, where the other 2 models were worse (CMC with $\sim$28, GEFS with $\sim$38). Overall, the ECMWF performs better when considering both metrics of testing colocation between spread and MAE.

The same centroid approach was performed between the 45 deepest and 45 weakest cyclones (45/44 for CMC). The models performed relatively better in centroid distance for the deepest cyclones in the short range, with the ECMWF performing significantly better than the other models.

The long range forecasts were similar between the weakest and deepest cyclones, for all 3 models. The only difference between the overlapping gridpoints and the centroid distance when comparing the deepest and weakest cyclones is that the ECMWF performs better in the long range for overlapping gridpoints in the 80-100% bins.

The analyzed 500 mb geopotential height patterns over North America between the best performing and worst performing cases, based on both centroid distance and overlapping gridpoints, were assessed. There are varying changes in the Rossby wave packet associated with mid-latitude cyclones around the east coast of the US. All of the differences were statistically significant using the difference of means test. The GEFS for both metrics showed a weaker trough in the eastern Pacific, weaker ridge in the western US, and weaker trough associated with the cyclone in the eastern US. The ECMWF also showed similar patterns between each metric, with a weakening of the amplitude of the eastern US trough and weaker ridge east of Newfoundland. The CMC was less clear; a slightly stronger ridge on the west coast of the US and stronger Arctic ridge was observed when looking at the difference between cyclones with the closest centroid distances and furthest centroid distances. Different synoptic patterns characterized each model's sorted centroid and overlap metrics. The similarity between each metric showed the patterns which each model performed better at representing the uncertainty with. The GEFS and ECMWF tended to perform better with less amplified patterns and the CMC performed better with higher Arctic heights and a more amplified ridge in the western US.

5.2 Standardized Spread Anomaly

The motivation to analyze the spread/error relationship for MSLP was due to the development of the standardized spread anomaly (SSA) tool for operations. Any tool which postprocesses model data will retain the biases and errors unless it explicitly address them. The SSA tool utilizes the GEFS Reforecast to compare the spread of similar ensemble mean anomalies at each gridpoint of a forecast. By doing so, it can identify the uncertainty of an event relative to similar historical forecasts. Unlike the ensemble spread, the SSA does not increase with lead time, which helps to provide context for each forecast hour. An ensemble spread forecast will increase with lead time due to the divergence of the ensemble members. The temporally averaged spread will also increase with latitude due to an increase in significant MSLP events associated with the polar jet stream. The SSA tool accounts for both of these with taking a gridpoint by gridpoint approach to the calculation as well as only comparing the forecast with the same lead time in reforecast data. One of the biggest benefits of the tool, though, is the ability to provide a statistical context to how significant the spread anomaly is. It is challenging (and subjective) for a forecaster to make a determination on spread anomaly with only an ensemble spread product available. Providing an objective measure to assess spread anomaly removes the risk of human error when considering the uncertainty of a forecast.

The GEFS and GEFS Reforecast, version 2 (GEFS-R), were compared. The goal of comparing the ensemble means, spreads, and MAE was to identify what relationship the two models had, as the GEFS-R is used as the M-Climate for the real time GEFS. The relationship would affect the interpretation of the SSA tool depending on the similarities or differences. For extreme cases such as mid-latitude cyclones, it was shown that the GEFS and GEFS-R for days 1-7 lead times

have both similarities and differences. The mean values of both spread and MAE, for instance, are statistically lower for the GEFS-R compared to the GEFS for a majority of, if not all lead times. An explanation behind the disparity in ensemble spread could be due to the lower number of ensemble members in the GEFS-R, though the MAE is also lower overall, which could signal that more ensemble members may not necessarily correspond to a better model solution. The percentiles and standardized anomalies for test cases throughout the development of the tool were noted to be very large which is likely due to the consistently lower spread of the GEFS-R.

Three different examples of the SSA tool were plotted, overlayed by the standardized MAE and juxtaposed against the ensemble spread to compare. The SSA subjectively shows a slight improvement with positioning of the MAE compared to the spread for all three sample cases examined. Areas which had significantly high spread anomaly also were spatially close to, or overlapping with significant MAE anomaly. The MAE was compared with the SSA by standardizing it with a similar approach for all 90 cyclones. The values correlated in the short range with r values of 0.546 and 0.537 for mean and median values for each cyclone, respectively. In the long range, the values correlated with r values of 0.641 and 0.634 for mean and median, respectively. The increase in correlation for the long range shows the efficacy of this tool for relating the spread anomaly to the resulting error when used in a forecasting setting, especially in longer range forecasts. A verification method that was utilized in the Ensemble Situational Awareness Table (ESAT) was replicated and found that the mean, along with the 10th and 90th percentile was slightly improved with the longer range than shorter range. This result diverges from the results in the ESAT verification table, but this is likely due to the extreme nature of the data in this thesis compared with the large datasets utilized for the ESAT verification plots.

5.3 Discussion and Future Work

The results in this thesis show evidence of varying degrees of correlation with ensemble spread and MAE for 3 global models. The ECMWF performed significantly better overall, which motivates the continuing improvement of the GEFS. The variability of colocation of spread and MAE for the cyclones in this study provides a more robust understanding of the relationship between the two metrics. Operational forecasters should be vigilant of the limitations of ensemble spread in predicting the location of the error. The forecasting enterprise should also be incredibly cautious if using spread as a predictor of magnitude of error, as spread for all lead times tended to be lower than MAE. The maximum values between spread and MAE were significantly different for all lead times, which shows the inability of spread to accurately reflect the potential of significant errors.

The SSA tool provides the forecasting enterprise a specialized tool to utilize for uncertainty. The methodology which the tool is based on is able to provide context of the spread both from a perspective of lead time as well as historically similar events. Each forecast hour has a unique map generated for it based on the M-Climate data for that hour. The tool only utilizes spread M-Climate datapoints within 10% of the ensemble mean, which filters out events in the M-Climate too far from the forecast value. For instance, when taking the subset spread M-Climate, an ensemble mean forecast of 996 hPa at a point would likely filter a value in the entire M-Climate at 1030 hPa.

It is important to note the caveats in this study, such as the methodology to determine the cyclone centers. The method that produced the ensemble member centroids could have been affected by members without a cyclone, only identifying the minimum MSLP in the domain. It is important to note that in an operational context, it is challenging to assess each ensemble member individually. A forecast with verifies with a cyclone that has ensemble members without a cyclone could

be considered overdispersive depending on lead time and the synoptic conditions. Regardless, it is important to note that the procedure to identify the centers could be affected by inaccurate changes in the centroid position. The size of the box for the cyclone relative approaches was set due to the incidence of unrelated maxima on the domain affecting the calculation of average spread, spread centroids, and MAE centroids, but cyclones do occasionally have a range of effects greater than $\sim$1100 km.

Other extreme events are important to assess such as heat waves and mesoscale severe weather events using the methodology in this study. It would be advantageous to apply a similar methodology to different variables such as 500 hPa geopotential heights. The SSA tool is also useful for multiple variables; an assessment of similar variables would provide further motivation to expand the tool's usage. The tool, as mentioned, is also only as useful as the ensemble spread of the model and its reforecast. This limitation motivates the need for a post-processing tool similar to the SSA for more models; to do that, more reforecast products are needed. It is hoped that this work will motivate the creation of more spread anomaly products as well as more reforecast products to utilize for comparison.

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